**Title: Genomic analyses reveal distinct demographic history and identify genetic changes underlying adaption, speciation and domestication in ducks**

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**Abstract**

Domestic ducks was considered to being tamed from the mallard or the descendant of the mallard and the spot-billed duck. Domestic ducks showed remarkable phenotypic variation in morphology, physiology and behavior. However, molecular genetics for ducks' origin and their phenotypic variation are still poorly studied.

Here we presented the mallard and the spot-billed genomes and performed whole-genome sequencing on eight domestic duck breeds and eight wild duck species. Surprisingly, Analyses on these data supports a model that domestic ducks diverged from their closely wild lineage (mallard duck and spot-billed duck) at the last glacial period (LGP, 100~300 kilo years ago (Kyr)). Wild lineage further speciated into mallard duck and spot-billed duck about 70 Kyr, whereas domestic lineage decreased their population size through the LGP. A scan of wild duck genomes by comparing with domestic duck genomes identified a number of locus that may have been affected by positive selection in ancestral wild ducks after their divergence from domestic lineages. Function analyses suggested that those genes usually affecting organs development and energy metabolism that may involving long-distance flight ability. Further selective sweep analyses identified two genes associated with egg production and three genes related to feeding modulation being under selection in domestic ducks.These analyses unravel a distinct evolutionary pattern of ducks and two wild duck *de novo* genomes also provide a novel resource for speciation study.

## Background

As two major concerns of evolutionary study, both domestication and speciation (or population divergence) benefited from the advance of sequencing technology(1-9). The genome assembly and characterizing of genetic variation in many domestic and wild animals have facilitated the revealing of their complex demographic history, which provides profound knowledges for both domestication and speciation studies (6-8, 10). We previously drew the draft of a Beijing duck genome and identified genomic variation within this assembly(11). However, as the most economical important waterfowl and one of the principal natural hosts of influenza A viruses, genetic variations of duck were not well characterized. And debates about domestic ducks being domesticated from the mallard duck or the descendant of mallard duck and spot-billed duck are still continuing.

Until now there are more than 140 wild duck species across the world and most of them can hybridize each other (12, 13). Phylogeographic structure of dabbling ducks using 5'-end sequencing of the mtDNA control region and ODC-6 of nuclear DNA demonstrated that the mallard duck diverged from its closely population (spot-billed duck) recently, implying they are good resource for speciation studies (14). Domestic ducks are considered to originated from East Asian and Europe and show broad phenotypic difference from these wild duck species, such as loss of flight and migration ability, reducing of brain size, increasing of body size and fertility(15-19). Previous studies based on similar outlooks habits and distributions provide two hypotheses about domestic ducks' origin. One hypothesis suggested that domestic ducks were originated from the mallards, while the other argued that they were tamed from descents of the mallard (*Anas platyrhynchos*) and spot-billed (*Anas zonorphymcha*)(20). Until now, evolutionary patterns of wild and domestic ducks in genome-wide scale are poorly understand.

## Results

Here, we report high-quality drafts of the mallard duck and spot-billed duck respectively, and compare them to the genome of Beijing ducks and other birds (SI Appendix; Fig. S1; Fig. S3-S11; Table S1-S9). We also generate a total of 1,219 Gb whole-genome resequencing data of 35 wild ducks from eight species (gadwall (n=1), baikal teal (n=1), one pintail (n=1), falcated teal (n=1), common teal (n=1), wigeons (n=3), spot-billed duck (n=13) and mallard duck (n=14)), 94 domesticated ducks from six Chinese breeds, 20 ducks from two commercial breeds (Cherry and Campebell ducks), together with 20 individuals from two outgroups (ten Muscovy ducks and ten Taihu geese)(Fig. S2; Table S10) (mean coverage for mallard, spot-billed and Beijing duck is 12x).

Variants were called using GATK for each population and different combination separately, applying several quality filters. 44,852,612 SNPs were identified in the 10 duck populations (eight domestic duck breeds and two wild duck populations) 20,676,377 and 42,012,846 SNPs were detected in domestic ducks and two wild duck populations (the mallard duck and spot-billed duck), respectively, with 18,093,115 SNPs shared (Fig. S12). Moreover, mallard and spot-billed shared comparative number SNPs with domestic ducks (SI Appendix).

**Domestic duck originated in a lineage distinct from the mallard duck and spot-billed duck**

We initially explored the genetic relationships of domestic and wild ducks by performed a principal component analysis (PCA) using genome wide SNPs from 27 Shaoxing, 27 Beijing ducks representing domestic ducks and 35 wild ducks. Expectedly, the domestic ducks (Shaoxing and Beijing ducks) clustered to mallard and spot-billed ducks, while separated clearly from other wild species (Fig. 1a). Such genetic relationships were also identified by further PCA analysis and tree analysis based on the IBS (identical by state) distance of those 89 individuals (Fig. 1b; SI Appendix; Fig. S13, S14). It supported that mallard duck and spot-billed duck had a more closely genetic relationship with domestic populations than other wild species.

We then asked whether domestic ducks were originated from mallard duck or the descents of the mallard and spot-billed ducks (20). We performed a 3-population test for admixture by taking the mallard duck and spot-billed duck as ancestral populations, Beijing and Shaoxing as target populations through a Block Jackknife analysis that corrected for linkage disequilibrium among SNPs (21, 22). Surprisingly, this effort obtained significantly positive scores for these four groups (f3 = 0.007279, Z score = 23.834; Table S11) and did not support that domestic ducks were descents of the mallard duck and spot-billed ducks.

We further performed genetic structure analysis using the ADMIXTURE to estimate ancestry among individuals of the four populations (Beijing, Shaoxing, the mallard duck and spot-billed duck) with *K* (the number of ancestry components) ranging from two to seven(23). This analysis indicated that domestic ducks had a homogeneous genetic background whereas the wild ducks showed a heterogeneous genetic background which is consistent with PCA (Fig. 1a,d). However, the wild ancestor duck started to divide into mallard and spot-billed when *K* were increased to four that is following the separation between the wild ducks and domestic ducks with when $K = 2$ which have lowest cross-validation error(Fig. 1d; SI Appendix; Fig. S15). These observations neither support domestic ducks were offspring of mallard and spot-billed ducks which means that traditional domestication hypothesis may not fit to genetic data whereas the $F_{ST}$ based phylogenetic tree indicate a population history compatible with the genetic structure analysis (Fig. 1 c-e).

To eliminate the effect of population drift within each domestic population which may also lead the separation between domestic and wild ducks previous than the two wild ducks' split in genetic structure analysis, and further confirm the tree topology of domestic duck and the two wild ducks, we randomly selected and polarized 20,333 unlinked SNPs from these 10 duck populations with two outgroups (Muscovy ducks) for a diffusion approach demographic simulation (∂a∂i)(Methods)(24). We estimated the divergent time of the domestic ducks (quantize all eight domestic breeds into one population) and two wild duck populations (mallard and spot-billed) using two simple but informative models, which one took and the other did not take migration into account (Fig. 2a; SI Appendix). One hundred bootstrap simulations of the two models support that both the divergent times between two wild duck populations and domestic duck were almost equal (100~300 Kyr) and longer than the corresponding one between mallard and spot-billed ducks (~70 Kyr, Fig. 2b-c; SI Appendix; Table S12, Fig. S16-17). These results further supported that the ancestor of domestic duck split from the ancestor of mallard and spot-billed ducks before the two wild ducks' divergence.

Moreover, we performed a haploid pairwise sequential Markovian coalescent (PSMC) analysis to reconstruct the demographic history and estimated the effective population size ($N_e$) change along time among the mallard, spot-billed, Beijing and Shaoxing duck. This analysis suggested that ancestors of these four populations hold a comparative level in effective population size before 110 Kyr ago (beginning of LGP). However, ancestors of the two wild ducks and domestic ducks seemed to be under differently demographic histories after LGP. During this period, the effective populations of Beijing and Shaoxing duck ancestors decreased dramatically and continually. It is contrast to the case in the mallard and spot-billed duck ancestor, where their effective populations didn't decrease or even experienced once or twice times of dramatic increasing (Fig. 2d; SI Appendix; Fig. S18). PSMC can also be informally used to infer divergent times when $N_e$ trajectories that are overlapping diverge moving forward in time towards the present (25, 26). Interestingly, the divergent time was roughly coincident with the one inferred from ∂a∂i. Such consistency further encouraged us to infer that the ancestor of domestic duck diverged from the ancestor of mallard duck and spot-billed duck at about the beginning of the LGP. Similar to $N_e$ estimated by the PSMC analysis, we found that populations size of domestic population decreased whereas these of both the two wild duck species increased after split between wild and domestic population when using ∂a∂i to simulated a more complex model (i.e.

populations' size change exponentially after split) of divergence between populations (SI Appendix).

**The early ancestors of mallard duck and spot-billed duck might benefit from adaptive selections**

It is interesting that different tendency of population size change occurred after divergence between wild lineage and domestic lineage during LGP. One possibility might be that nature selection granted the mallard duck and spot-billed duck ancestors, but not domestic duck ancestors, new advantage traits to adjust environment change after their split. Consequently, environment changes lead the domestic ancestor population size decreased, whereas wild ancestor population size even increased (Fig. 2d; SI Appendix).We detected positive selections in the early ancestors of these two wild populations after their divergence with domestic lineage, and identified a total of 424 putative selective sweep regions, with a total size of 10.86 Mb, using thresholds of 4 standard deviations away from the mean (Fig. 3a-b; Fig. S19; Methods)(27, 28). Among them, 168 selective sweeps harbored 202 protein coding genes, while the remained ones don't contain annotated protein coding genes (SI Appendix; Table S13). Recent studies indicated that positive selections in intergenic and intragenic regions were associated with phenotypic variation (4, 7, 27). We then extended 180 kb in both upstream and downstream the regions containing no annotated genes. This effort identified 162 closest genes in 156 of these extended regions (Table S13).

After that, we performed gene ontology (GO) analysis of 306 genes (12 of them detected in domestic selection were removed, Methods) in or closest (extended 180 kb in both upstream and downstream) to selective sweeps regions (SI Appendix) and highlighted 343 enriched categories related to 236 biological process (Table S14, S15). Interestingly, the most remarked categories were related to a series of organs and system, such as morphogenesis and development of lung, heart, metanephros, aortic and cardiac muscle, and functions of blood circulatory system, respiratory system, peripheral nervous system and ventricular system development, rhythmic excitation (Table 2). Since long distance flight need adaptive changes in body shape, energy metabolism and spatial navigation, we proposed that selection on those genes associated with above functions might enhance the ability to fly long distance in the ancestor of the mallard duck and spot-billed duck (SI Appendix).

We further investigated functions of genes in putative selective sweep regions by surveying published literature and listed several genes might be related to the flight activity. For example, *KDR* associated with lymph vessel development were required for lung inflation at birth and enhance the function of respiratory system (29). Selection of two genes including *DNAJA1* stimulates ATP hydrolysis(30) and *MRPS27* help in protein synthesis within the mitochondrion(31) might contribute to supply high energy for sustained flight. We also noted that *PARK2* were under selection when use a lower threshold (S=-3). *PARK2* promotes the autophagic degradation of dysfunctional depolarized mitochondria which might suggest an adaptation to facilitate degradation of damaged mitochondria caused by products of elevated metabolic rate. Previous studies found some genes play important roles in energic metabolism as a ferrous iron transporter (i.e *SLC39A14*) or positive regulation of fever generation (i.e. *PTGS2*)(32, 33). Positive selection on these genes in the early mallard and spot-billed duck ancestor might enhance their ability to transport oxygen in blood and adapt to cold and hypoxic environments of high altitude during flying. Interestingly, we found some genes involving retina and forebrain regionalization may imply functions related to magnetic navigation ability.

In addition, we examined functional activity of four mutations locating in intron, UTR or downstream of *EFNA5*, *ISL1* and *MRPS27*, respectively. Three of the four sites with wild type showed significantly higher gene expression than domestic genotype in luciferase activity assay (Fig. 3d; Table S22; Fig. S29).

**The genomic islands of mallard and spot-billed duck divergence**

We estimated that the mallard ducks diverged from the spot-billed ducks about 70 Kyrs using two split models (Fig. 2a-b). This divergent time was consistent with that one counted using mtDNA sequences, and even less than the corresponding one (about 340 Kyrs~2 Myrs) of two flycatcher species, which were studied extensively in speciation divergence(5, 10, 14). Fixed difference ($d_f$) for 40kb bins, an indication of species divergence, were used to detect 'divergence peak' regions which show highly elevated divergence up to 50 times than genomic median, presented a genomic landscape that is sharp contrast to the case of flycatchers, where using the same criterion. Fewer $d_f$s (4,386) with larger proportion (80%) contained in "divergence peak" is distributed in smaller genomic regions (0.6% of duck genome) than flycatchers (SI Appendix). This difference might due to that the two wild ducks is in an earlier stage of speciation than flycatchers(10). Moreover, almost all of these "divergent peaks" were located in scaffolds homologous to the chicken "Z" chromosome. This may indicate an analogous situation to flycatchers' speciation (Discussion Section; Fig. S22).

Other genomic parameters ($F_{ST}$ (fixation index), dxy (sequence divergence), $\pi$ (nucleotide diversity) and Tajima's D) for 40kb bins shown a similar pattern to many speciation studies. Observations revealed highly heterogeneous genomic landscapes of differentiation when using elevated $F_{ST}$ (4-fold standard deviations away from the mean, "differentiation islands") as an indication of genomic divergence (5, 9, 34-36)(Fig. 4, Fig. S23; Table S16), and significant difference between Z chromosomes and autosomes, nucleotide diversity and SNP density of differentiation islands were reduced (Fig. 5c-f; SI Appendix; Table S17). However, dxy were not elevated and even reduced in most of differentiation islands, which is consistent with the observation in flycatchers whereas different with the case of Darwin's finches(9, 34). Moreover, Tajima's D of high differentiation were not reduced in both mallard and spot-billed, that is different from observations on other species(9, 34)(Discussion Section; Fig. 5f).

**Identification of selective sweep regions in domestic duck**

To detect genomic regions under domestic selection, we searched the duck genome for regions with reduced heterozygosity (Hp) and increased fixation index ($F_{ST}$) using methods similar to the chicken and dog genome (Fig. 3c;SI Appendix)(3, 4). We Z-transformed the Hp and $F_{ST}$ distributions of autosomes and sex-chromosomes separately, and focused on putatively selective sweep regions falling at least four standard deviations away from the mean (Z(Hp)<-4 and Z(Z($F_{ST}$)>4)(Fig. 3c;Methods;Fig. S24-S25). However, we found only small proportion selective sweep regions were predicted using both the two signals (Fig. S26). This may attributed to long time different evolutionary history since their split (100~300 Kyr). We therefore focused on these 137 regions with extremely low levels of Hp (average length = 63.6230 kb, average Hp = 0.2100 for autosomes, 0.2470 for sex-chromosomes)(SI Appendix). We found that 131 genes and 289 miRNAs contained in 80 of these CDRs, while 47 genes closest to one of 33 CDRs when we extended 180 kb in both upstream and downstream of each these 57 CDRs containing none annotated protein coding genes (Table S18, S20). Further gene ontology analysis indicated that 162 genes in or closest to these 137 CDRs were enriched to 145 biological process categories (p-value <0.05) representing developmental signaling and nervous system, lipid, carbohydrate and protein metabolic process, cell division and immune response (Table S21). Among them, two genes (*NLGN1* and *PER2*) involve into circadian sleep/wake cycle or is associated with eggs production(37), three genes (*MRAP2, MC4R* and *RMI1*) are associated with the modulation of feeding behavior or reducing of food intake in response to dietary excess, and four genes (*HSD17B4, HEXB, B4GALT1* and *CYP19A1*) are related to sexual characteristics, male courtship behavior or androgen metabolic process (38, 39)(Table S18-S21).

Among these candidate domestication genes, we examined functional activity of two mutations locating in introns of *GRIK1* and *LYST*, respectively. Domestic genotype of *LYST* mutation showed significantly higher gene expression acidity than wild genotype in luciferase activity assay (Fig. 3d; Table S21)

**Reanalysis provide further evidence on evolutionary relationship of domestic duck,**

**mallard wild duck and spot-billed wild duck**

As the genomic average genetic distance between mallard and spot-billed (mean $F_{ST}$: 0.0637) is significantly lower than it in mallard/spot-billed vs domestic pairs (mean $F_{ST}$: 0.2832, 0.2498)(Fig. 4b). To further investigate whether this higher genomic ”similarity” between mallard and spot-billed is caused by hybridization or phylogeny as we inferred above. We stratified the genomic divergence values of each windows by Δ DAF (or absolute ΔDAF) (derived allele frequency difference between the two populations) bins. According to recent research on introgression issue in chimpanzee, the chance of derive alleles being introduced through gene flow increases with the frequency in the donor population. Gene flow should introduce into receptor populations at low frequencies (40). So, we may expect lower divergence value in these higher ΔDAF (absolute ΔDAF) bins as it represented more probability of introgression. We did find decreased genetic distance ($F_{ST}$, dxy) between mallard/spot-billed and domestic ducks in higher ΔDAF bins (Fig. S30b). However, both $F_{ST}$ and dxy between mallard and spot-billed is increased with the ΔDAF (Fig. S30a). This observation suggested that this higher genomic “similarity” between the two wild duck species maybe not caused by introgression. Furthermore, we found that the mean ΔDAF value didn’t show significant difference between high differentiation regions and genomic background, which is consistent with recent research that gene flow is not the major factor in the formation of the genomic islands (Table S23) (34).

To further deciphering the introgression effect and It’s influence on evolutionary relationship inference of domestic and the two wild ducks, we performed ABBA-BABA test (assume that mallard, spot-billed, domestic duck is P1, P2, P3 respectively) using Muscovy duck as outgroup. It also suggested that genetic relationship between mallard and spot-billed ducks are closer than it between the two wild species and domestic(Fig. S31a). We also found evidence of gene flow between domestic and wild ducks which is consistent with above analysis. (Fig. 1d, D-statistics: -0.2972; jackknife std: 0.4189). In order to characterized the gene flow effect on the phylogenetic tree reconstruction, we used the same idea, ΔDAF, to test is there evidence that these derived allele sharing is caused by introgression (Fig. S31a). We stratified the number of BBAA (ABBA, BABA) sites by the ΔDAF bins between mallard and spot-billed ducks populations (spot-billed and domestic, mallard and domestic, respectively). The most majority proportion of BBAA sites were distributed in lower ΔDAF bins (-0.4 ~ 0.4), imply the major reason for excess derived allele sharing due to phylogeny rather than introgression (Fig. S31b left). As the majority number of SNPs distributed at lower ΔDAF bins. We further measured the proportion distribution of the BBAA sites among all SNPs (i.e. polymorphic in the compared two populations. i.e. Polymorphic sites in spot-billed and domestic, mallard and domestic, mallard and spot-billed for ABBA, BABA, BBAA, respectively). If the introgression exists between mallard and spot-billed (mallard and domestic, spot-billed and domestic), we may expect the proportion of BBAA sites would increase with the ΔDAF (or absolute ΔDAF in negative). We did find that the proportion of BABA, ABBA sites increased with ΔDAF, which indicate that the derived allele sharing between wild (mallard or spot-billed) and domestic populations is partly due to the introgression. Whereas the distribution of BBAA proportion across ΔDAF bins showed a different pattern that the BBAA rate has only little difference across ΔDAF bins, which indicate that absent evidence of introgression between mallard and spot-billed (Fig. S31b).

**Discussion**

How domestic animals originate and lead to striking morphological and behavioral changes are hot topics of evolutionary biology. Recently, the rapid development of high-throughput sequencing technologies has significantly boosted knowledges for origination and domestic process in dog, pig and rabbit, as well as chicken (6, 7, 25, 41, 42). Here we demonstrate the domestication process of duck and surprisingly found that domestic ducks were neither originated from the mallard ducks nor tamed from descents of mallard and spot-billed ducks. Instead, compelling evidence indicate that ducks were domesticated from an ancient wild duck lineage, which was close to the ancestor of mallard and spot-billed ducks. And we also found evidence shows that, compare to domestic duck, the ancestor of wild duck might acquire enhanced flight ability through early selection. It may contribute to their long-distance migration ability and the preservation of *Ne* through the LGP climate changes which may leading to the decrease of domestic lineage. In the context that accumulated studies have revealed that extant wild animal may not be the wild progenitors of its domestic forms (dog, horse, pig)(6, 43-45). We propose that our conclusion may help to explain the extinction (or hardly to find) of those progenitors and highlight the importance of the genetic architecture of wild for the understanding of remarkable phenotype changes in domestication (18, 25, 43, 46, 47). Meanwhile, we found that duck domestication was accompanied by selection at genes affecting neuronal activities (e.g. *SH3GL2* (42)), lekking behavior (e.g. *HEXB, B4GALT1* and *HSD17B4*). Moreover, Genes associated with lipid, carbohydrate, protein metabolic process and food intake control were under selection (e.g. *RMI1*, *MRAP2* and *MC4R* (8)). In summary, an adaption to food-rich and artificial feeding environment with development of agriculture may contribute to duck domestication.

The rapid speciation in mallard and spot-billed, which species may occur in only 70 kyr, provide excellent resource to speciation studies(34, 48, 49). As recent studies of this area have focused on high differentiation islands, our observation also shown the obvious extraordinary features of high differentiation regions. Think about the dynamic changes of these islands across speciation process may help to understand the drive process of speciation and the evolves of genomic differentiation landscape. In our observations, all "divergent peaks" were located in scaffolds homologous to 'Z' chicken chromosome (Fig. S22). Studies on flycatchers suggest that both species-specific male plumage traits and species recognition are located on the Z chromosome(50, 51). Similar with flycatchers, male plumage traits of mallard and spot-billed duck are quite different. This may indicate that "divergent peaks" on Z chromosome is associated to the difference in male-plumage traits between two wild duck species and their speciation. Although it's still unclear what trigger the speciation process, we propose that those limited few genes contained in these high "differentiation islands" and "divergent peaks" may provide critical clue due to they may in a very early speciation stage. Linked selection is suggested to dominantly drive the evolution of the genomic differentiation rather than gene flow(9). Not elevated dxy of "differentiation islands" in observations again, support this suggestion (Fig. S23). Furthermore, Tajima's D is not reduced in differentiation islands could also be best explained by new strong positive selection (Fig. 4). Tajima's D is used to detect selection of new mutations and correlated to difference between $\pi$ and the expectation of $\pi$ which is related to number of SNPs (52). Due to short divergence time, the mallard and spot-billed ducks did not deposit much new mutations to increase ($\theta_w$) after selection (new mutations increase diversity slowly and in a low frequency, mainly increase population mutation rate ($\theta_w$) rather than nucleotide diversity ($\pi$)), thus results in allele frequency spectra didn't skew towards rare alleles. Whereas collared and pied flycatcher (about 340 Kyr~2 Myr) and Darwin's finches reduce Tajima's D in their genomic divergent islands due to a long time after initial selection may associated with the driving force of speciation (10, 34). In the end, we argue that our article has presented distinct genetic resource between wild and domestic duck. Genetic resource of wild duck may have been significantly shaped through the global climate change during LGP, which making wild duck evolved new property to adaption to environment change, also consist of the stability of the ecology system. We also note evidence in genetic structure showed that mallard duck and spot-billed duck (especially spot-billed duck) have interspecific assignment to domestic ducks (Fig. 1d; Fig. S15. Especially when K=3,4). This may pose a significant threat to genetic diversity of wild stocks.

## Methods

### Sequence assembly and annotation

Sequencing was carried out with Illumina technology using genomic DNA of a wild female mallard and a wild female spot-billed collected by Aoji Duck Farm (Ningbo, Zhejiang) with hunting permission granted by the local government. Sequences from paired-end reads of 17 libraries for each individual with insert size ranging from 250 bp to 20,000 bp were assembled with SOAPdenovo and SSPACE software (Table S1). The super-scaffolds were positioned along the chromosomes based on the duck cytogenetic map and the comparative genomic map between duck and chicken (11, 53). A reference gene set for the mallard genome and the spot-billed genome were retrieved through merging the homologous set, the transcript set (predicted with 21 duck transcriptomes) and the *de novo* set(11).

**∂a∂i simulation and PSMC analysis**

For ∂a∂i simulation analysis, SNPs were sampled randomly each per 100 kb from the SNPs identified in wild populations (mallard and spot-billed) and domestic populations (Cherry Valley, Campbell and 6 Chinese duck breeds). SNPs were further filtered according to the coverage of resequencing data at their position with thresholds of 20-fold for pooled data and total number of genotyped alleles more than 50, 36, 26, 28 for Beijing, Shaoxing, mallard duck and spot-billed duck populations, respectively. We then inferred ancestral alleles of the above sampling filtered SNP using Muscovy ducks as outgroups using the following thresholds: (1) more than 30-fold coverage two outgroups; (2) the allele was fixed and same in outgroups. After that, we quantized the frequency of each SNP of 8 domestic duck populations into one population and simulated the divergence among spot-billed, mallard and “domestic” using split without migration, split with migration and IM models 50 bootstraps each (Fig. 3a; SI Appendix). *Nref* was calculated with the following format:

$Nref = \frac{\text{Watterson's estimator } \Theta}{4*\text{mutation rate} * \text{L}}$; L was length of the SNPs from and diluted by rate of SNP sampling.

For the PSMC analysis(54, 55), SNPs were filtered according to the coverage of resequencing data at their position with thresholds of 10- to 100-fold for Beijing, spot-billed and mallard, or 5- to 80-fold for Shaoxing. Effective population sizes were assessed by PSMC model analysis. We assumed that mutation rate per year to be 9.97e-10 and that the generation time to be one year for both $\partial a \partial$i simulation and PSMC analysis(56).

**Positive selection in early ancestor of mallard and spot-billed ducks**

To detect selection that occurred in wild duck linage (mallard and spot-billed ducks) after their divergence with domestic lineage, we selected SNPs according to the inferred ancestral alleles and using strict thresholds (28). We then calculated the derived allele frequency of 18,083,967 selected SNPs in both wild lineage and domestic lineage, and defined selective sweep regions of the early mallard duck and spot-billed duck ancestors using the thresholds of -0.2819 (4-fold standard deviations away from mean) for autosomes and -0.3089 for sex-chromosome (1.5-fold standard deviations away from mean), respectively.

As expected, the derived allele frequencies in both lineages are highly correlated (Fig. S18). This correlation was used to calculate the expected derived allele frequency in the domestic lineage as a function of the derived allele frequency in wild lineage. We then calculated the observed derived allele frequency in domestic lineage, and the log-transformed value of the ratio for the observed/expected derived allele frequency in sliding 20-kb windows with a step size of 10kb. After Z-transform the genome-wide distributions to arrive at a measure S. As mentioned previous, this method naturally disregards regions of recent selection or strong purifying selection in wild lineage. However, recent selective sweeps regions in domestic lineage could also reduce S value, thus we exclude regions that show lower zHp (CDRs) in domestic lineage.

**Availability of supporting data and materials**

**Data availability**

The genome assembly has been deposited in GenBank PRJNA392350. The duck, Taihu goose and Muscovy duck genome resequencing data under the bioproject PRJNA315043.

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**Author contributions**

Y.H.H and N.L. designed the project. Z.L.L, H.F.L., L.C., H.B.C., W.T.S. and E.G.R collected and purified DNA samples. Q.W.L., Y.Z, G.Y.F., H.F.L., Y.S.S., W.B.C., X.L.

and X.X. performed genome assembly, gene annotation and gene family evolution of the spot-billed and mallard genome. R.L performed phylogenetic evolution, selective sweep and speciation analysis. E.G.R. and X.X.W. carried out luciferase activity analysis. Y.H.H., R.L. and Q.W.L. wrote the manuscript. Y.H.H. and R.L. revised the manuscript.


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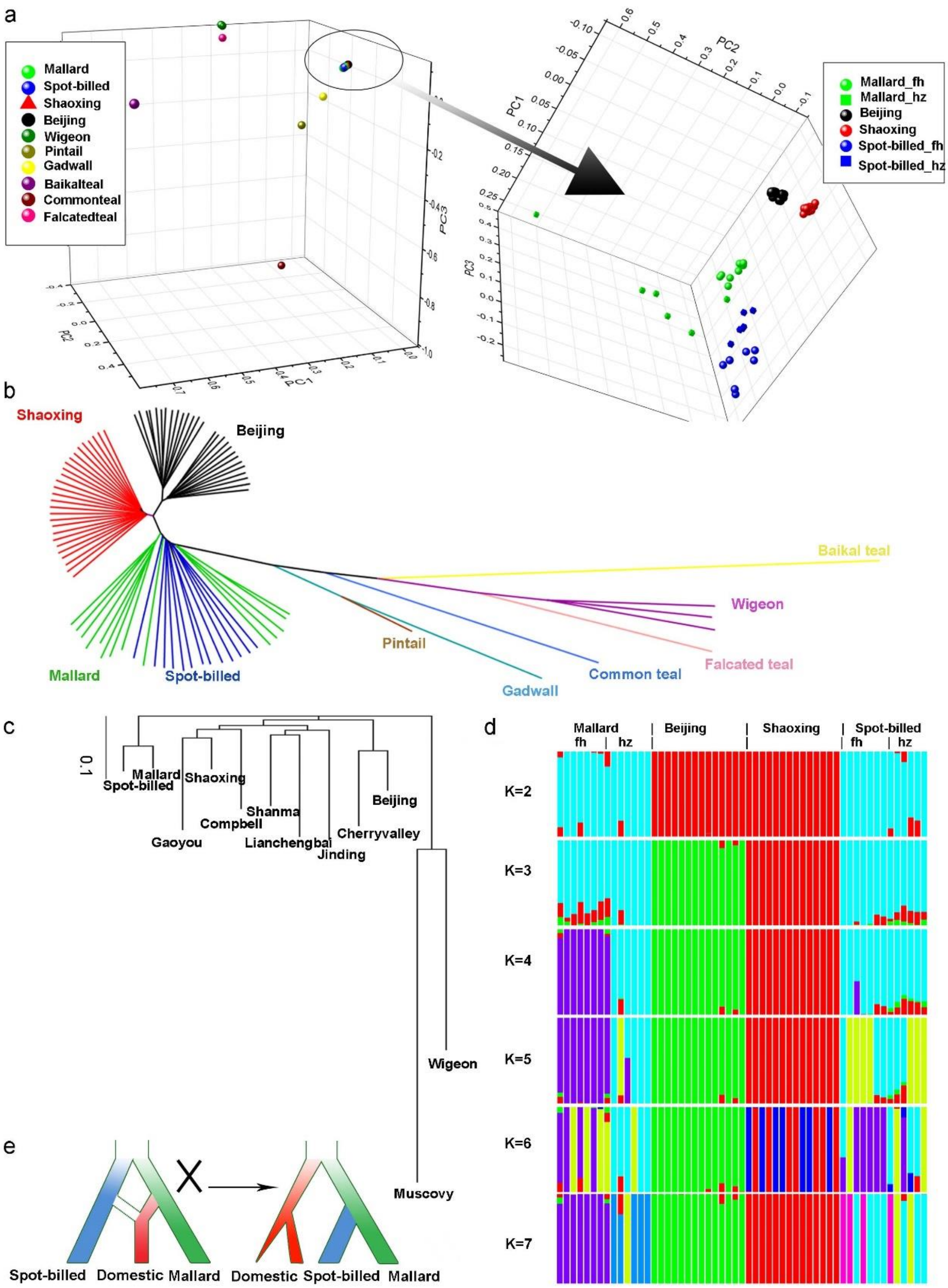
566 **Figure legends**




567

568 **Figure 1: Inference of phylogenetic topology in wild and domestic ducks.** Beijing

569 and Shaoxing ducks representing domestic ducks were used the Principal Component

Analysis (PCA), phylogenetic analysis based on identity-by-state (IBS) and genetic structure analyses. "_fh" and "_hz" mean spot-billed and mallard collected from Fenghua and Hangzhou respectively. (**a**) PCA plot constructed for Principal Component 1 (PC1), PC2 and PC3 using 10,525 high-quality filtered SNPs from 27 Beijing and 27 Shaoxing ducks and 35 wild ducks (one bailkal teal, one gadwall, one pintail one falcated teal, common teal, three wigeon, 13 spot-billeds and 14 mallards) shows that domestic ducks are close to spot-billed and mallard wild ducks. (**b**) Neighbor-joining tree based on IBS distance of 80,505,576 SNPs. (**c**) Neighbor-joining tree based on genome wide pairwise fixation index ($F_{ST}$) of 44,852,612 SNP. (**d**) Genetic structure analyses of 13 spot-billed and 14 mallard wild ducks, together with 14 Beijing and 14 Shaoxing domestic ducks, using admixture ranging *K* from two to seven. (**e**) The inferred demographic models for wild and domestic ducks inferred based on genetic analysis of **a**-**d** (right). The traditional demographic model, hypothesized that domestic ducks were originated from mallard or the descendant of a mixture of spot-billed and mallard (left) are incompatible with observations in neighbor-joining trees (**b**-**c**) and genetic data(**d**).

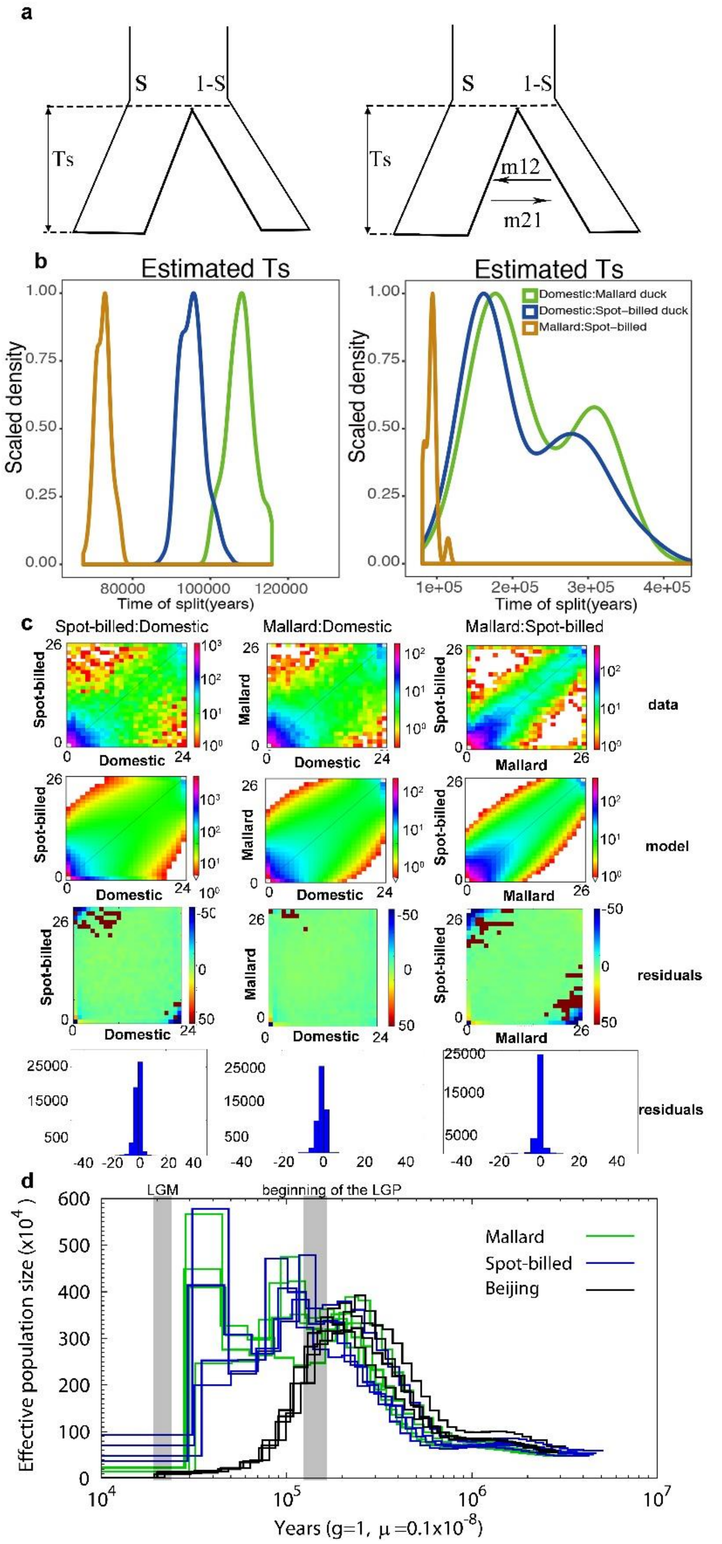



**Figure 2: Inference of joint demographic history of ducks.** Wild ducks included spot-billed and mallard ducks. Domestic ducks included Beijing, Cherryvalley, Compbel, Gaoyou, Jingding, Lianchengbai, Shaoxing and Shanma. (**a**) Illustration of

the two models (with/without migration) used in the history inference through the ∂a∂i analysis. Parameters corresponding to the code that were provided in supporting information. (**b**) Distribution of estimate Ts (divergence time) of each model in 100 simulations between spot-billed and mallard (orange), mallard and domestic lineage (green), spot-billed and domestic lineage (blue), respectively. (**c**) Joint Site Frequency Spectrum (SFS) for observed data, simulation data and residuals. (**d**) Demographic history inference of mallard duck, spot-billed duck and Beijing duck through the PSMC analysis.

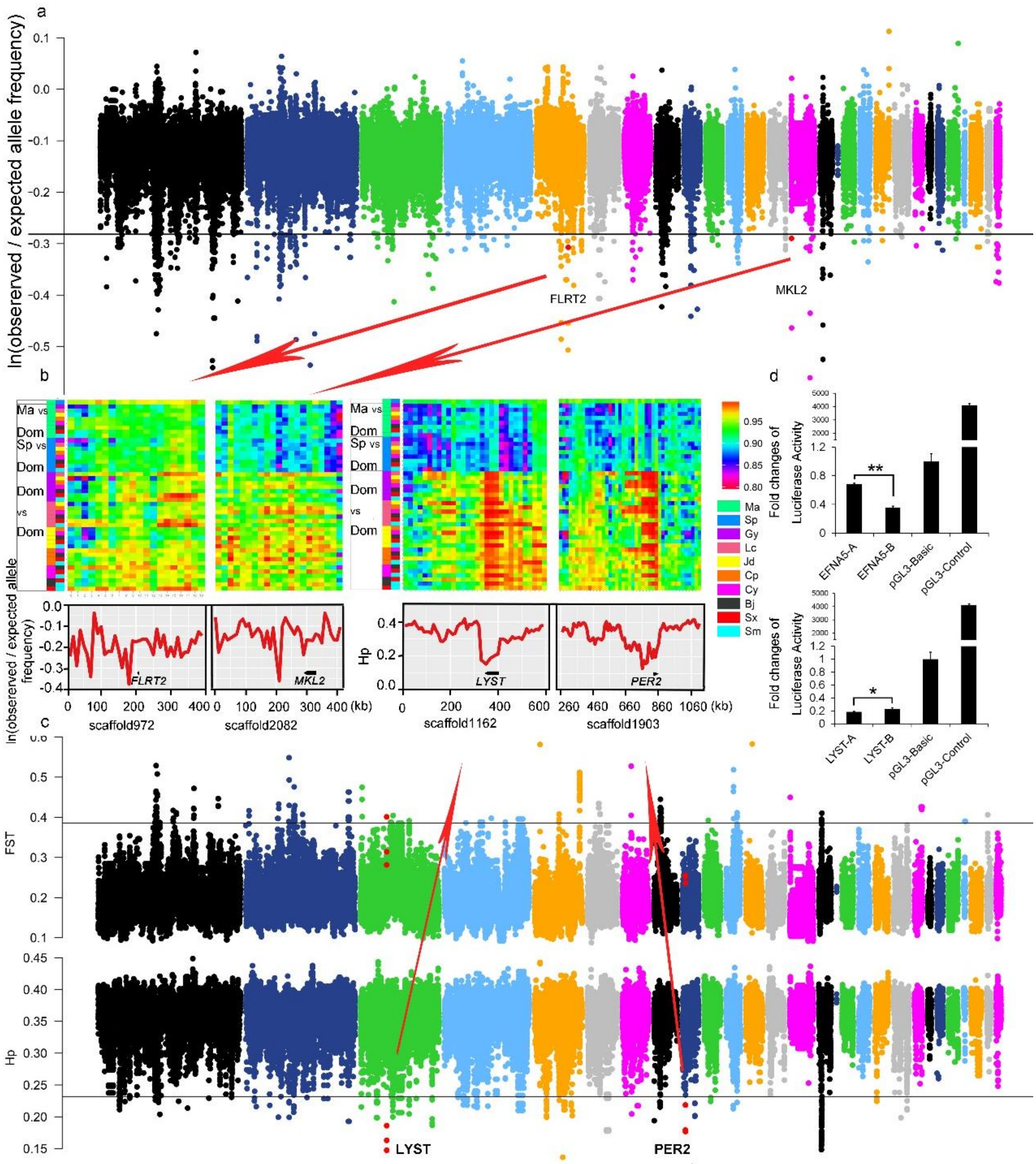



**Figure 3: Summary of selective sweep analyses of the early wild duck lineage and domestic duck.** (**a**) Distribution of ln(observed/expected derived allele frequency) that is calculated in 20 kb window with a sliding size of 10 kb along scaffolds that is arranged according to their alignment to the chicken autosomes. The horizontal black line indicates the threshold at ln(observed/expected derived allele frequency) = -0.3136 (mean of ln values - 4 standard deviations). Outlier regions that were detail visualized

by identity scores (IS) and re-computing values in more smaller window size in (**b**) were highlight with red dots. (**b**) Details of four putative selective sweep regions around the *FLRT2, MKL2, LYST* and *PER2* genes. Upside of each panel shown degree of haplotype sharing in pairwise comparisons among populations. Color boxes (left) indicate the comparison performed on that row. “Ma”, “Sp”, “Dom”, represent mallard, spot-billed and domestic duck respectively. Y-axis (downside) of *FLRT2,MKL2,* which is associated with retina/heart morphogenesis that were under seletion in wild ancestor, are ln(observed/expected derived allele frequency), whereas y-axis of *LYST,PER2* that were under selection in domestic duck are hp. (**c**) Distribution of $F_{ST}$ between the wild and the domestic duck, and Hp in the domestic duck that is calculated in 40 kb with a sliding size 20 kb along scaffolds that is arranged according their alignment to the chicken autosomes. The horizontal black line indicates the threshold at $F_{ST} = 0.3858$ or Hp=0.2319 (mean of ln values - 4 standard deviations). Red dots indicate the location of outlier genes (4 standard deviations away from hp means) that were detail visualized in (**b**). (**d**) Comparison in luciferase activity of major alleles of MRPS27 and EFNA5 in wild (spot-billed and mallard) and domestic (Beijing) duck in DF1 (chicken embryonic fibroblast) cells. “**A**” and “**B**” represent wild duck and domestic duck, respectively. “*” $P \leq 0.05$.

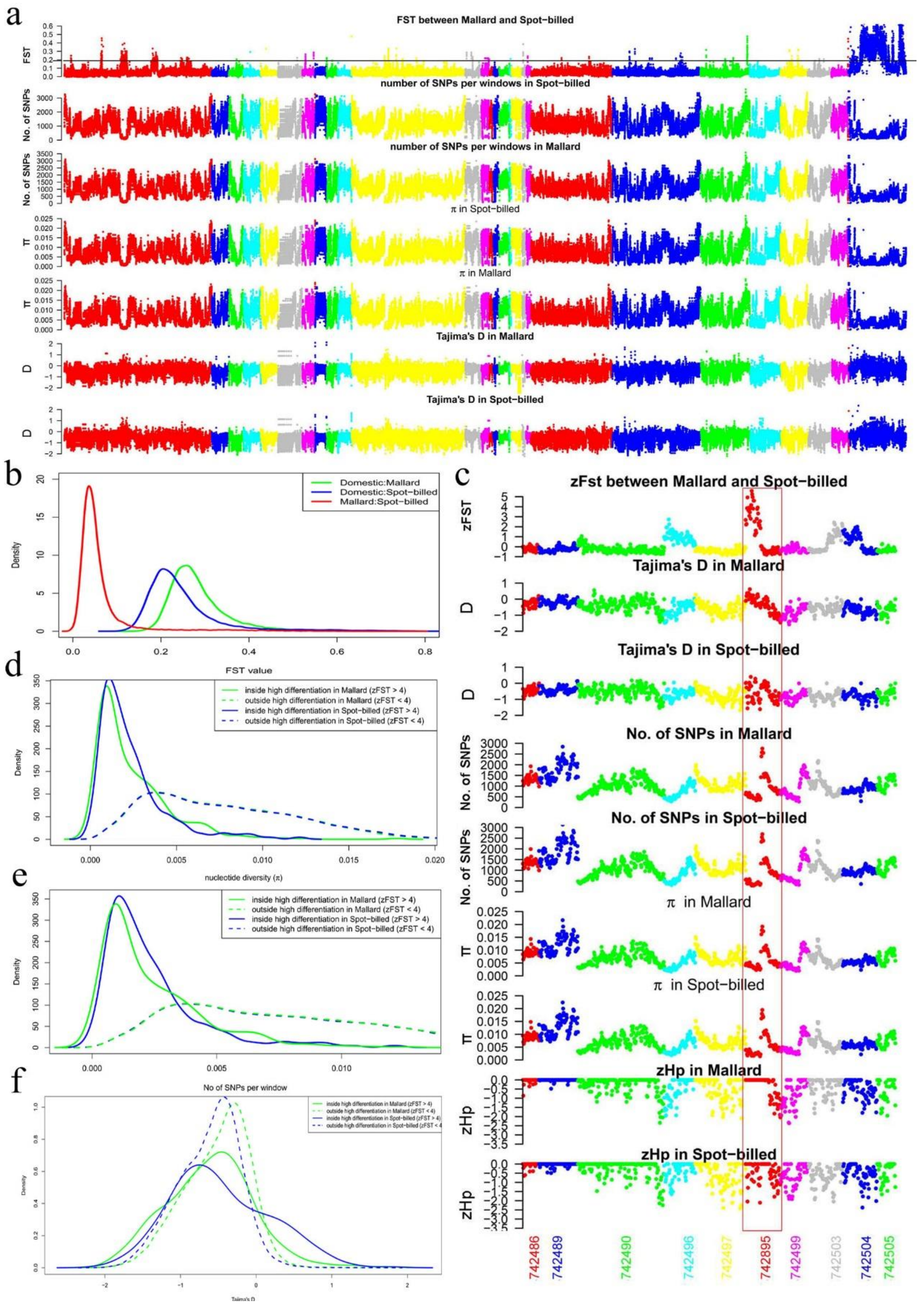


**Figure 5: The genomic landscape of species divergence between spot-billed and mallard ducks.** (**a**) Distribution of population genomic parameters of the Beijing duck assembly, in 40 kb windows sliding 20 kb steps. The blue scaffolds in the right were

referred as duck Z scaffolds. The black horizontal line indicates the threshold at $F_{ST}$ = 0.1888 (mean of $F_{ST}$ + 4 standard deviations, referred it as $Z(F_{ST}) = 4$). The plots show the $F_{ST}$ between spot-billed and mallard, number of SNPs per window, nucleotide diversity ($\pi$), Tajima's D for spot-billed and mallard respectively. (**b**) Distribution of pairwise $F_{ST}$ for 40 kb windows sliding 20 kb steps. The red line shows the plot in the first row of (**a**). The blue and green line show the pairwise $F_{ST}$ between spot-billed/mallard and domestic ducks (combination of 8 domestic breeds). (**c**) Distribution of population genomic parameters along ten scaffolds. (**d-f**) Distribution of nucleotide diversity ($\pi$), number of SNPs per window and Tajima's D inside/outside high differentiation regions in Beijing duck assembly.

## Tables

**Table 1. Comparison in features of the mallard, spot-billed and Beijing duck genomes**

| Genomic features | Mallard | Spot-billed | Beijing duck |
|---|---|---|---|
| Assembled genome size (Gb) | 1.27 | 1.31 | 1.11 |
| Contig N50 (bp) | 38,271 | 33,061 | 26,114 |
| Scaffold N50 (bp) | 2,489,142 | 2,054,876 | 1,233,631 |
| GC content (%) | 41.90 | 41.90 | 40.90 |
| Repeat rate (%) | 10.71 | 10.55 | 7.31 |
| Heterozygosity rate ($\times10^{-3}$) | 3.60 | 3.54 | 2.40 |
| Predicted protein-coding genes | 21,056 | 21,123 | 20,629 |

**Table 2. Top 32 enriched biological process terms among genes under positive selection in the early spot-billed and mallard ancestors**

| **Gene ontology term** | **P-value($10^{-3}$)** | **Number** | **Size** |
|---|---|---|---|
| positive regulation of tyrosine phosphorylation of Stat3 protein | 6.29E-02 | 4 | 12 |
| regulation of activin receptor signaling pathway | 0.000382 | 2 | 2 |
| positive regulation of cytosolic calcium ion concentration involved in phospholipase C-activating G-protein coupled signaling pathway | 0.000387 | 3 | 8 |
| detection of virus | 0.001131 | 2 | 3 |
| D-aspartate import | 0.001131 | 2 | 3 |
| metanephros morphogenesis | 0.002232 | 2 | 4 |
| L-glutamate import | 0.002232 | 2 | 4 |
| negative regulation of extrinsic apoptotic signaling pathway in absence of ligand | 0.003441 | 3 | 16 |
| positive regulation of granulocyte macrophage colony-stimulating factor production | 0.003672 | 2 | 5 |
| **respiratory system development** | **0.003672** | **2** | **5** |
| **determination of left/right symmetry** | **0.003701** | **5** | **53** |
| spinal cord motor neuron differentiation | 0.004119 | 3 | 17 |
| **lung morphogenesis** | **0.004119** | **3** | **17** |
| ionotropic glutamate receptor signaling pathway | 0.004872 | 3 | 18 |
| **ventricular cardiac muscle tissue morphogenesis** | **0.004872** | **3** | **18** |
| **dorsal/ventral pattern formation** | **0.005176** | **4** | **36** |
| negative regulation of peptidyl-threonine phosphorylation | 0.005437 | 2 | 6 |
| **serotonin metabolic process** | **0.005437** | **2** | **6** |
| **lung epithelium development** | **0.005437** | **2** | **6** |
| cytokine-mediated signaling pathway | 0.006291 | 4 | 38 |
| nucleotide-excision repair | 0.006613 | 3 | 20 |
| neutrophil chemotaxis | 0.006613 | 3 | 20 |
| negative regulation of epithelial cell proliferation | 0.006904 | 4 | 39 |
| **positive regulation of heart rate** | **0.007514** | **2** | **7** |
| inositol phosphate-mediated signaling | 0.007514 | 2 | 7 |
| **bud elongation involved in lung branching** | **0.007514** | **2** | **7** |
| **limb morphogenesis** | **0.009839** | **3** | **23** |
| long-chain fatty acid metabolic process | 0.00989 | 2 | 8 |
| dicarboxylic acid transport | 0.00989 | 2 | 8 |
| interleukin-6-mediated signaling pathway | 0.00989 | 2 | 8 |
| thrombin receptor signaling pathway | 0.00989 | 2 | 8 |

| | | | |
|---|---|---|---|
| **response to wounding** | **0.012416** | **3** | **25** |

Full lists of enriched terms are presented in Table S15. Terms that considered to be associated with enhancement of long range flight ability were in bold. Size represents total number genes of this term.

# Supporting information for: Genomic analyses reveal distinct demographic history and identify genetic changes underlying adaption, speciation and domestication in ducks (SI Appendix)

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## Contents Tables

## Supplementary text

### 1. Genome sequencing and assembly

We generated 262.45 Gb (~209-fold) pair-end reads of a female mallard duck and 275.20 Gb pair-end reads (~220-fold) of a female spot-billed duck, respectively (Table S1,Fig. S2-S3). Using methods similar to those applied to the Beijing duck genome (1), we generated a high-quality mallard assembly consisting of 61,591 scaffolds and covering 1.27 Gb, and estimated this assembly having a heterozygosity rate $3.61\times10^{-3}$ higher than the corresponding in Beijing duck genome ($2.40\times10^{-3}$). The contig N50 and scaffold N50 values of the mallard genome assembly were 38.27 Kb and 2.49 Mb, respectively (Table 1; Table S2). We also generated a high-quality spot-billed genome assembly containing 67,685 scaffolds and covering 1.31 Gb. Similarly, we found that spot-billed assembly had a high heterozygosity rate ($3.54\times10^{-3}$). The contig N50 and scaffold N50 values of the spot-billed assembly were 33.06 Kb and 2.05 Mb, respectively (Table 1; Table S2). We then estimated coverage of the mallard duck and spot-billed duck assemblies with alignment to seven finished BACs (completed independently using Sanger sequencing technology) and 319,996 duck ESTs assembled in our previous duck genome project using BLAT (1, 2). This analysis suggested that 7 BACs covering 640 Kb on chromosomes 1, 3 and 4 were aligned over more than 92% of their lengths (Fig. S4-S5, Table S3), and greater than 98% of 319,996 ESTs were aligned to both the mallard duck and spot-billed duck assemblies (Table S4).

We annotated transposable elements (TE) of the mallard duck and spot-billed duck assemblies using a combined pipeline. This effort found that the mallard duck and spot-billed duck assemblies contained ~136 Mb and ~139 Mb TE sequences, respectively, accounting for about 10% of their assemblies, respectively (Table S6). Among those TE elements, the LINE (long interspersed elements) type CR1 (chicken repeat 1) transposons are of the most abundance, a situation similar as ones in Beijing duck and chicken. Phylogenetic analysis of CR1 in mallard duck, spot-billed duck, Beijing duck and chicken revealed two duck-specific CR1 clades (clade 1 and 3), in which large variation of copy numbers were shown between the two wild ducks and Beijing duck. This implies that, as a main class of TE, CR1 evolved rapidly among different avian species and distinct duck lineages. (Fig. 1a; Fig. S6).

We predicted 21,056 and 21,123 protein-coding genes, which constitutes approximately 2.21% of the mallard genome and 2.13% of the spot-billed genome, using the BGI pipelines (Table S7). These numbers are slightly larger than the corresponding in the domestic duck (Beijing duck) genome (20,629 protein-coding genes) (Table 1). Of the 21,056 mallard genes, 12,817 were mapped to categories established by the Gene Ontology (GO) project, 15,878 had orthologs in the Kyoto Encyclopedia of Genes and Genomes (KEGG) database, and 14,894 were supported by the duck ESTs produced in our previous duck genome project (Fig. S7). These numbers are similar to these of the spot-billed, where 12,815 genes were mapped to GO project, 15,803 had orthologs in KEGG database and 14,827 were supported by the duck ESTs (Fig. S8).

We then constructed gene families using the above mallard duck and spot-billed reference gene sets, our previous Beijing duck gene set, together with a combined outgroup gene sets from chicken, turkey, flycatcher, ostrich, mouse and human (Fig. S9, Table S8). Interestingly, two wild ducks were similar and their ancestors were clustered to the domestic (Beijing) duck (Fig. 1b). We further examined large-scale differences in gene complements between one domestic duck and two wild ducks. We found that 1,305 gene families were specific to wild duck and this number was larger than the corresponding of Beijing duck (497 gene families) (Fig. S10). Among 19,405 gene families detected in the above three duck genomes, 25 were significantly expanded in wild duck. GO enrichment analysis showed that those expanded genes were significantly over represented in biological processes like chromatin organization, cell-cell adhesion, ATP synthesis and immune response. This implies enhanced genome regulation, energy metabolic activity and response to stimulation along with the speciation of the two wild ducks, which might be beneficial to their adaption to new environments (Table S9; Fig. 1c-d).

## 1.1 Genome sequencing

We extracted genomic DNA with standard molecular biology techniques from blood and random fragmented DNA for library construction. For short insert size libraries, 5 μg of DNA were fragmented to 250-800bp, end-repaired, A-tailed and ligated to Illumina paired-end adapters (Illumina). The ligated fragments were size selected at 250, 350 and 800 bp on agarose gel and amplified by PCR to yield the corresponding short insert size libraries. For mate-pair library construction, 20-40μg DNA were sheared to the desired insert size using nebulization for 2 kb or HydroShear (Covaris) for 5 kb, 10 kb and 20 kb. Then the DNA fragments were end-repaired using biotinylated nucleotide analogues and circularized by intramolecular ligation Circular DNA molecules were sheared with Adaptive Focused Acoustic (Covaris) to an average size of 500 bp. Biotinylated fragments were purified with Dynabeads® M-280 Streptavidin beads (Invitrogen), end-repaired, A-tailed and ligated to Illumina paired-end adapters, PCR amplification and agarose gel electrophoresis for size-selection. Then sequence all these libraries with Illumina Hiseq2000 platform.

## 1.2 Raw data filtering and genome size estimation

To reduce the effect of sequencing error or low-quality reads for genome assembly, we took a series of checking and filtering steps on raw reads. We filtered: 1) Reads having an 'N' over 10% of their length; 2) Reads with more than 40 bp low quality base (quality score $\leq$ 7); 3) Reads with more than 10 bp adapter sequences (allowing $\leq$ 3 bp mismatches); 4) Short insert size paired-end reads that were overlapped ($\geq$ 10bp); 5) Redundant duplicated reads generated by PCR amplification in the library construction (Read1 and Read2 of two pairs of paired-end reads were identical, respectively). In total, 275.20 and 262.45 Gb of data were kept for D2B and D2L *de novo* genome assembly respectively.

Genome size can be estimated using K-mer analysis method, the principle is briefly depicted as follows: A K-mer is an artificial sequence division of K bp iteratively from sequencing reads. A read with L bp in length composes (L-K+1) K-mers. The frequency of each K-mer can then be calculated from the input data. The resulted K-mer frequencies should generally follow a Poisson distribution, except for a high proportion of low frequencies due to sequencing errors. The genome size G can be estimated as G= K_num/K_expect, where K_num is the total number of K-mer, and K_expect is the expected value of the poisson distribution. We performed K-mer analysis (K=17) using about 21-27X filtered data, and estimated the mallard and spot-billed genome size to be about 1.15Gb and 1.14Gb, respectively (Figure S1).

### 1.3 Genome assembly and super-scaffold construction

At first, we assembled the two genomes using SOAP *de novo*(3) for contig and scaffold construction. Detailed assembly steps and the data used in each step: 1) Constructed *de Bruijn* graph with reads from the short insert size libraries using K-mer size 29 bp, which means the parameter –K is set as 29. 2) Simplified the *de Bruijn* graph for contigs construction by removal tips, merging bubbles, removal the low coverage of the connection and resolve the small repeats. 3) Constructed scaffolds with short pair-end and mate-pair libraries step by step. 4) Filled in gaps inside the constructed scaffolds with gap closure step.

### 1.4 Quality evaluation of the genome assembly

We made use of known EST and BAC sequences to evaluate the accuracy of genome assembly. We mapped 7 BACs(2) and 319,996 ESTs of Beijing duck onto the two assemblies using BLAT (4) with default parameters. We found that, for those 7 BACs totally spanning 640 Kb, more than 92% of the spanning were aligned well to the assemblies (Figure S3 and S4). In addition, around 94% of the ESTs were mapped to individual scaffold of the assemblies (Table S6).

Excessive heterozygosity at regional homologous chromosomes usually lead to failure of collapsing the two haplotype sequences, and thus results in redundant sequences in the assembly. To determine whether there is any case of redundant sequences in our assemblies, we performed a self-to-self alignment of the assemblies. Then found the total length of the scaffolds with the coverage rate is more than 50% is 11.3Mb and 10.3Mb in female mallard and a female spot-billed genome, respectively. That analysis indicated there was almost no redundant sequence in the two assemblies and we masked these scaffolds with coverage rate more than 50% for further deep analysis.

## 2. Repeat prediction, gene structure and gene functional annotation

### 2.1 Transposable elements annotation

We annotated transposable elements (TE) using the following steps:

1) We first scanned for repeat elements using LTR_FINDER(5) with the default parameters. The resulted TE sequences were used as TE database for RepeatMasker(6) search in step 2.

2) We then used RepeatMasker to identify repeat elements against the Repbase database(7) as well as the TE sequences founded in step1.

3) We further searched the assembly against the protein repetitive database provided in

RepeatMasker using RepeatProteinMask.

4) At last, we combined all the three set of predictions (two from step2 and one from step3) to generated an integrated repeat annotation result for each genome assembly.

5) Tandem repeats were identified by Tandem repeat finder(8) using the defaults of "Match=2, Mismatch=7, Delta=7, PM=80, PI=10, Minscore=50, and MaxPeriod=12".

### 2.2 Variation of CR1 (chicken repeat 1) transposon among ducks

From the result of repetitive elements annotation in section 2.1, we found that CR1 (chick repeat 1), a subtype of LINE retrotransposon, accounts for both the majority of both TE content and variation of TE content among mallard, spot-billed duck, Beijing duck and chicken assembly. In fact, we found there were more than 90Mb (7.4%) of CR1 subtype repeat in two wild ducks, but only approximately 60Mb (5.5%) in Beijing duck. Therefore, we further studied the variation of CR1 among those four species in a perspective of phylogeny. We downloaded non-LTR retrovirus reverse transcriptase (RT) of chicken retrotransposon CR1, complete consensus sequence from NCBI. We extracted the 252 amino acids (aa) long RT_nLTR_like protein sequences from the consensus, and then mapped it to the four genome assemblies using tblastn. We identified 2579, 2294, 617, 1737 RT_nLTR_like domains in the four genome assemblies (evalue < 1e-5, identity >50%, alignment length >200bp), respectively. We then performed multiple sequence alignment of all the identified RT_nLTR_like domains using MUSCLE(9) (version 3.8.31). Based on the alignment, a phylogenetic tree was subsequently reconstructed using FastTree.

### 2.3 Gene structural annotation

First, we applied the homolog based gene prediction method using protein datasets of *Anas platyrhynchos, Gallus gallus, Struthio camelus, Meleagris gallopavo, and Ficedula albico*. All the protein sequences were firstly mapped to the assembly using BLASTN with the parameters as "-e 1e-5 –F F –m 8". Then the aligned protein sequences were realigned to the assembly using GeneWise(10) for accurate spliced alignments.

Second, we used Beijing Duck transcriptome reads to polish our target gene annotation. We mapped the transcriptome reads to the two wild duck genome assemblies using TopHat(11) with its default settings and then constructed transcripts using Cufflinks. The process is the same as one preformed in the Beijing Duck genome project(12).

Finally, we combined all the gene models (five homologous gene models sets and the transcript gene models set) by GLEAN(13) to generate an consensus gene set.

### 2.4 Gene functional annotation

We aligned protein sequences of the final gene set to various protein databases with known functional knowledge, including InterPro(14) (iprscan_4.7) (profilescan, blastprodom, hmmsmart, hmmpanther, hmmpfam, fprintscan, patternScan), Gene Ontology(15), Swiss-Prot(16), TrEMBL(16) and KEGG(17) (release76), to identify conserved domains, putative biological functions and molecular pathways they may involve. In total, there are more than 87% of the annotated protein-coding gene models had at least one hit with known function (Table S5).

## 3. Evolutionary analysis

Mallard and spot-billed ducks, together with five other avian species (*Gallus gallus, Struthio camelus, Meleagris gallopavo, Ficedula albico and Anas platyrhynchos*, which is the Beijing duck) and two mammals (*Homo sapiens*, *Mus musculus*), were used in evolutionary analysis.

### 3.1 Gene family clustering

Protein-coding genes for *Gallus gallus, Struthio camelus, Meleagris gallopavo, and Ficedula albico* were downloaded from Ensembl release 80. The gene sets for *Anas platyrhynchos* was obtained from the BGI inner database. For gene loci with alternative splicing isoforms, only the transcript with the longest translation product was retained. We carried out an all-to-all alignment of all the collected protein sequences using BLASTP (E-value < 1E-7), and conjoined fragmental alignments using Solar(18). Then a simplified version of Treefam(19) methodology was used to cluster individual sequence into families based on the conjoined alignment (Table S11).

### 3.2 Phylogenetic tree reconstruction and divergent time estimation

After gene family clustering, single copy genes were selected to reconstruct phylogenetic tree of those nine species. Multiple sequence alignment of each gene family was performed by MUSCLE(9) (version 3.8.31). Phase1-base degenerate sites were extracted and concatenated to generate a super alignment matrix. We built phylogenetic tree using MrBayes(20) which takes advantage of both codon-based and amino acid-based algorithms and adjusts them to the topology of the species tree, to form a more accurate consensus tree according to phase1-base degenerate site. Divergence time of species were estimated using molecular clock model implemented by PAML mcmctree(21).

### 3.3 Segmental duplications

Segmental duplications (SDs) are duplicated blocks of genomic DNA typically ranging in size from 1–200 kb(22). To explore any difference in the content of SDs among two wild duck and Beijing duck genome, we detected SDs in their genome assembly using a whole-genome-alignment based method. We performed self-to-self whole genome alignment of these three assemblies using BLASTZ with the parameters: T=2 C=2 H=2000 Y=3400 L=6000 K=2200. Repetitive elements were masked in advance. We then filtered out self-alignment results (>0.85 similarity and >1000 aligned bp) and obtained more credible align block pairs. Finally, we realigned and filtered these block pairs (>0.9 similarity and >1000 aligned bp), producing a final version of SD regions in each genome. In summary, we identified 59.64 Mb (coving 4,367 genes) and 66.05 Mb (coving 4,864 genes) segmental duplications in mallard and spot-billed genome respectively, which are much larger than Beijing duck's 8.39 Mb (coving 1,133 genes). We further did pair-wise alignment among SDs detected in all these three assemblies to identify SDs specific to each of them.

## 4. Resequencing and population genomic analyses

### 4.1 Genome sequencing, Polymorphism identification, accuracy verification and its combine between populations

Genomic DNA libraries were prepared according to the manufacturer's instructions (Illumina). One DNA sample or pools of DNA samples from 10 individuals for each six duck breeds were used to generated pair-end reads with the Illumina technology (Table S1). Pair-end reads were mapped to the Beijing duck assembly (BGI_duck_1.0) using the bowtie2 with default parameters. After removing the reads which map to multiple place, we called SNPs using the Genome Analysis Toolkit (GATK) and filtered SNPs according to the coverage of resequencing data with thresholds of ≤ one third of the mean or ≥ three-fold of the mean and MQ value ≤ 28(23)

We produce BAM file for each individual sample or population pool according its sequencing strategy (Table S11). VCF files for each population were using "UnifiedGenotyper" of GATK by taken all processed BAM files of individual samples as input, or taken pool BAM file of a population as input. And all BAM files were processed by "REMOVE_DUPLICATES", "use unique mapped reads" and "IndelRealigner". In the functional validation section, 10 SNP sits were selected to be used to experimental verification its functional activity. 9 of these 10 were true SNPs but one was INDEL instead. We think that these results could reflect the high accuracy of SNP calling in some extent. In the following selection analyses or ∂a∂i simulation, all SNPs are from regions that homologous to chicken autosomes. We combine the VCF files by using a way like outer join of SQL (Structured Query Language) (https://github.com/vitogump/lifelit). For the variations that present in one VCF but absent in the another one, we fill it as it fixed as ref allele when the coverage (information from BAM) of this site reached the threshold.

### 4.2 Domestic duck shared a comparative number of SNPs with the mallard duck and spot-billed duck

Detailed analyses suggested that domestic ducks shared large numbers of SNPs (18,093,115) with wild duck, and hold comparative numbers of common SNPs with the mallard (15,904,435) and spot-billed (16,062,465) ducks. Moreover, we found that the numbers of mallard specific SNPs (heterozygous in mallard and homozygous in spot-billed) are comparative to the numbers of spot-billed specific SNPs (homozygous in mallard and heterozygous in spot-billed) in each of eight tested domestic duck breeds (Fig. S12).

### 4.3 Estimation of nucleotide diversity and population mutation rate

We estimated nucleotide diversity ($\pi$) and Watterson's estimator $\Theta$ using the PoPoolation package, which corrects biases causing by pooling and sequencing errors, in 40 kb non-overlapping windows(24, 25). We required per position a minimum coverage of 13,13,13,6 and 6 a maximum coverage of 150,150,500,100 and 100 for mallard, spot-billed, Beijing, Shaoxing, and other 8 domestic population each, and set min-count and min-covered-fraction to be 3 and 0.6 for all.

### 4.4 Analysis of linkage disequilibrium (LD), identical by state (IBS), principal component analysis (PCA), f3-statistics and population structure

LD level in Beijing, Shaoxing, mallard duck and spot-billed duck populations were measured through calculating the correlation coefficient ($r^2$) of alleles using the PLINK(26).

The parameters were set as follows: --ld-window-kb 500 --ld-window 99999 --ld-window-r2 0 --r2. To reduce effect of LD, SNPs were pruning by LD and subsequently used for the PCA, f3-statistics and population structure analyses (Fig. S11). We constructed phylogenetic trees by using neighbor-joining method with MEGA based on IBS distance matrix data of all individuals calculated by the PLINK using the above sampling SNPs(27). Principal component analysis (PCA) was performed using the GCTA64(28). The f3-statistics were calculated using the ADMIXTOOLS and genetic structure was inferred using the admixture program.

### 4.5 PCA, genetic structure and PSMC analysis

SNPs used in both PCA and genetic structure analysis were filtered using PLINK(26) (--indep-pairwise 500 50 0.4 --maf 0.05). First, we want to know the genetic relationship of all 8 wild duck species and these domestic populations as mentioned in main text Fig. 2a, which provided evidence that mallard duck and spot-billed duck having a far more closely genetic relationship with domestic duck than other wild species. To eliminate the unequal simple size effect, assembles difference effect and autosomes/sex difference effect, we supplemented PCA analysis by using same sample size for each population, with both mallard duck assemble and Beijing duck assemble (Figure S13). These results further supported our conclusion.

Second, in our genetic structure analysis, it is interesting that domestic lineage separated from wild duck first, and even subdivided into Beijing and Shaoxing previous than the mallard duck and spot-billed duck's subdivision. To further investigate this phenomenon and eliminate the coverage effect. We perform genetic structure analysis by using equal sample size and same coverage of each population with different populations combinations (Figure S14). These conclusions are consistent.

Third, in our PSMC analysis, we cutting the coverage of Beijing individuals into the same with Shaoxing individuals. They show the same curve with Shaoxing ducks (Figure S17). So, we propose that all domestic ducks should have the same PSMC curves.

Cutting sequence coverage was performed by samtools(29) (samtools view -bhs).

### 4.6 ∂a∂i simulation

There are two ways to compare the two phylogenetic trees (spot-billed, (mallard, domestic)) and (domestic, (mallard, spot-billed)). First, as some literates did, simulated the three populations split model and calculated the likelihoods to compare the fit of each model (30).

We think this is unsuitable to our date. Because this would take more parameters into model that would significantly increase the difficult to simulate and is very time consuming. Furthermore, the likelihoods are influenced by multiple parameters. We hardly give a property demographic model and constraint the parameters, since there is lack of reliable evidence (beside genetic data) about the real evolutionary history of ducks. Moreover, both PCA and structure analysis shows that wild duck populations are structured that is challenging with ∂a∂i because the dadi model treats populations each as distinct homogenous entities. So, the best way is simplifying the model and making it easy to be get convergent parameters.

We used an alternated way by simulating paired divergence and comparing the distribution of divergence time. If (Fig. 1e left) is the correct evolutionary relationship, then we expected longer divergence time between mallard duck and spot-billed than the ones between the two wild ducks and the domestic ducks. Otherwise if (Fig. 1e right) is the correct tree topology. We would expect the divergence time between two wild duck populations and domestic population were similar and longer than the one between mallard duck and spot-billed duck.

As we only focus on divergence time, we first used an extreme simple model with only two parameters (Fig. 2a left). This way would eliminate the influence between different parameters. Especially taking migration rate into simulation would have a significate impact in divergence time which could be purely due to mathematical property of ∂a∂i software.

The distribution of each estimated parameter and the confident intervals were presented in Figure S15 and Table S12, respectively. And the comparison of divergence time between the domestic and mallard/spot-billed duck, mallard duck and spot-billed duck were shown in Fig. 2b left.

We then add migration parameters into model (Fig. 2a right). The distribution of each estimated parameter was presented in Figure S16 that show bimodality distributions rather than normal, so we don't calculate the confident intervals. And the comparison of divergence time between the domestic and mallard/spot-billed duck, mallard duck and spot-billed duck under this model were shown in Figure 3b right. The maximum likelihood Ts between domestic and mallard/spot-billed duck, mallard duck and spot-billed duck are 333,504 years ago, 334,738 years ago and 74,915 years ago respectively. Combined the two models' result, we inferred that the ancestor of wild duck diverged from domestic duck about 100Kyr~300Kyr, and subsequently diverged into mallard duck and spot-billed duck about 70Kyr.

In conclude, dadi simulation on both models, considering or not considering the migration, are support the phylogenetic tree that is inferred from our previous population structure analysis. Although dadi software cannot distinguish short-divergence with low migration from long divergence with high-migration, this usually cause the estimate Ts not convergent or convergent into two peaks (smaller Ts accompanied by high migration or bigger Ts coupled with low migration rate). In our second simulations (with migration), estimate Ts does increased, when compared with first model, due to introduce migration. But the Ts still convergent into two peaks, higher estimated Ts with higher estimated migration rate whereas shorter Ts with lower migration rate. a reasonable explanation could be: although dadi have difficult to determine longer or shorter Ts, thus having simulation on both situation and produce two peaks of estimate Ts, the Ts between mallard and spot-billed is still significant smaller than Ts between wild and domestic one in any case.

In additional, we add parameters to allow population change exponentially (IM_2 model). We found that many of simulations with IM model can't get the convergent estimates or get estimates with some parameters hit the boundary, especially for divergence between mallard duck and spot-billed duck. This may be because as the number of parameters (2 for "split_no_M" model, 4 for "split_M" and 7 for "IM_2" model) increased, the correlation between parameters (which is a purely mathematical property of) will significantly increase the difficulty to getting the correct estimates. And the two wild populations are highly structured and exist many fixed difference sites which is particular challenging with ∂a∂i. Furthermore, the distribution of each parameters shown two or more peaks. So, we hardly estimated results. But we found that 29 of 39, 53 of 68 bootstrap fits yield increased domestic population size change after they split with mallard duck/spot-billed duck. All bootstrap fits yield increased wild duck population size change.

(data not show)

```
def split_nm(params,ns,pts):
    s,Ts=params
    xx=dadi.Numerics.default_grid(pts)
    phi=dadi.PhiManip.phi_1D(xx)
    phi=dadi.PhiManip.phi_1D_to_2D(xx,phi)
    phi=dadi.Integration.two_pops(phi,xx,Ts,nu1=s,nu2=(1-s))
    fs=dadi.Spectrum.from_phi(phi,ns,(xx,xx))
    return fs
```

model code1: split without migration

```
def split_m(params,ns,pts):
    s,Ts,m12,m21=params
    xx=dadi.Numerics.default_grid(pts)
    phi=dadi.PhiManip.phi_1D(xx)
    phi=dadi.PhiManip.phi_1D_to_2D(xx,phi)
    phi=dadi.Integration.two_pops(phi,xx,Ts,nu1=s,nu2=(1-s),m12=m12,m21=m21)
    fs=dadi.Spectrum.from_phi(phi,ns,(xx,xx))
    return fs
```

model code2: split with migration

```
def IM_2(params,ns,pts):
    s,nu1,nu2,TS,m12,m21=params
    xx=dadi.Numerics.default_grid(pts)
    phi=dadi.PhiManip.phi_1D(xx)
    phi=dadi.PhiManip.phi_1D_to_2D(xx,phi)
    nu1_func=lambda t: s *(nu1/s)**(t/TS)
```

```
    nu2_func=lambda t: (1-s) *(nu2/(1-s))**(t/TS)
   phi=dadi.Integration.two_pops(phi,xx,TS,nu1=nu1_func,nu2=nu2_func,m12=m12,m21=m21)
    fs=dadi.Spectrum.from_phi(phi,ns,(xx,xx))
    return fs
```

model code2: split with migration and population exponentially change.

full program is provided in https://github.com/vitogump/lifelit.

### 4.7 Selection analyses

In the $d_f$ calculation, we required all sampled individuals were sufficient covered for each df sites.

Whole genome alignment to chicken were performed by LASTZ(31).

#### 4.7.1 Detail functional analysis on candidate selected genes in early ancestor of mallard and spot-billed ducks.

We detected positive selections in the early ancestors of these two wild populations using methods invented in early modern humans and applied in European and Asian wild boards pig (3, 32). As the long time passed from the ancient selection occurrence in wild lineage, the accumulation of recombination, mutation and the other population effects may obscure the selection regions. We use a relative narrow window width to detect ancient selection. We filtered variation that passed the following criteria:

1) No CpG sites
2) SNPs were fixed in the tested Muscovy ducks with at least 20-fold coverage;
3) At least 5 individuals were observed in each population or 10 reads were observed for pool populations
4) At least 20 reads were observed when the site was fixed in a population
5) Derived allele was detected in both wild and domestic lineage

For windows with less than 10 variation sites were discarded.

In addition to the functional analysis presented in main text. An overrepresented term of positive selection related to "dorsal/ventral pattern formation", "determination of left/right symmetry" also imply selection on genes which making their body more suitable for flying. overrepresented on terms "response to wounding", " egg activation", "acute inflammatory response to antigenic stimulus", "detection of virus" may indicate that selection on genes increased mallard and spot-billed ducks' survivability in wild.

In addition, we found that at least 32 genes (e.g. *IL11RA*, *PRKAA1, LPAR1,* and *NS1BP*) involved into immune response to infections and three of them (i.e, *ADAP, RIG-I* and *CX3CR1*) played critical roles against influenza A virus infections (Table S15; Fig. S21)(33, 34). Therefore, it seemed that positive selections on immune genes might help the early mallard and spot-billed duck ancestor keep evolutionary equilibrium with influenza A viruses.

Some cryptochrome-containing cells within the retina are active at night when the birds perform magnetic orientation(35). In some migratory birds, a distinct part of the forebrain, where primarily processes input from the eye, is highly active at night(36). We also found three genes associated with retinal ganglion cell axon guidance (*EFNA5*, *ISL1* and *EPHA7* (S=-3)) and two genes (*CHRD* and *WNT7B*) related to forebrain regionalization or development were under selection in the early mallard duck and spot-billed duck ancestor (Fig. 3d; Fig. S20)(37, 38). Gene family that function on short wavelength-sensitive opsin (Cryptochromes, which have been suggested to form the basis of light-dependent magnetic compass in birds, is excited by blue (short wavelength) light) expanded in wild ducks may also indicated an enhancement in (Table S9)(36).

**4.7.2 Detail analysis on genomic islands of mallard and spot-billed duck divergence.**

To elucidate genomic properties and speciation of these two wild duck species, we firstly calculated how many sites all mallard ducks homozygous for one allele and all spot-billed ducks homozygous for another (which we referred the density of these fixed difference sites as $d_f$) in 40kb sliding windows with 20kb step size along duck scaffolds (Methods). This effort found that a small number of dfs (4,386) and only 26 SNPs of these located within protein-coding regions distributed in small genomic regions (758 of totally 52,215 windows) between mallard and spot-billed. We refer the windows showing highly elevated divergence up to 50 times higher than the genomic mean as "divergent peak". Divergent peaks contain most (3,522, 80%) of these dfs that distribute in 348 (0.6% of the whole duck genome) windows. This is sharp contrast to the case in Ficedula flycatchers, where using the same criterion, that 25% (59,936) dfs were distributed in the large genomic divergent islands covering 2.7% of the flycatcher genome. Such large difference in number of dfs and the pattern of genomic divergent islands in two these species might due to that the wild duck (the mallard and spot-billed duck) has a shorter divergence time and is in an earlier stage than the Ficedula flycatcher. As our sequence depth (~10X) is higher than the flycatchers' research (~5x), which may leading to the reduce of the dfs. We also random sample the reads into about 5-fold coverage of duck genome and the distribution of dfs remain almost the same (data not shown).

In order to study the genomic divergence in duck autosomes, we then estimated the population differentiation, nucleotide diversity through calculating $F_{\mathrm{ST}}$ (fixation index), dxy (sequence divergence), π (nucleotide diversity) and Tajima's D values in sliding window that homologous to chicken autosomes. (Methods; Fig. 4a; Table S16). Scaffolds homologous to chicken Z chromosome showed higher level of $F_{\mathrm{ST}}$ (5.5-fold) and Tajima's D (1.5-fold), but smaller number of SNP density (0.5-fold) and lower level of π (0.6-fold) than autosomes (Table S16, Fig. S23). These observations are consistent with change of $F_{\mathrm{ST}}$, Tajima's D and SNP density in flycatcher and wild ducks (Mexican and mallard duck)(39, 40). Analysis on autosomal differentiation regions shows that the distribution of $F_{ST}$ between the mallard duck and spot-billed ranged from -0.0157 to 0.7785 with an average of 0.04930346 and distributed in a narrow peak, lower than $F_{ST}$ between the wild ducks and domestic ducks ranged from 0.0880 to 0.8912 with an average of 0.2100 and distributed in a wide peak (Fig. 4b). Detailed analysis indicates that small part of genomic regions of the mallard duck and spot-billed duck have $F_{\mathrm{ST}}$ values being away from the mean $F_{\mathrm{ST}}$ at least 4 standard deviations (we referred these regions as high differentiation islands). These higher $F_{\mathrm{ST}}$ regions showed lower level of nucleotide diversity (π mallard: $0.0024 \pm 0.0022$; π spot-billed: $0.0023 \pm 0.0020$) and

SNP density (360 ± 305 and 334 ± 280 SNP per window in the mallard duck and spot-billed duck respectively) than genomic backgrounds (π mallard: 0.0076 ± 0.0042; π spot-billed: 0.0077 ± 0.0043. 1134 ± 569 and 1161 ± 597 SNP per window in the mallard duck and spot-billed duck respectively) in both mallard duck and spot-billed duck (Fig. 4c-f; Fig. S23; Table S17). This is similar to the case in flycatchers, Darwin's finches and ducks, where high differentiation islands are characterized by reduced levels of nucleotide diversity and polymorphisms(32, 39-41). A noteworthy consequence of these coinciding features was that both Tajima's D of these high differentiation islands was not reduced in both mallard duck and spot-billed, whereas it is different with the case in flycatchers that Tajima's D of high differentiation islands was significantly reduced (Fig 4f; Table S17). Moreover, sequence divergence between mallard duck and spot-billed didn't exceed background levels in high differentiation islands that is consistent with the observation in flycatchers, whereas Darwin's finches' study show elevated dxy in high differentiation islands (Discussion Section; Fig. S23)(41).

**4.7.2 Detail analysis on identification of candidate domestication regions**

Limited by many factors, we used very conserved strategy and has been succeed applied in previous reports, may not the very advance in methods, to detect genomic regions under domestic selection. We first used the methods that combine the Z-transformed distributions of Hp and $F_{ST}$ signals of autosomes and sex-chromosomes separately. And focused on putatively selective sweep regions falling at least four/two standard deviations away from the mean of the Z(Hp) and Z($F_{ST}$) of autosomes and sex-chromosomes separately. We applied this thresholds because Z including a reduction in effective populations size and recombination rate as previous discussed(42). Furthermore, we perform the statistic test on this outlier approaches using 4 times of z$F_{ST}$ and zHp to define selective region, 4 times represent the extreme ends of the distribution of z $F_{ST}$ and zHp, and P-value is less than 0.05, z$F_{ST}$ (P(z$F_{ST}$ >=4) = 0.00548), zHp(P(zHp<=−4) = 0.00322). two signal combined identified 137 regions with extremely low levels of Hp (average length = 63.6230 kb, average Hp = 0.2100 for autosomes; Hp = 0.2470 for sex-chromosomes) and 145 regions with significantly elevated $F_{ST}$ values (average length = 58.7670 kb, average $F_{ST}$ = 0.5390 for autosomes, average $F_{ST}$ = 0.394 for sex-chromosomes). However, only a small proportion putative selective sweeps regions werer predicted using both extremely low levels of Hp and significantly elevated $F_{ST}$ values. Such observation might be partly attributed to that the domestic lineage and wild lineage (mallard duck and spot-billed ancestors) were under different evolutionary patterns after they had diverged about 100-200 Kyr. For example, the wild lineage seemed to benefit from positive selection and increased their effective population sizes once or twice times, while the domestic lineage dramatically declined their effective populations sizes during LGP (Fig. 2d,3a; Fig. S17). This complex demographic history of the domestic and wild lineage might affect the $F_{ST}$ distribution. We counted the heterozygosity of these populations (Hp) in 40-kb windows (windows containing less than 10 SNPs were not included) in eight duck breeds respectively. We further calculated the Hp mean of these breeds for each window. Similarly, we calculated fixation index ($F_{ST}$) between each domestic duck breeds and the wild duck population (combine spot-billed and mallard) using an estimator introduced in dog in 40-kb windows, and counted the $F_{ST}$ of eight domestic duck breeds for each window(24). After that, we Z-transformed the Hp and $F_{ST}$ to Z(Hp) and Z($F_{ST}$), and defined regions with Z(Hp) or Z($F_{ST}$) being four standards deviations for autosomes or two standard deviations for sex-chromosomes from the mean of the genome-wide (Z(Hp) as putative selective sweep regions.

In Hp calculation, we didn't require min SNP numbers because the region with extreme little SNP could be Runs of homozygosity (ROH) that is also our interest.

For the regions under early stage selection in wild lineage or under recent selection in domestic lineage. We performed more detail analysis to visualize the genetic variations in Figure S23, S24.

The comparison between the details of selection regions in Figure S23, S24 reflect that the early stage selections mostly couldn't be visualized by current variations directly. But regions under recent selection in domestic lineage showed clear trace.

GO enrichment analysis were performed by using Fisher's Exact Test (see https://github.com/vitogump/lifelit).

#### 4.7.3 Evolutionary relationship of domestic duck, mallard wild duck and spot-billed wild duck

Such inference was supported by both phylogenetic comparative analysis and population genetic analysis. Despite the incomplete lineage sorting (ILS) and gene flow may confound the analyses, the different SNP sets used in genetic structure analysis, f3-statistics, SNPs distributions analysis and tree construction that was sampled from genome-wide would avoid these confounding effects in phylogenic inference(43). Furthermore, Genetic structure show each individual of both the mallard duck and spot-billed duck almost consistent of purely one ancestry component. This is different from the case of admixture, where individuals of one population with components of other populations' ancestry (44-46), thus indicating no evidence of introgression between the mallard and spot-billed duck (Fig. 1d; Fig. S14). Subsequent ∂a∂i and PSMC simulations that rely on different data sets and theory reconstructed the perfect matched demographic history further support the robustness of our inference. In addition, the subsequent observations of $F_{ST}$ on domestication and speciation analyses are consistent with or could be best explained by this phylogenic inference (Fig. 4b).

### 4.8 Estimation of genetic diversity and distance

Nucleotide diversity (π) was calculated using the vcftools(47). Tajima's D was calculated using the Popgenome R package(48) Pairwise $F_{ST}$ was calculated in 40-kb windows with a sliding size of 20-kb. Neighbor-joining tree based on pairwise $F_{ST}$ values were constructed with phylip-3.696 "neighbor" function by taking the distance matrix of pairwise $F_{ST}$ and integrating neighbor-joining trees of 27,374 windows.

### 4.9 select candidate casual mutation to verify its functional activity

For the sake of simplicity, we catalog the selection regions by its overlap with CDS, UTR, intron, intergenetic region. And rank SNPs by the ΔAF between wild lineage and domestic lineage. we selected the SNPs that located in UTR, intron, intergenetic region and closed to the genes that may play important functional in the duck evolution or domestication.

## Supplementary Figures

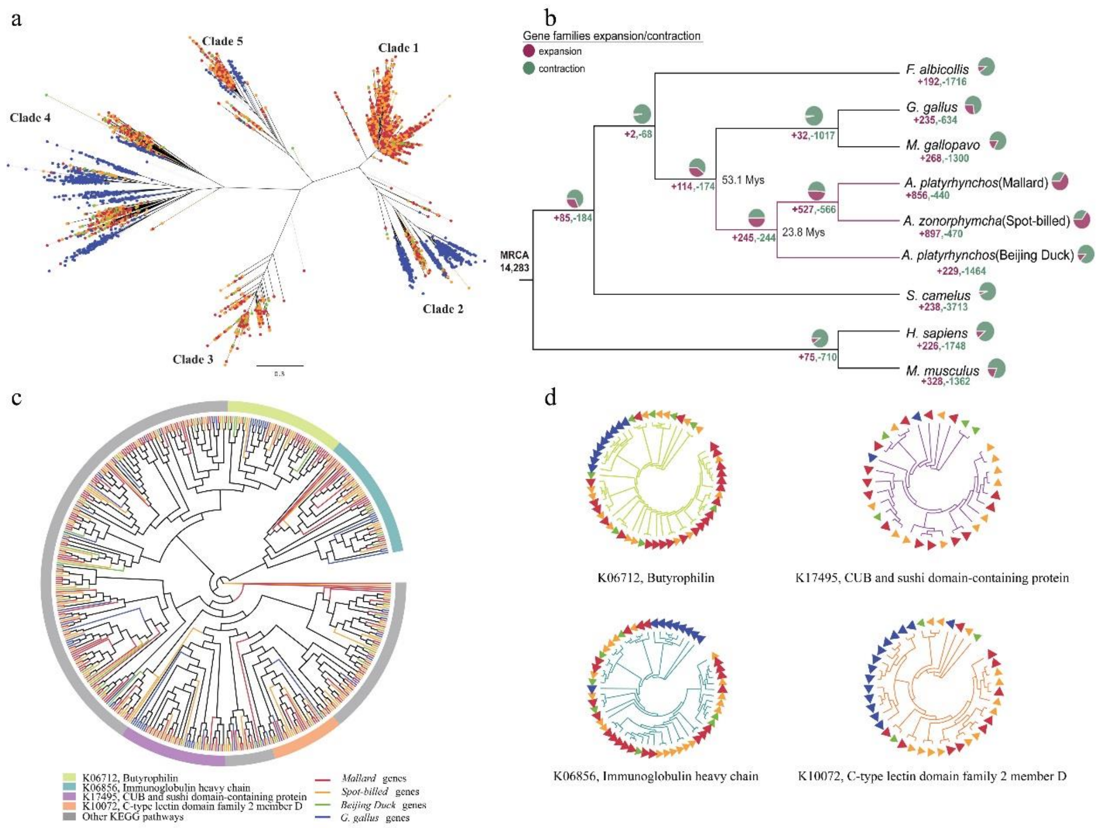



**Fig. S1: Expansion and contraction of transportable element and gene in mallard, spot-billed and Beijing duck genome.** Green, red, yellow and blue in a and c panel represent Beijing duck, mallard, spot-billed and chicken, respectively. (**a**) Phylogeny of CR1 retrotransposons in four bird genomes. (**b**) Numbers of gene family losses and gains across ten vertebrates. Pie chart and numbers nearby indicate the number of expanded and contracted gene families in each branch. (**c**) Phylogeny of gene families expanded in spot-billed and mallard. Top four families ranked by the number of gene gained are indicated in color arcs. KEGG pathway annotation of those four families are listed in the bottom. Other families are indicated in grey color. (**d**) Detailed phylogeny of four gene families expanded in spot-billed and mallard duck as indicated in (**c**).

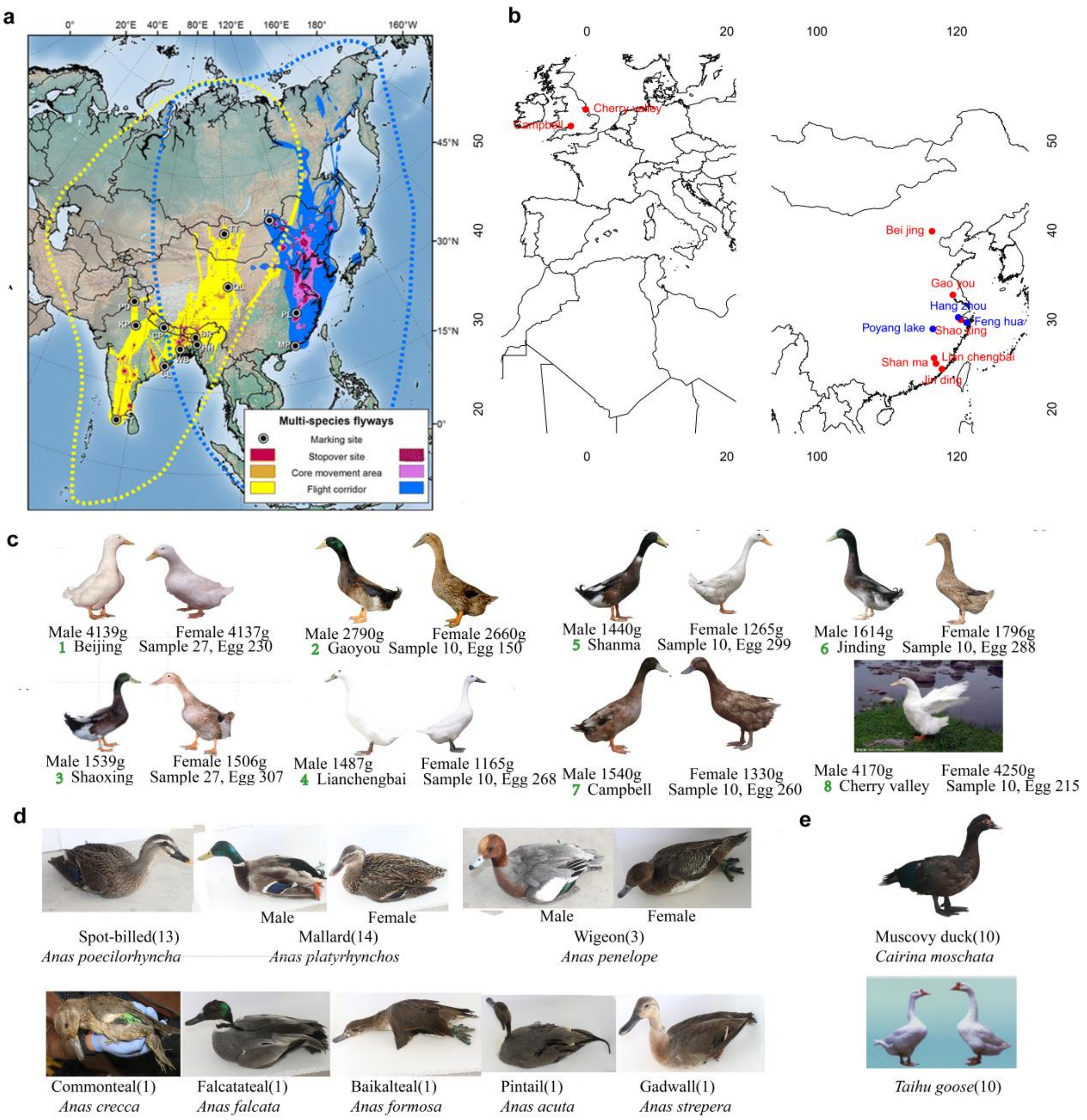



**Fig. S2. Sample distribution and phenotypic variation in domestic and wild ducks.** Photos were taken from the sequenced samples except Taihu goose and Cherry valley being download from web. (**a**) Estimated migration routes of Anatidae in Central Asian Flyway (CAF) and East Asian-Australasian Flyway (EAAF)**(49)**. Regions in the flyway of CAF displayed in yellow-red and for EAAF displayed in blue-purple. From darkest to lightest, colors represent 50%, 75% and 99% cumulative probability contours. (**b**) Geographical distribution of wild duck and domestic duck samples. Blue dots represent the place where wild duck samples collected from. Compare to **a**, our wild duck samples were collection from the main migration route of the Anatidae flyway. (**c**) Phenotype of the eight domestic duck breeds. Number of individuals, average weight (male/female), average eggs production per year were marked. (**d**) Photos and sample numbers of the eight wild duck species. (**e**) Photos and sample number of Muscovy duck and Taihu goose used as outgroup.

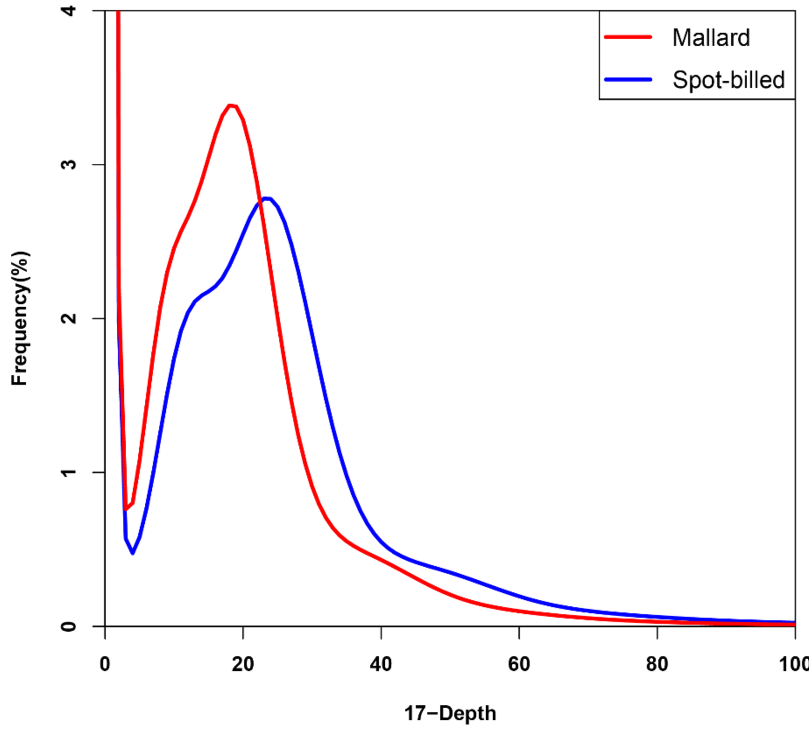



**Fig. S3. Distribution of 17-mer frequency in the corrected pair-end reads of the mallard and the spot-billed genome.**

Only the reads from short-size libraries (< 500 bp) were included in this analysis. We identified 22,312,838,068 kmers using 21-fold data of mallard and 28,137,616,208 kmers using 27-fold data of spot-billed. The D2B and D2L genome size can be estimated as G= K_number (total kmer number) / K_depth (the volume peak), which is estimated to be 1.14Gb and 1.15Gb, respectively.

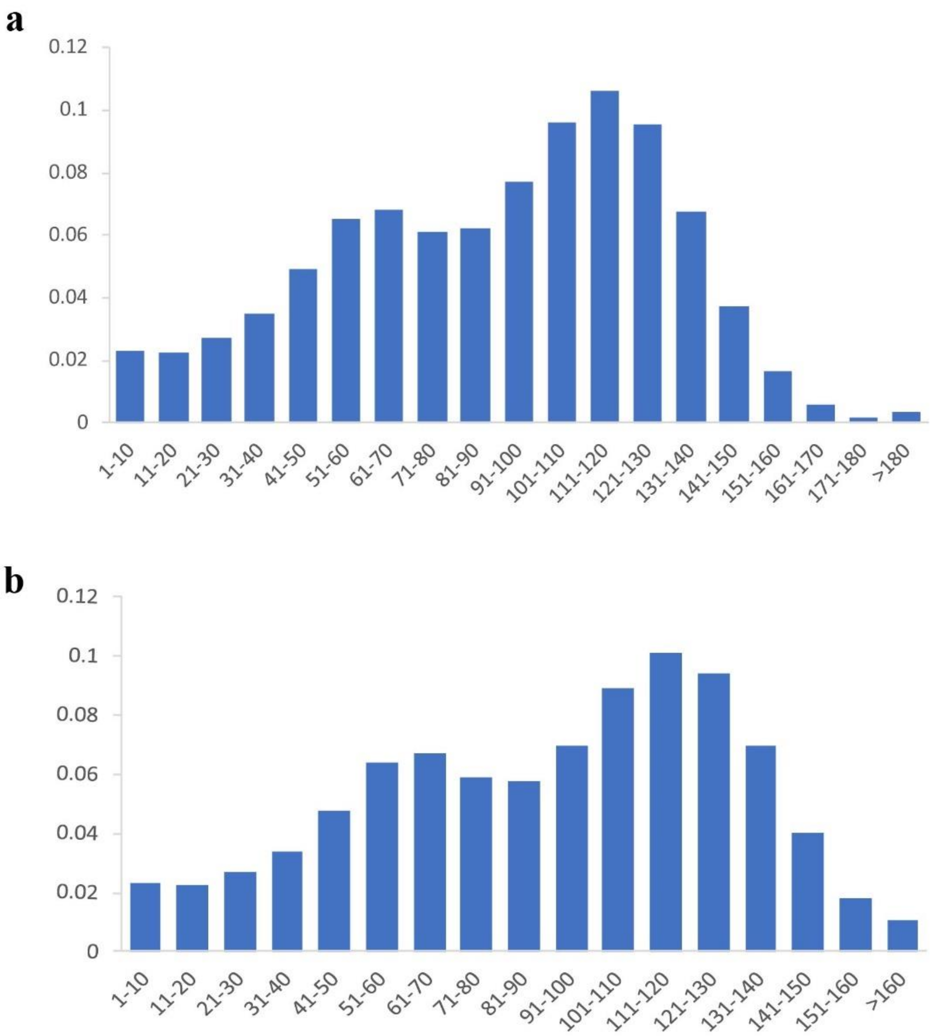



**Fig. S4. Distribution of the sequencing depth of the mallard and the spot-billed assemblies.**

(a) the mallard assembly. (b) the spot-billed assembly.

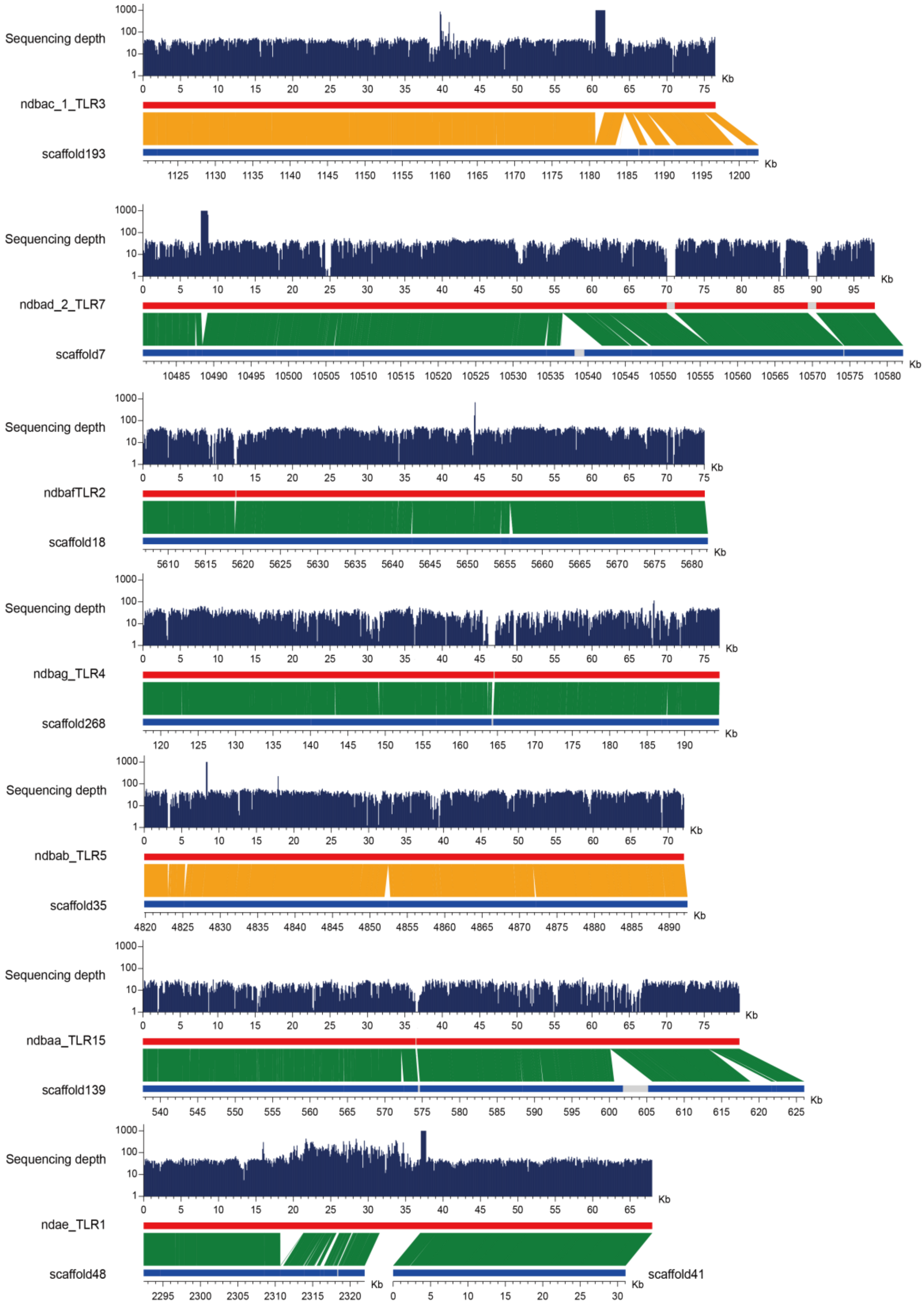



**Fig. S5. Evaluation of the mallard assembly with sequences of seven BACs.**
Seven BACs were from chromosome 1, 3 or 4. The BAC sequences and assembly scaffolds are shown in red and dark blue, respectively. Orange represents positively and green was reversely aligned, the remaining unclosed gaps on the scaffolds are marked as white blocks.

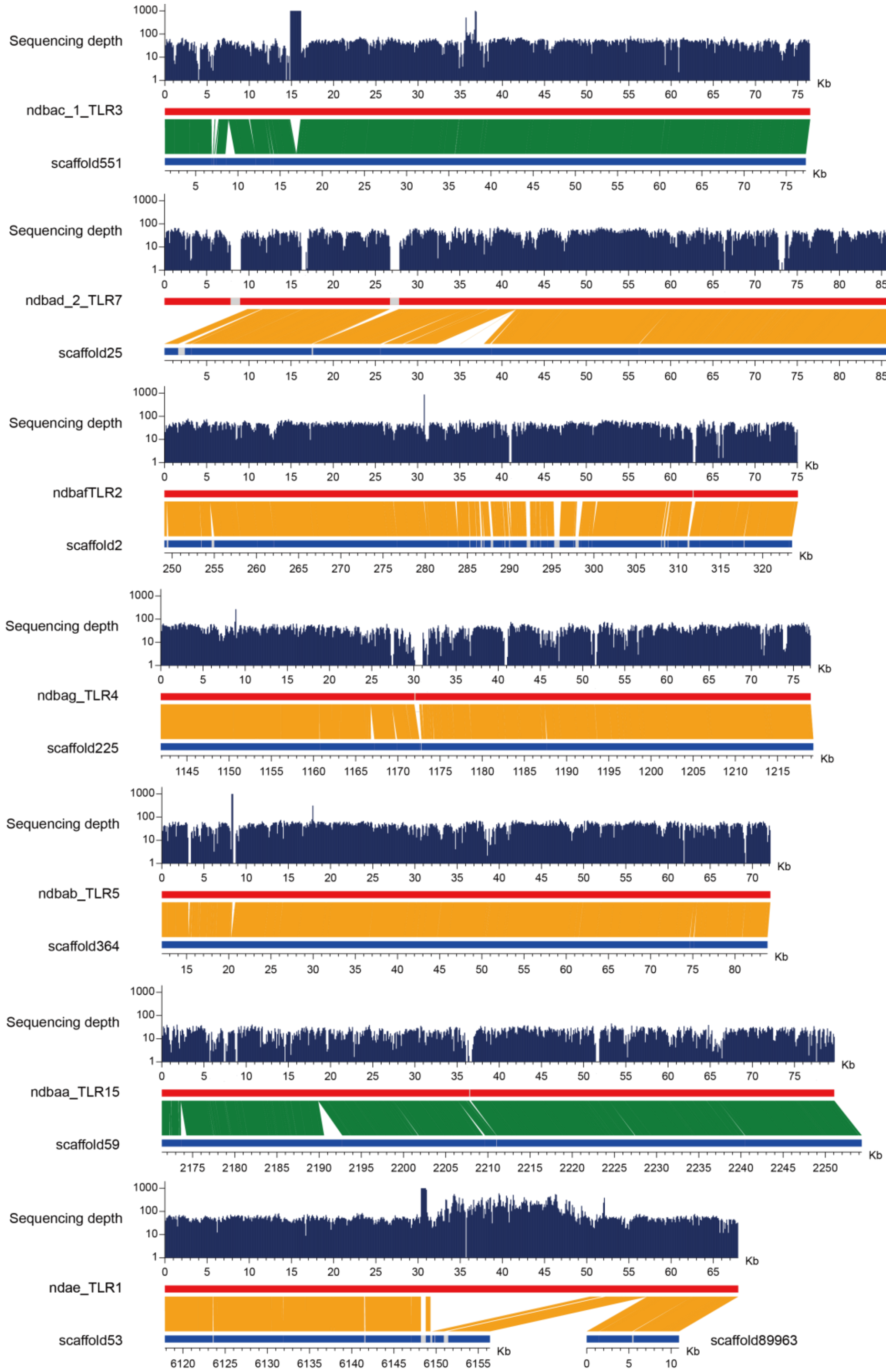



**Fig. S6. Evaluation of the spot-billed assembly with sequences of seven BACs.** Seven BACs were from chromosome 1, 3 or 4. The BAC sequences and assembly scaffolds are shown in red and dark blue, respectively. Orange represents positively and green was reversely aligned, the remaining unclosed gaps on the scaffolds are marked as white blocks.

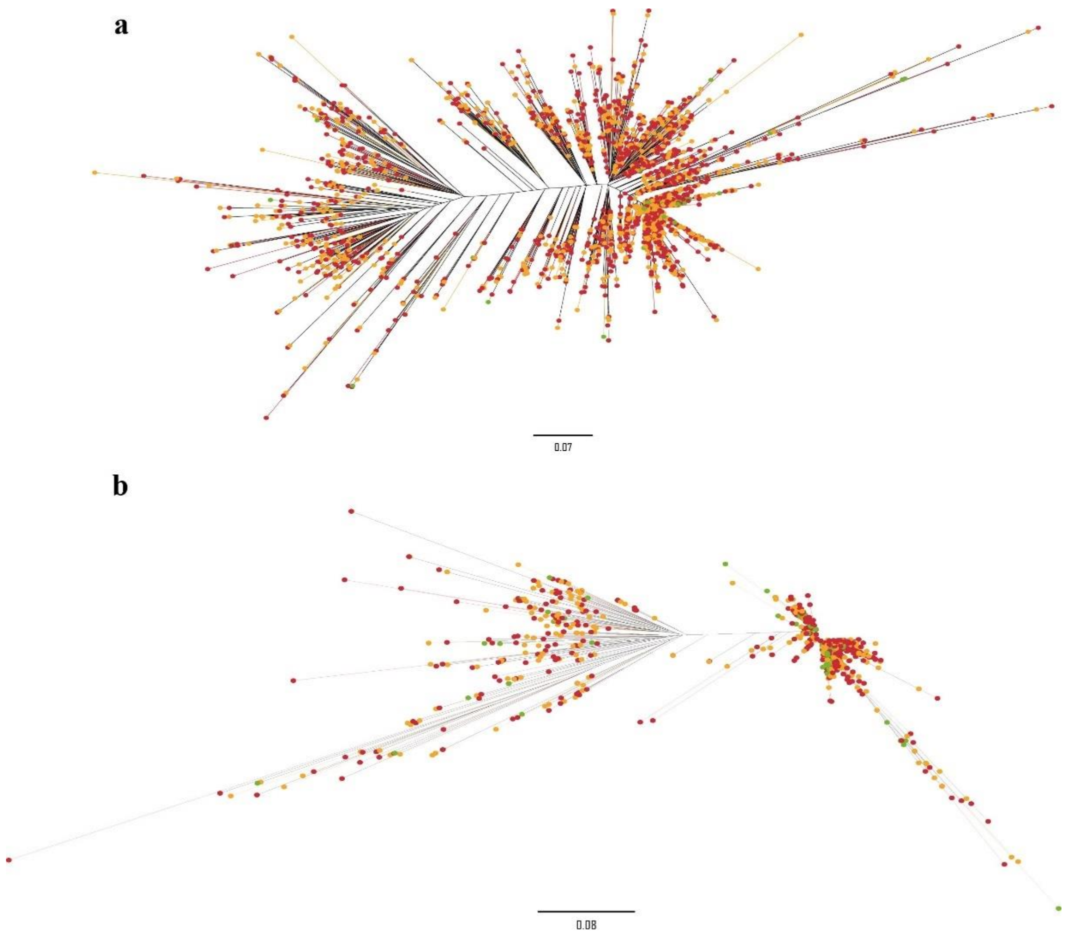



**Fig. S7. Phylogeny of two CR1 retrotransposon clade in four bird genomes.**
Green, red and yellow represent Beijing duck, mallard and spot-billed, respectively.

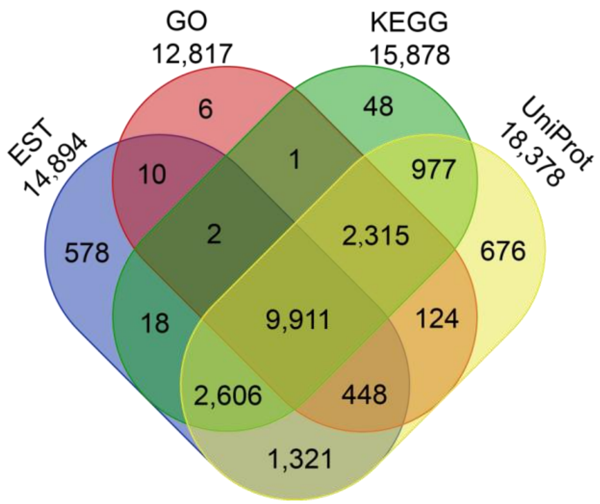



**Fig. S8. Venn diagram showing the mallard reference genes annotated using three databases or supported by EST.**
Total 21,056 mallard genes supported by 319,996 ESTs of Beijing duck and functioned annotation with three databases. And 19,041 genes were supported by one of them.

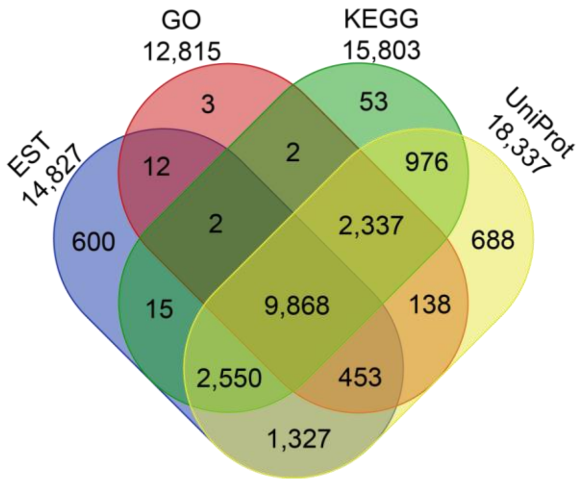



**Fig. S9. Venn diagram showing the spot-billed reference genes annotated using three databases or supported by EST.**

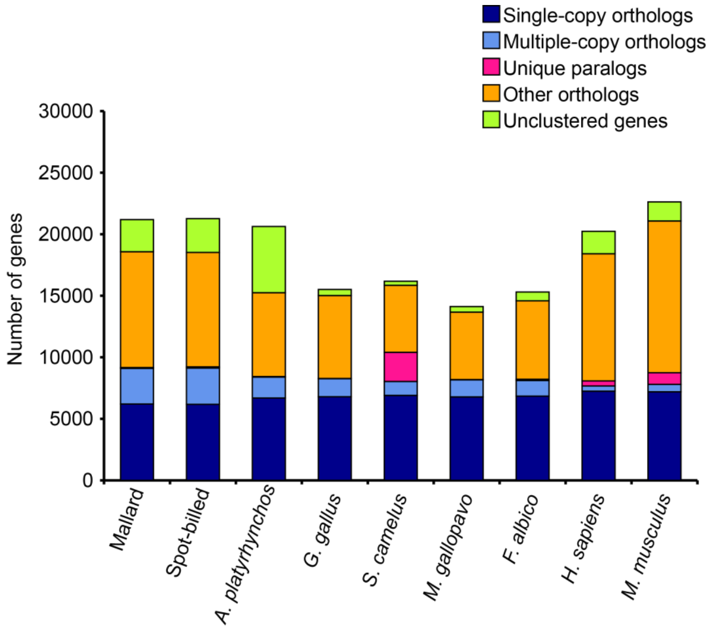



**Fig. S10. Protein homologous comparison among the reference gene sets of nine species by TreeFam.**

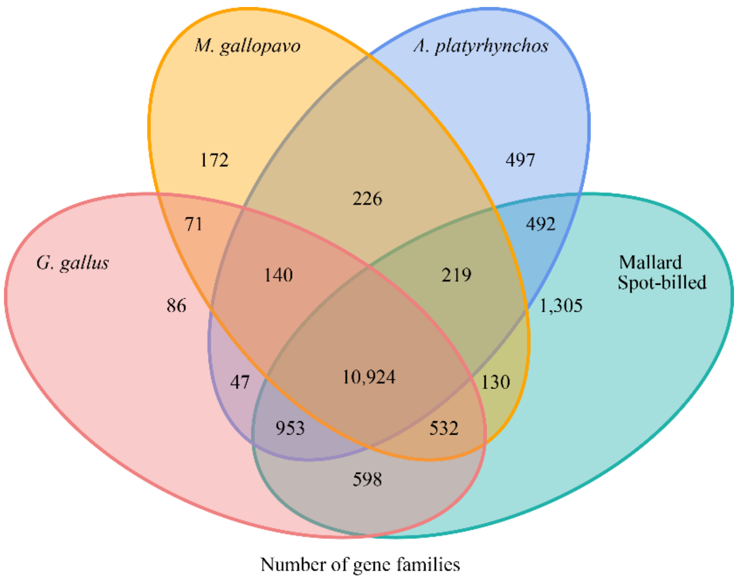



**Fig. S11. Venn diagram shows gene family clusters in five species by TreeFam.**
The number of unique and shared gene families among five genomes.

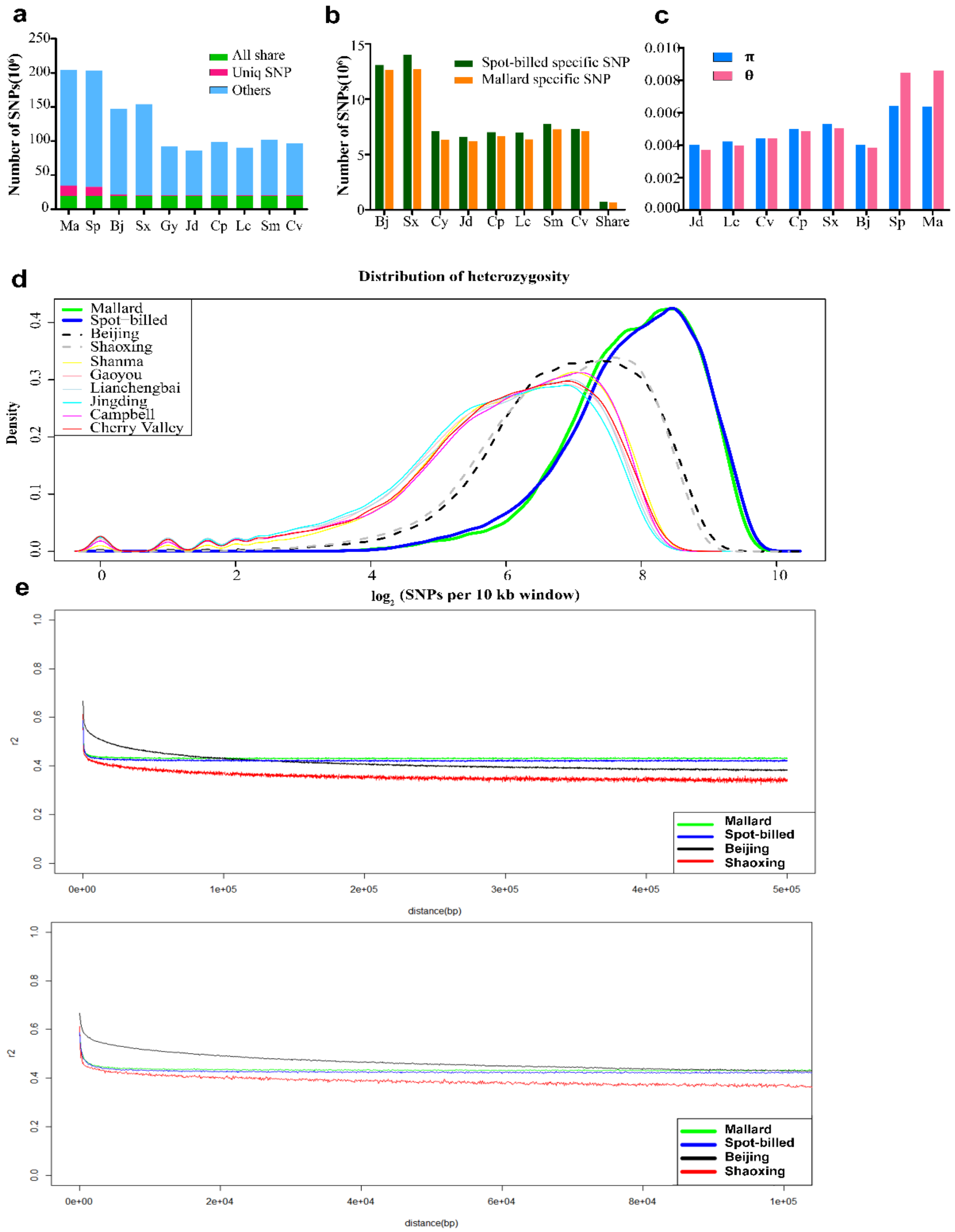



**Fig. S12. Distribution of genetic variation and characteristics for two wild and eight domestic duck populations.**
Mallard, Spot-billed, Beijing, Shaoxing, Gaoyou, Jinding, Campbell, Lianchengbai, Shanma and Cherry valley are represented by Ma, Sp, Bj, Sx, Gy, Jd, Cp, Lc, Sm, Cv. (**a**) Total number of SNPs. (**b**) Total number of mallard and spot-billed specific SNPs segregating in each domestic population. (**c**) Genetic diversity was measured by nucleotide diversity $\pi$ and Watterson's estimator $\Theta$. (**d**) Distribution of heterozygosity. (**e**) LD decay measured by $r^2$ estimator against distance (50 kb up and 10 kb down) between the SNPs in mallard, spot-billed, Beijing and Shaoxing populations.

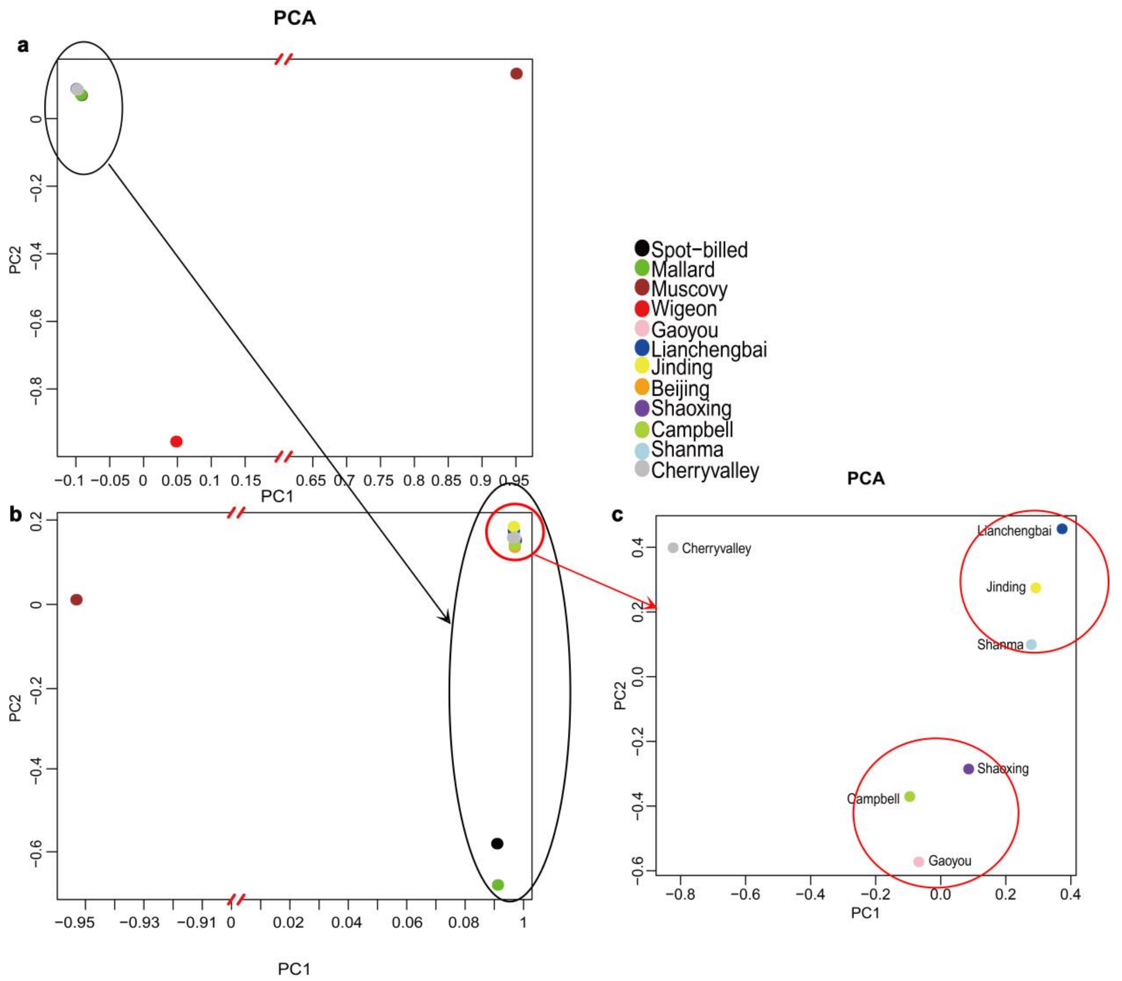



**Fig. S13. Principal component analysis of wild and domestic ducks using population as unit.**

(**a**) PCA analysis with genome wide SNPs set from 8 domestic duck breeds and three wild duck species, that having population data, together with Muscovy ducks. Two wild duck (mallard and spot-billed) and eight domestic duck populations were clustered in the top left, while the wigeon populations (consist of 3 individuals) and the muscovy duck population showing large distance with eight domestic duck populations. (**b**) PCA analysis on eight domestic duck populations and two wild duck (mallard and spot-billed) populations along with one muscovy duck population. It shows that 8 domestic duck populations cluster together and clearly separated from Mallard and Spot-billed in the second component. (**c**). PCA analysis on seven domestic duck populations. It shows that these populations clustered according to their geographical distributions.

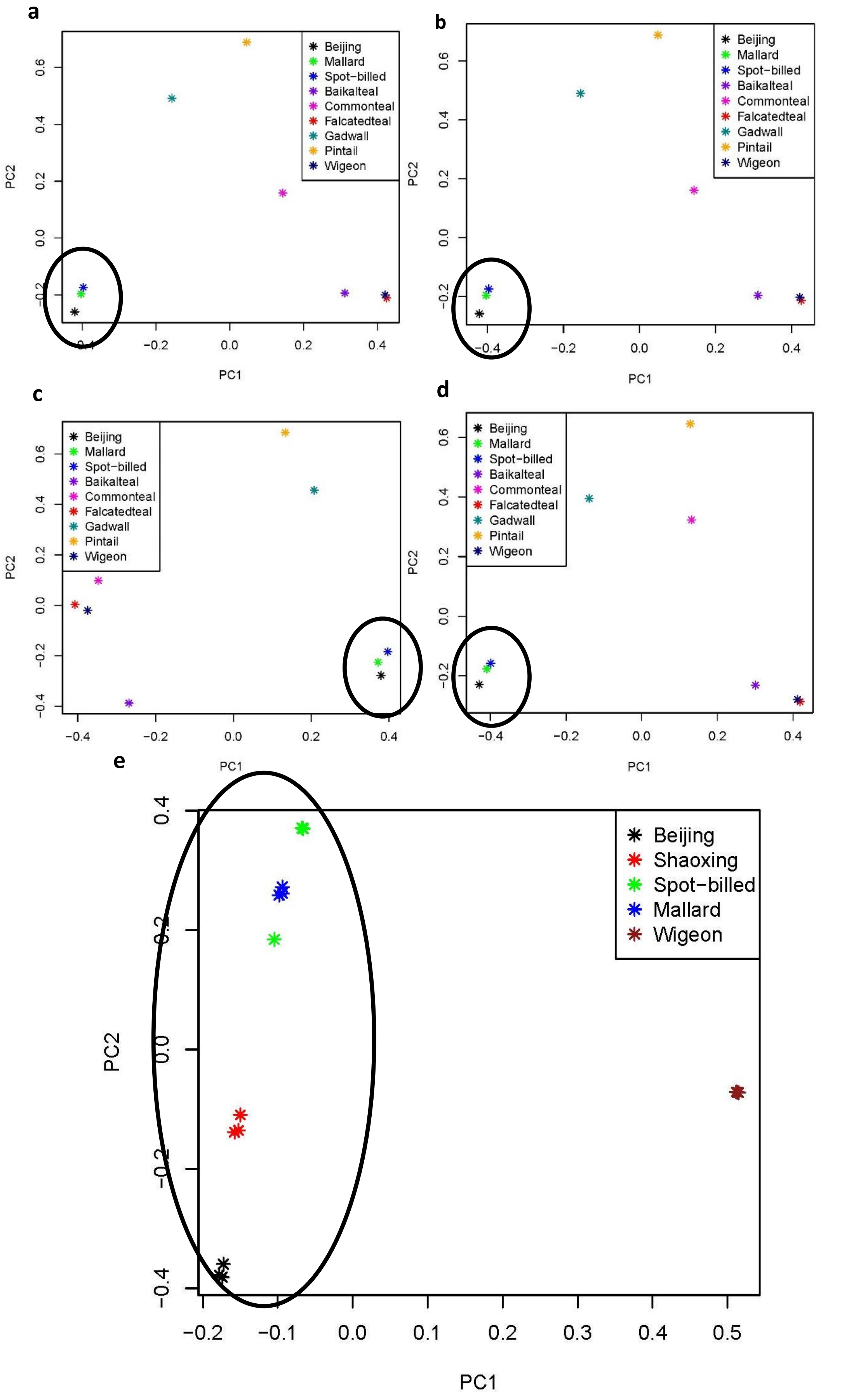



**Fig. S14. Principal component analysis of wild and domestic duck with same**

**coverage and sample size for each population.**

(a-d) shows PCA analysis on eight wild species (one individuals each, random sample reads into 12× that is equal to Beijing individual) and a Beijing individual with different data process strategy. a) using LD pruned SNPs from all scaffolds of Beijing assemble. b) using LD pruned SNPs from the scaffolds regions which were homologous to chicken autosomes. c) using LD pruned SNPs from the scaffolds regions which were homologous to chicken Z chromosome. d) using LD pruned SNPs from all scaffolds of mallard duck genome assemble. e) 3 individuals for each of the 5 populations (mallard duck, Spot-billed, Beijig, Shaoxing and wigeon, random sample reads into 3× for each individuals).

Black cycles indicate spot-billed, mallard duck and domestic duck are clustered together.

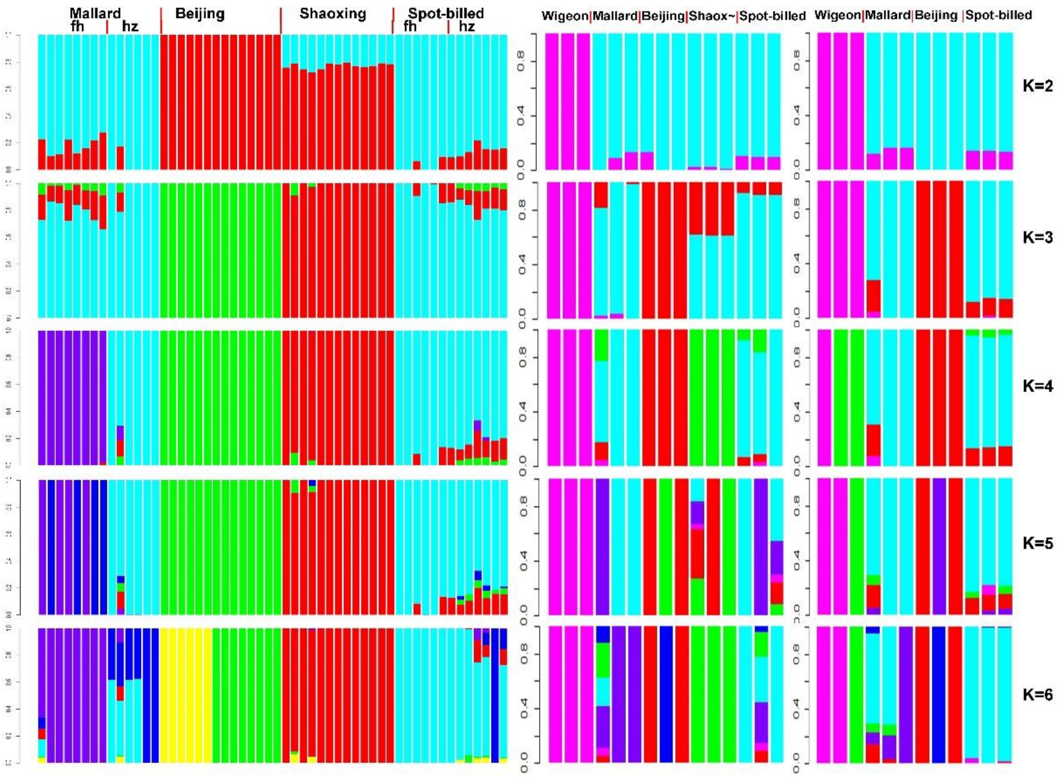



**Fig. S15. Genetic structure analysis with equal sample size and same coverage of each population for different populations combinations.**

Samples in left panel were random sampled to 3-fold coverage. Beijing duck samples were from a designed population, which were from a same population and formed a high growth line and low growth line after several generations selection. Samples in middle panel were random sampled to 3-fold coverage and three samples for each of the 5 populations. Samples in the right panel were random sampled to 12-fold coverage and three samples for each of the 4 populations. K range from 2 to 6.

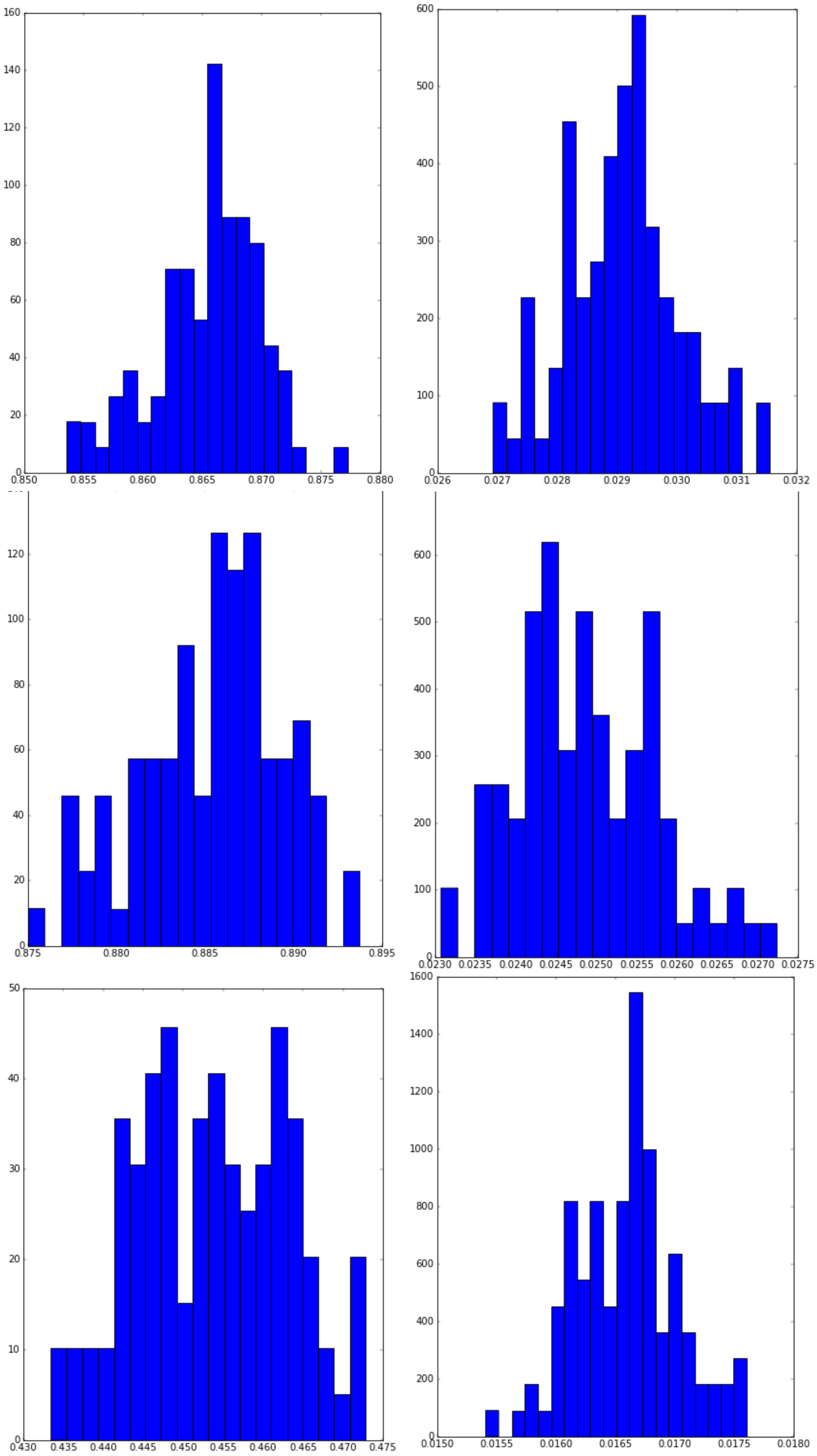



**Fig. S16. Split without migration model conventional bootstrap results.**
Panels from up to down corresponding to simulations of mallard and domestic duck divergence, spot-billed and domestic duck divergence, mallard duck and spot-billed duck divergence, separately. Left picture of each panels shows the distribution of s, whereas right one shows the distribution of Ts.

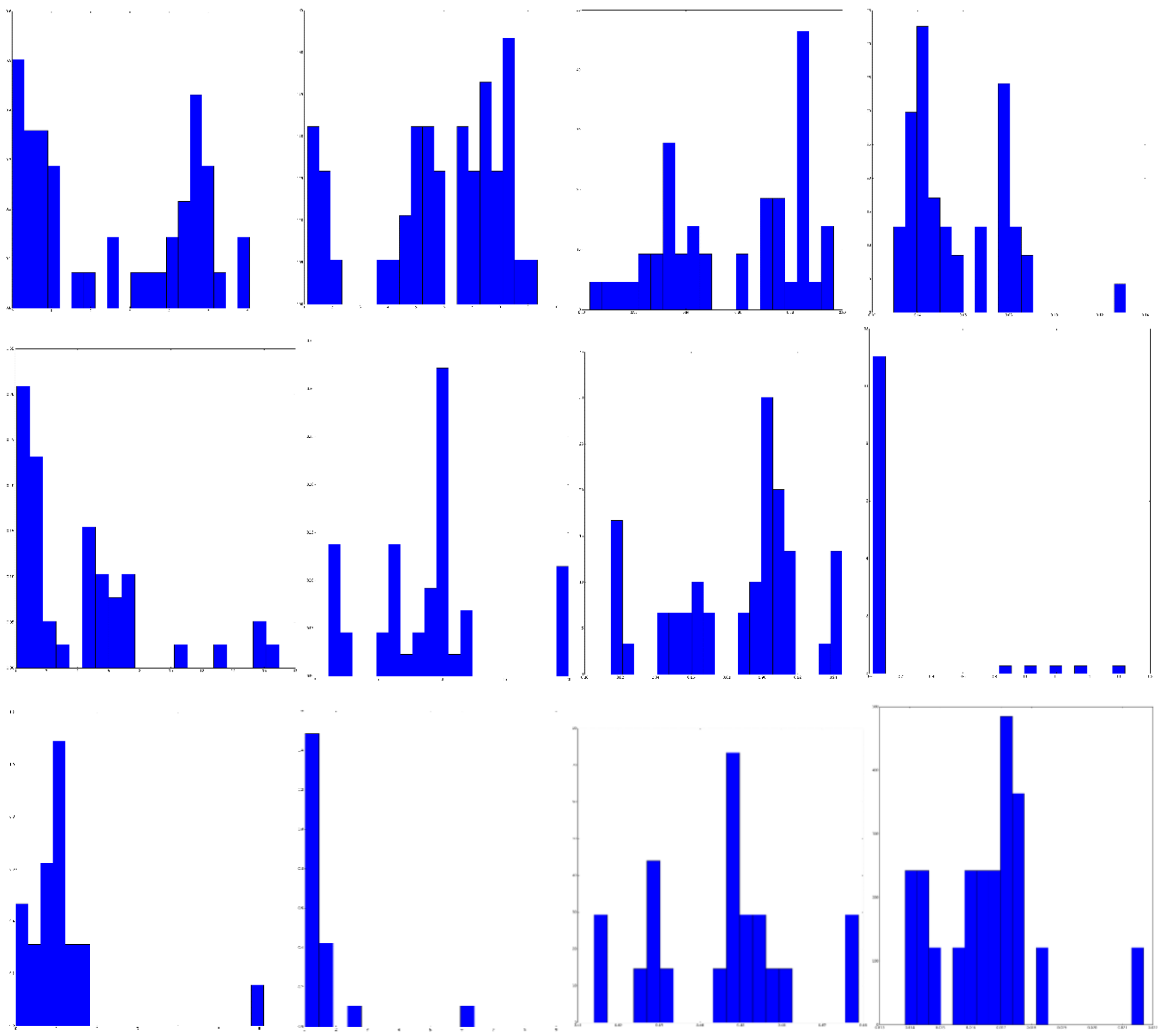


**Fig. S17. Split with migration model conventional bootstrap results.** Panels from up to down corresponding to simulations of mallard and domestic duck divergence, spot-billed and domestic duck divergence, mallard duck and spot-billed duck divergence, separately. Columns from left to right corresponding the distributions of m12, m21, s, Ts parameters.

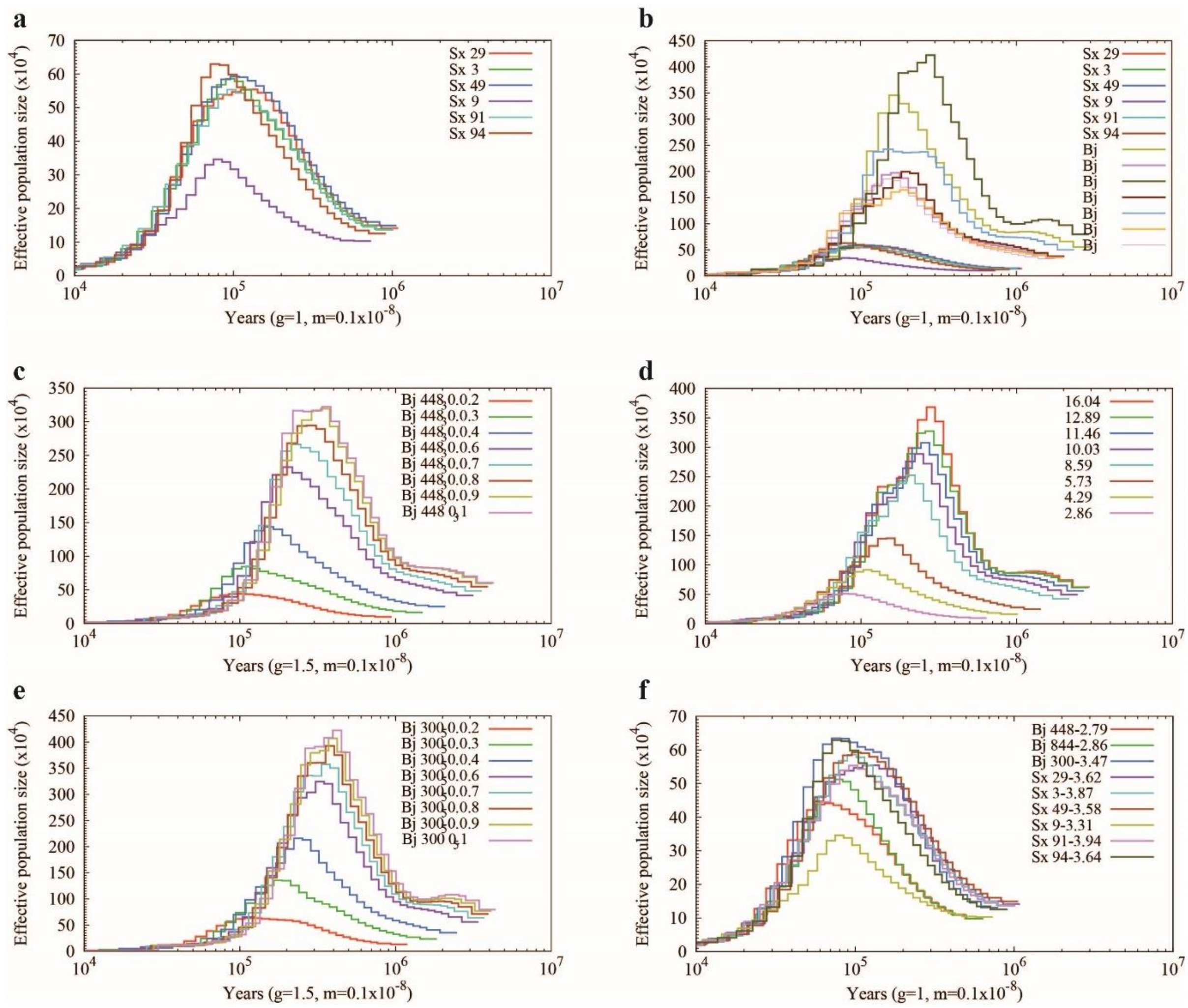
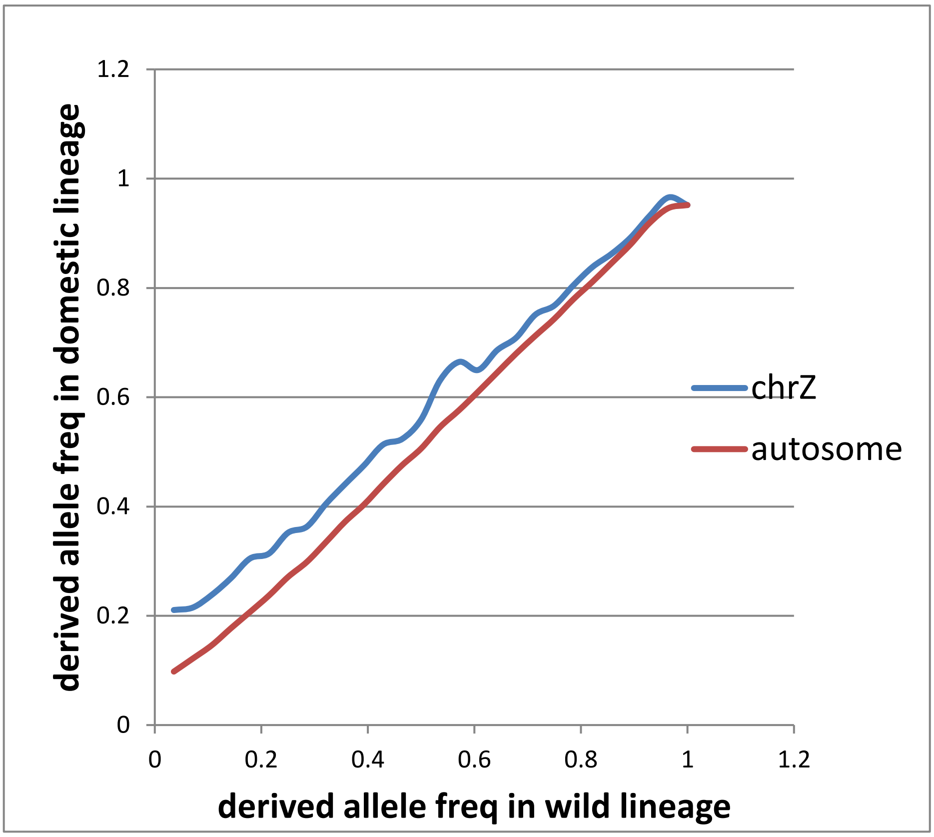


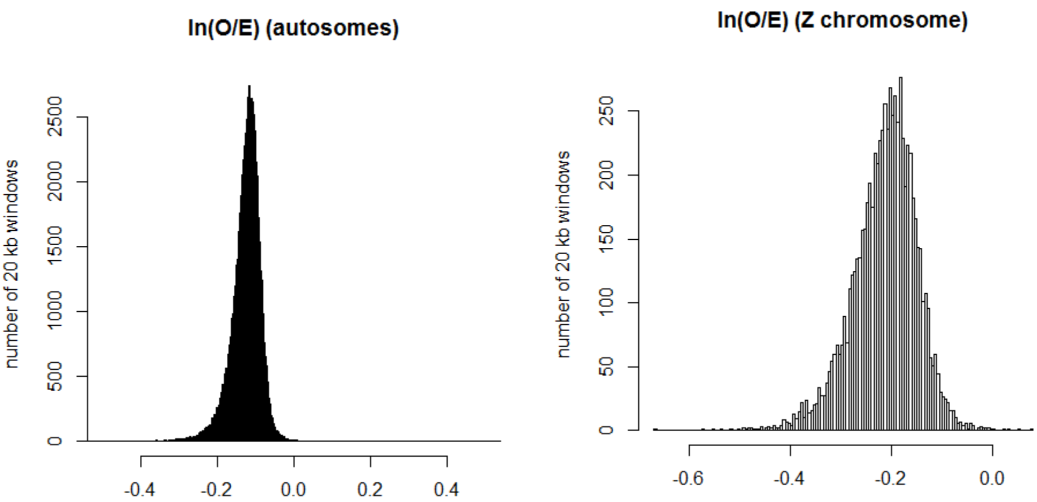

**Fig. S18. Comparison of PSMC between Beijing (Bj) and Shaoxing (Sx) ducks with different coverage data.**

(**a**) Demographic history of six Shaoxing ducks with a coverage of 2- to 3.5-fold. (**b**) Comparison demographic history of six Shaoxing (2- to 3.5-fold) and Beijing (13.5-fold) ducks. (**c-e**) PSMC analyses of Beijing or Shaoxing ducks based on different coverage of random sample reads. Sample ID in **c**, **d** and **e** are 44448_30, 43844_32, 41300_50, respectively. Coverage are represented in different color lines in legend. (**f**) PSMC curve of Beijing and Shaoxing ducks inferred with comparable coverage reads are similar.

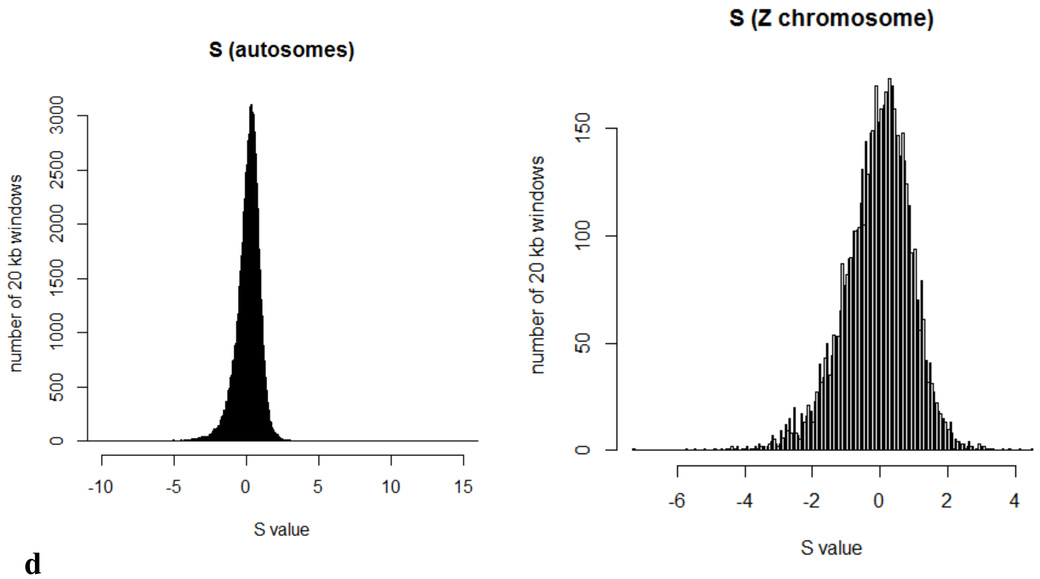
**a**


derived allele freq in domestic lineage
1.2
1
0.8
0.6
0.4
0.2
0
0
0.2
0.4
0.6
0.8
1
1.2
derived allele freq in wild lineage
chrZ
autosome

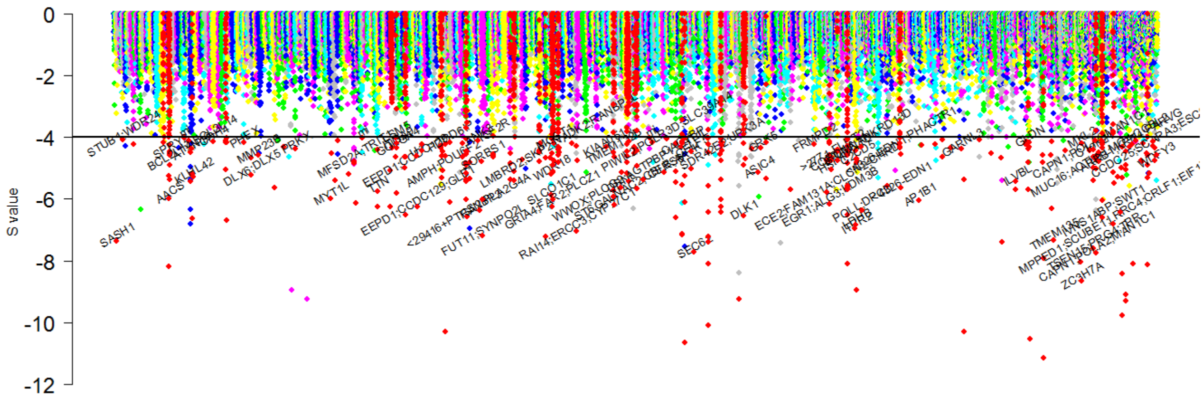
**b**


ln(O/E) (autosomes)
number of 20 kb windows
0
500
1000
1500
2000
2500
-0.4
-0.2
0.0
0.2
0.4
ln(O/E) (Z chromosome)
number of 20 kb windows
0
50
100
150
200
250
-0.6
-0.4
-0.2
0.0

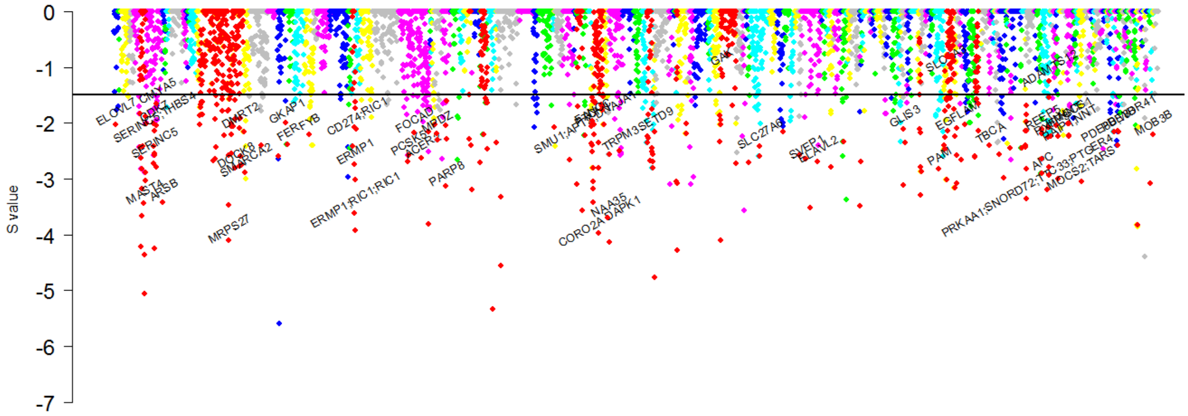
**c**


S (autosomes)
number of 20 kb windows
S value
S (Z chromosome)
number of 20 kb windows
S value

**d**


S value

**e**


S value

**f**

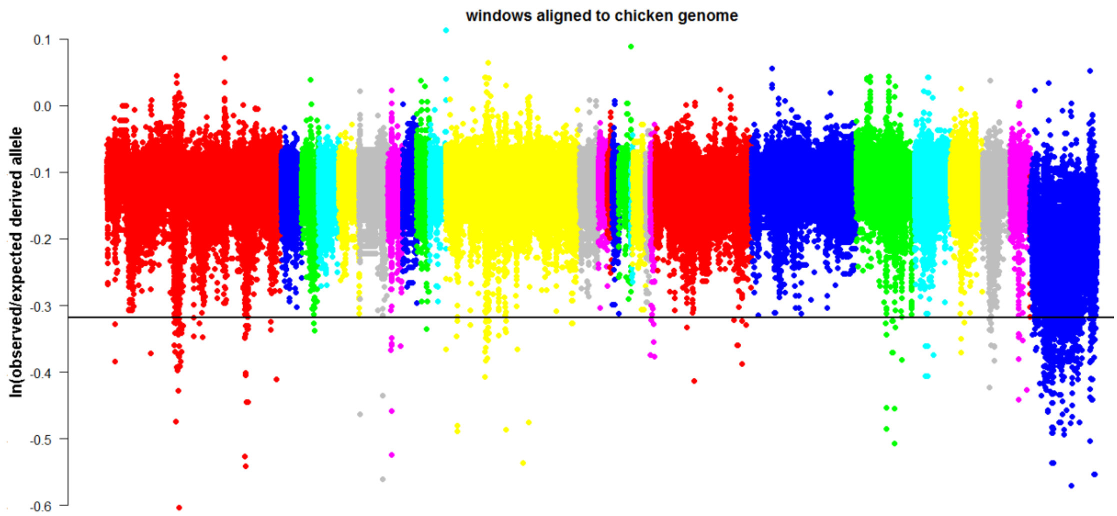
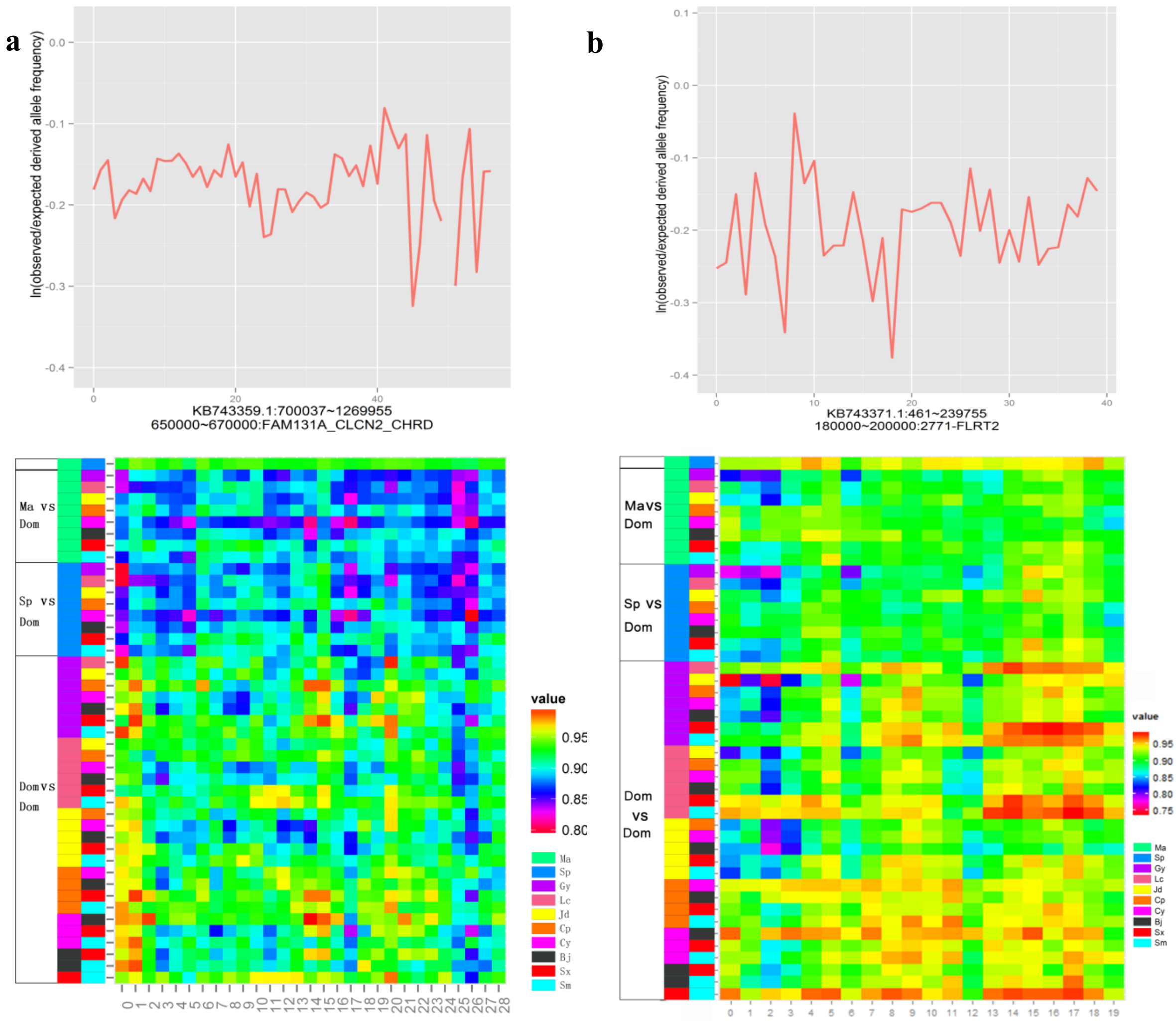



**Fig. S19. Whole genome screen for S score in wild duck with 20 kb window sliding by 10 kb.**
(**a**) Correlation of observed derived allele frequency between wild (combine mallard and spot-billed) and domestic duck (combine all 8 domestic duck breeds) lineage. (**b**) Distribution of the ln(observed/expected derived alleles frequency) for autosomes and Z-chromosome. (**c**) Distribution of the S score for autosomes and Z-chromosome. (**d**) The negative end of the S score distribution presented along duck autosomes. Genes that located in or near selected regions were showed its symbol. (**e**) The negative end of the S score distribution presented along duck Z-chromosome. (**f**) The S score distribution presented along scaffolds that is arranged according to their alignment to the chicken genome.


a
ln(observed/expected derived allele frequency)
KB743359.1:700037~1269955
650000~670000:FAM131A_CLCN2_CHRD
Ma vs Dom
Sp vs Dom
Dom vs Dom
value
Ma
Sp
Gy
Lc
Jd
Cp
Cy
Bj
Sx
Sm
b
ln(observed/expected derived allele frequency)
KB743371.1:461~239755
180000~200000:2771-FLRT2
Ma vs Dom
Sp vs Dom
Dom vs Dom
value

1484

c d

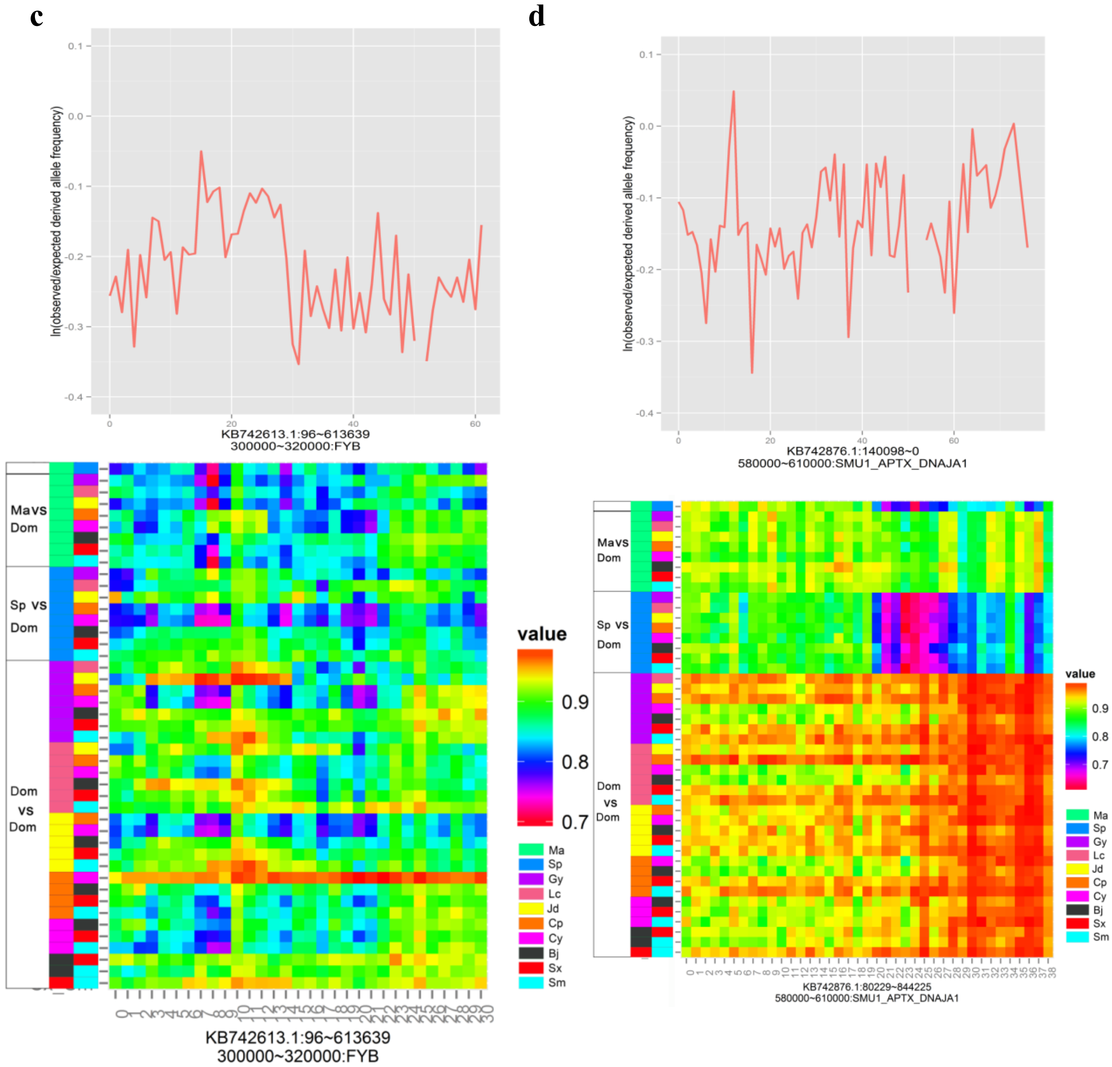



**Fig. S20. Details of four putative selective sweep regions in the early stage of the wild duck lineage.**
Each panel shows the distribution of the log-transformed value of the ratio for the observed/expected derive allele frequency using non-overlap 10 kb window consist (up) and scores between pairwise comparisons in 20 kb non-overlap windows (low). (**a**) *CHRD* locus. (**b**) *FLRT2* locus. (**c**) *FYB* locus. (**d**) *DNAJA1* locus.

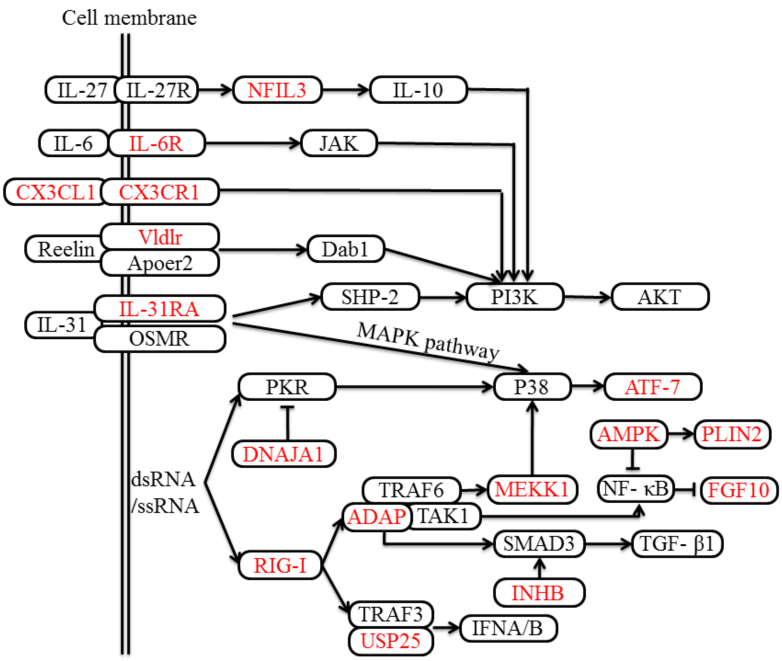



**Fig. S21. The RIG-I pathway preferring to be under positive selection.**

Sixteen immune genes detected to be under positive selection were in red. Three of them (RIG-I, DNAJA1 and ADAP, MAP3K1) are involved in the host immune response to influenza A viruses.

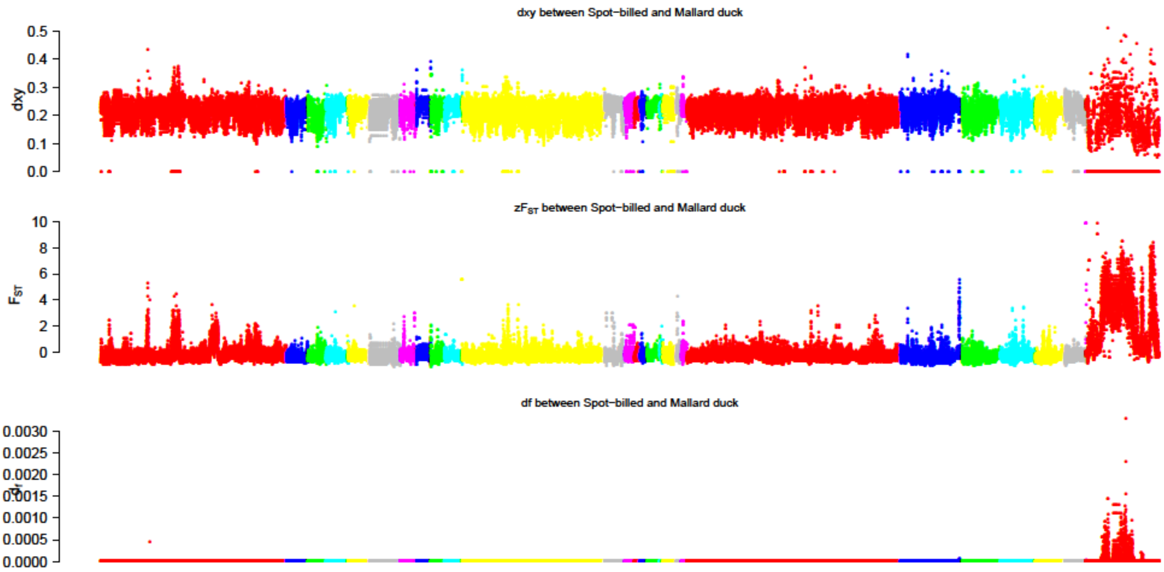
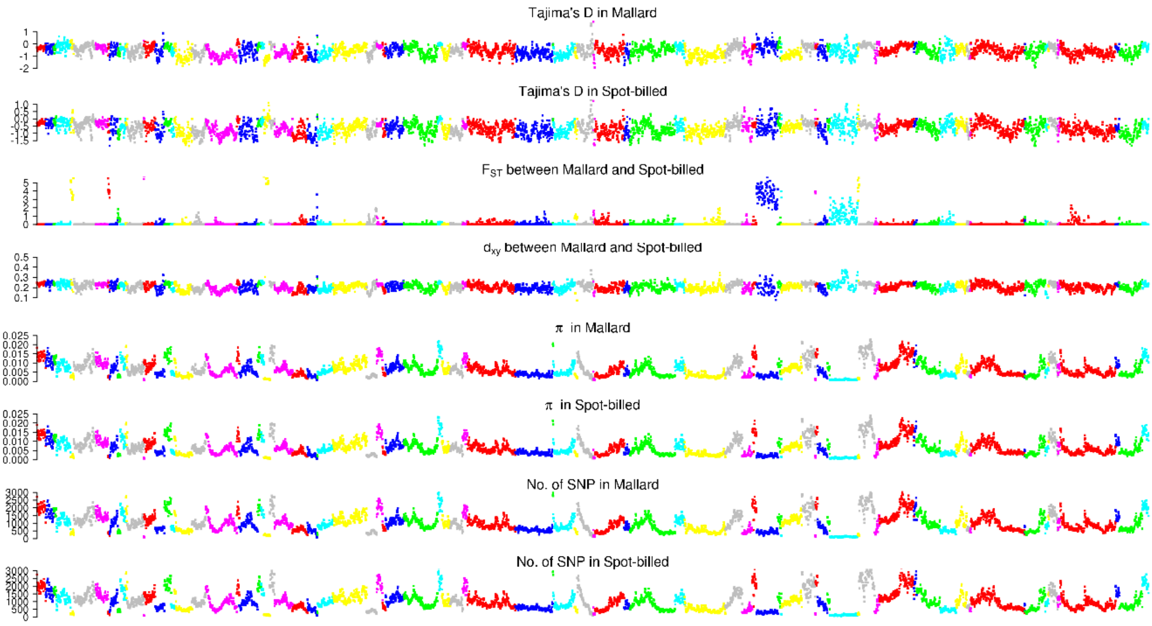


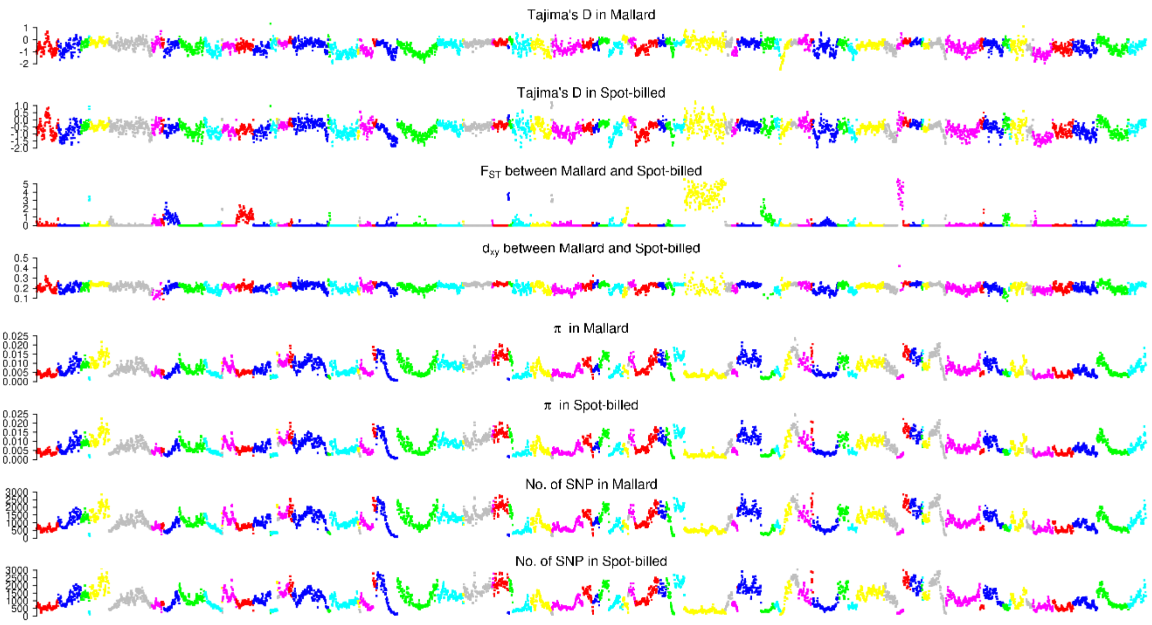

**Fig. S22. Distribution of $F_{ST}$ (fixation index), dxy and dfs (fixed difference sites) along Beijing duck scaffolds between mallard duck and spot-billed.**

$F_{ST}$ and dfs were counted with 40 kb window by step-size of 20 kb. The duck scaffolds that is shown in red and arranged at the right most are homologous to chicken Z chromosome and referred as duck Z chromosome.

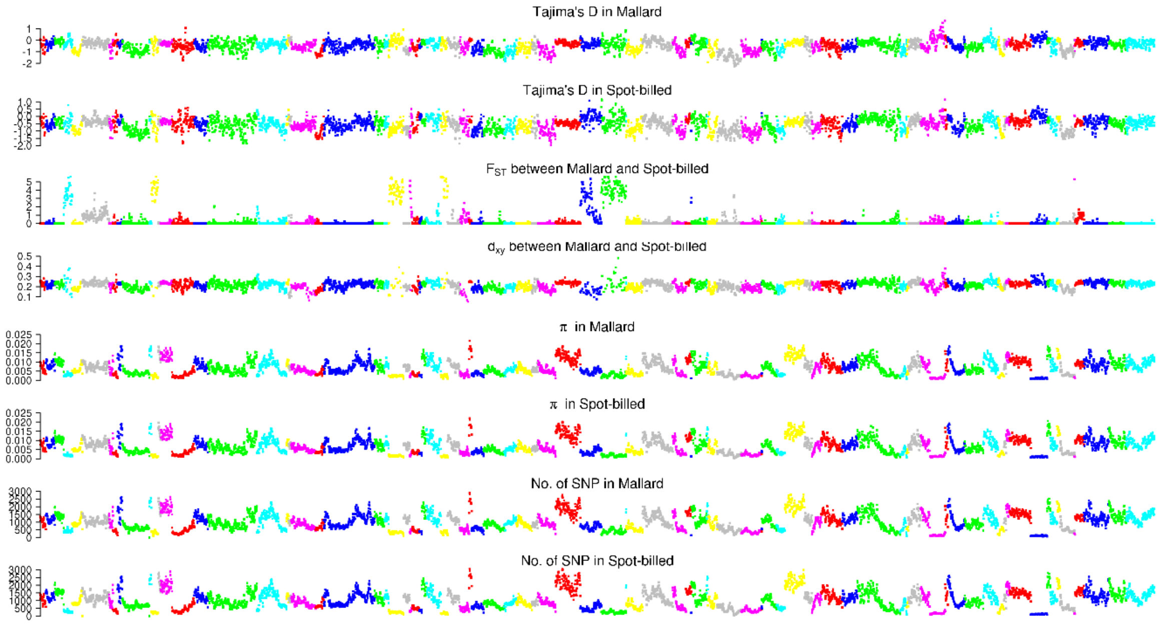
Scaffolds from KB742382.1 to KB742485.1



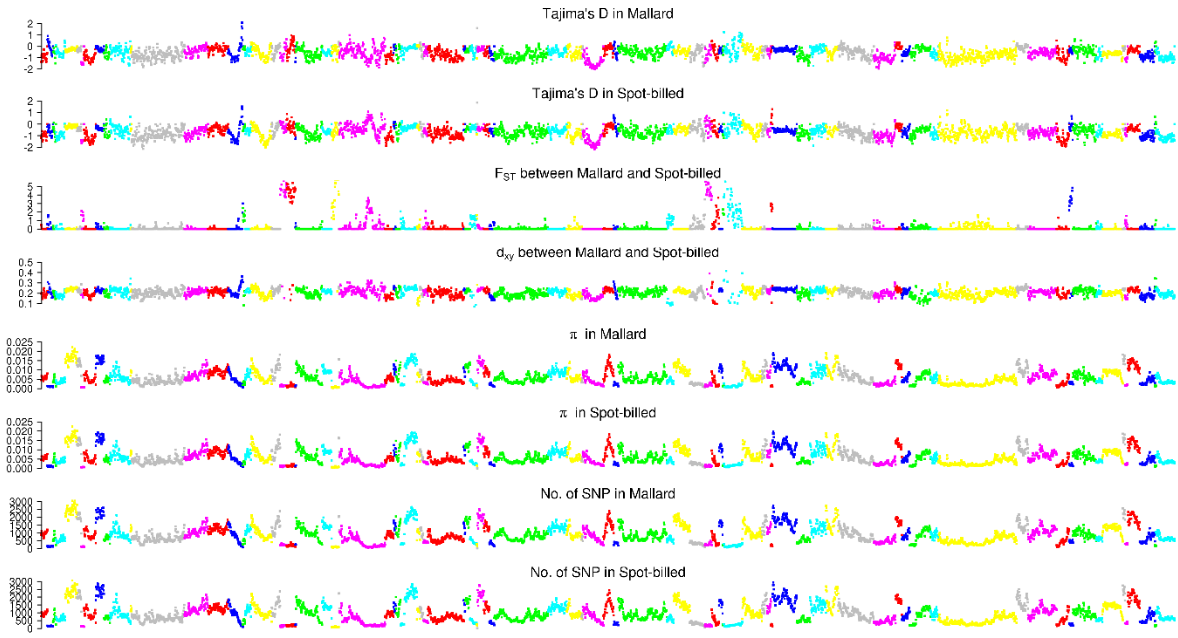

Scaffolds from KB742486.1 to KB742605.1



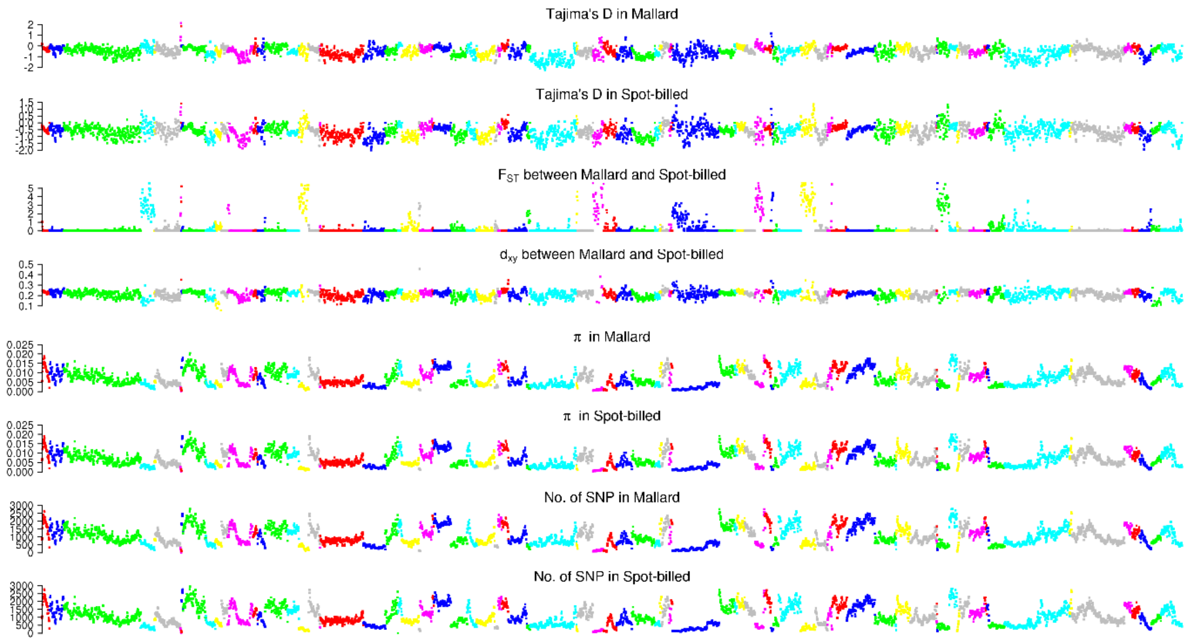

Scaffolds from KB742608.1 to KB742730.1



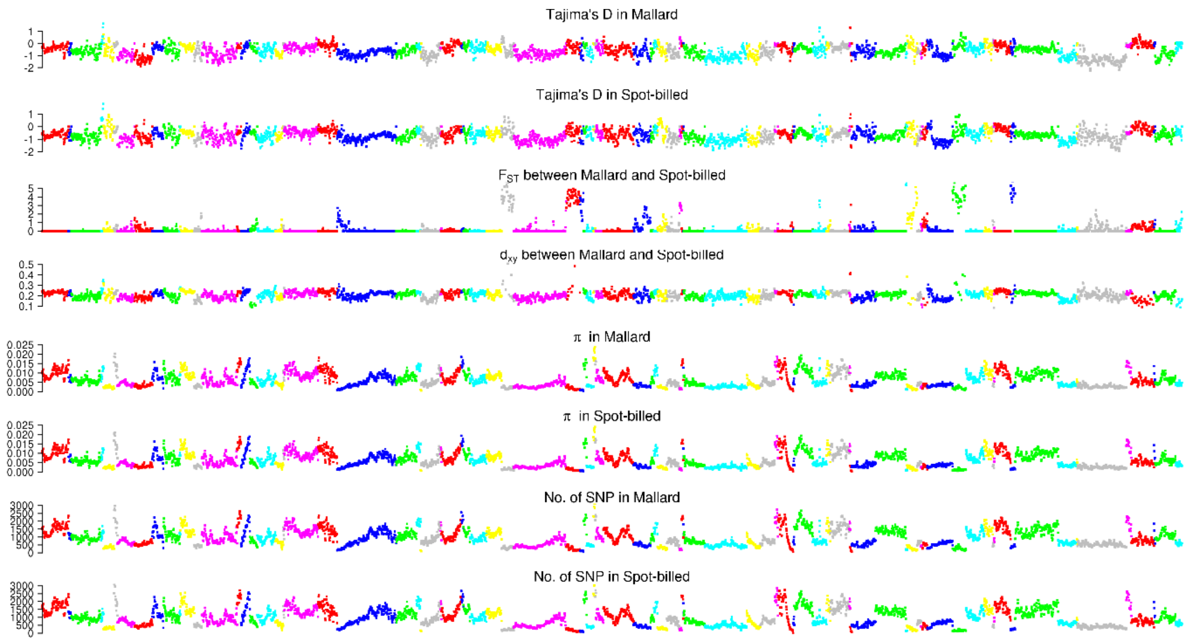

Scaffolds from KB742733.1 to KB742856.1



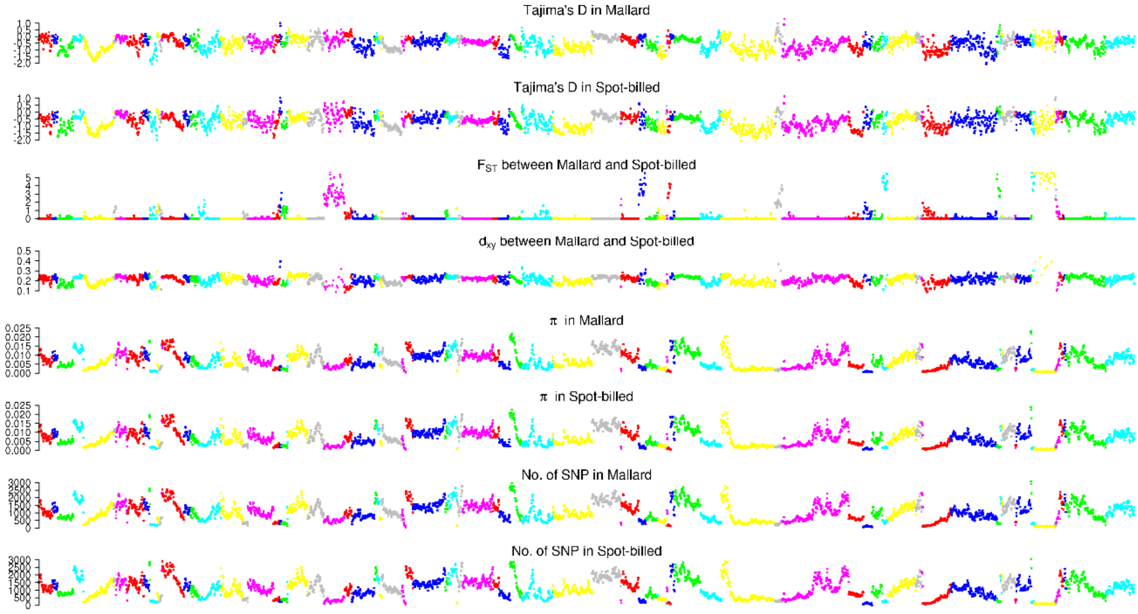

Scaffolds from KB742860.1 to KB742974.1



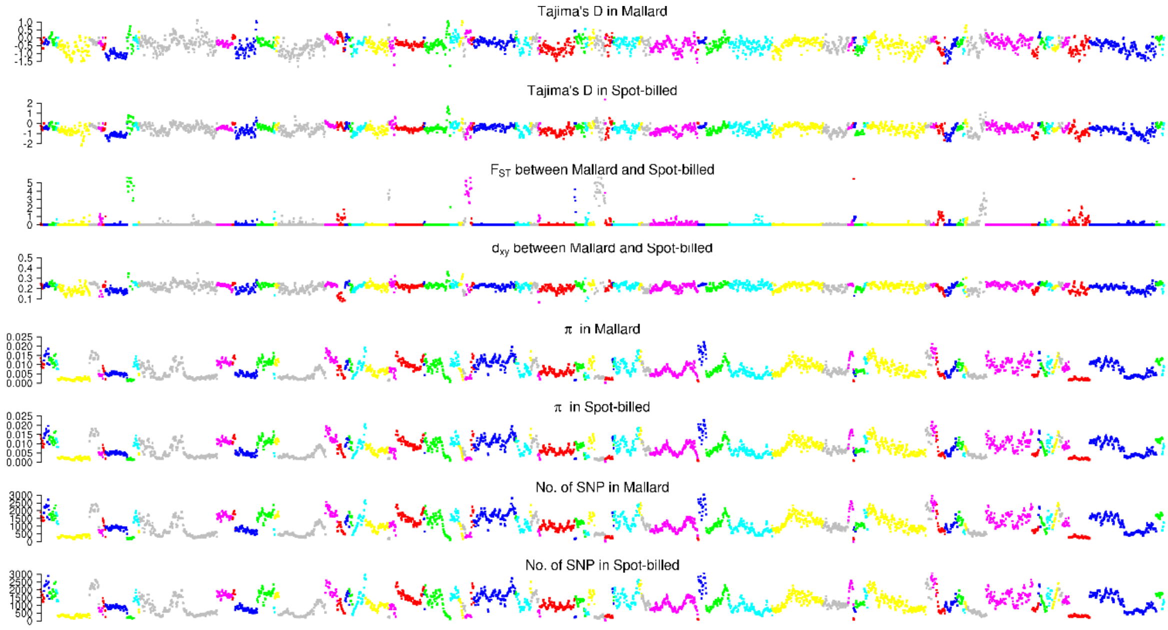

Scaffolds from KB742976.1 to KB743103.1



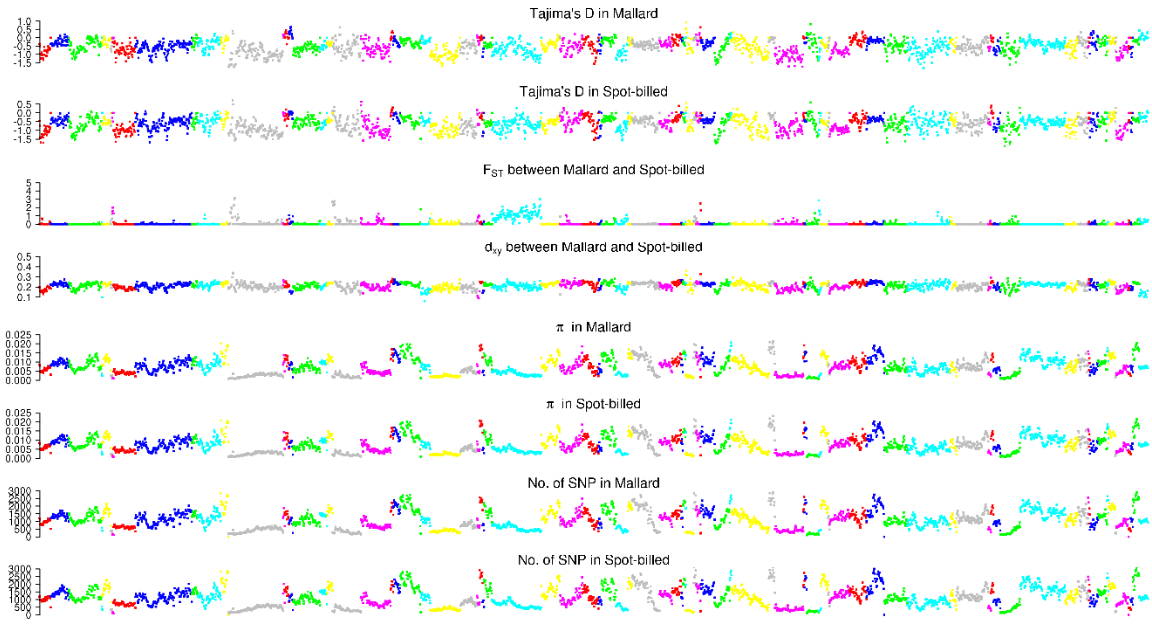

Scaffolds from KB743106.1 to KB743245.1



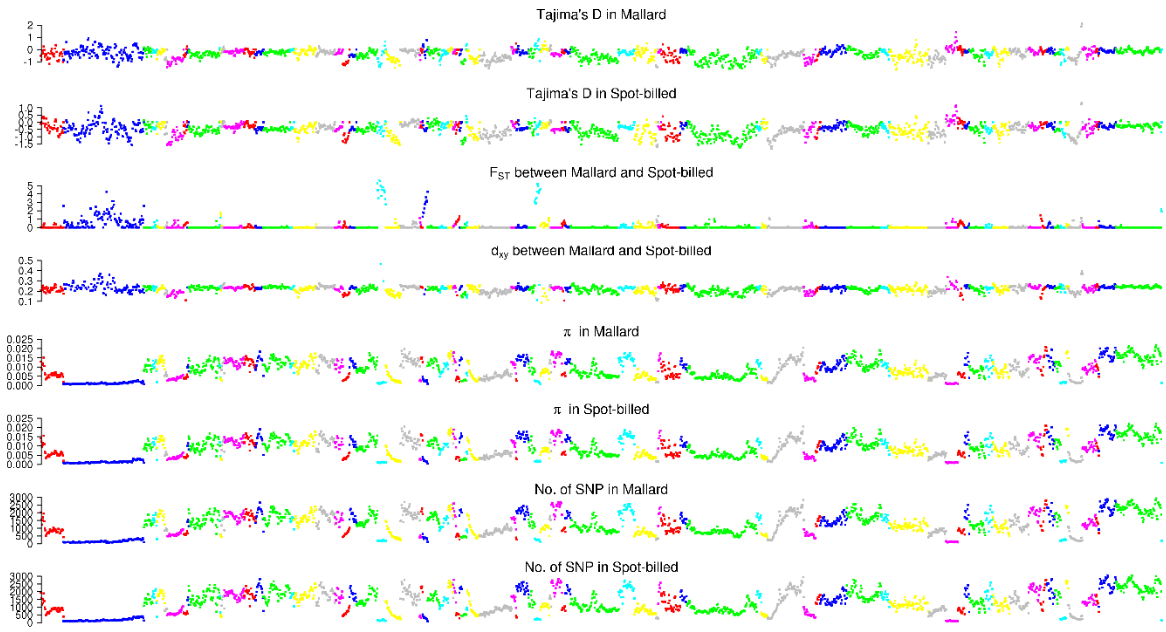

Scaffolds from KB743248.1 to KB743387.1



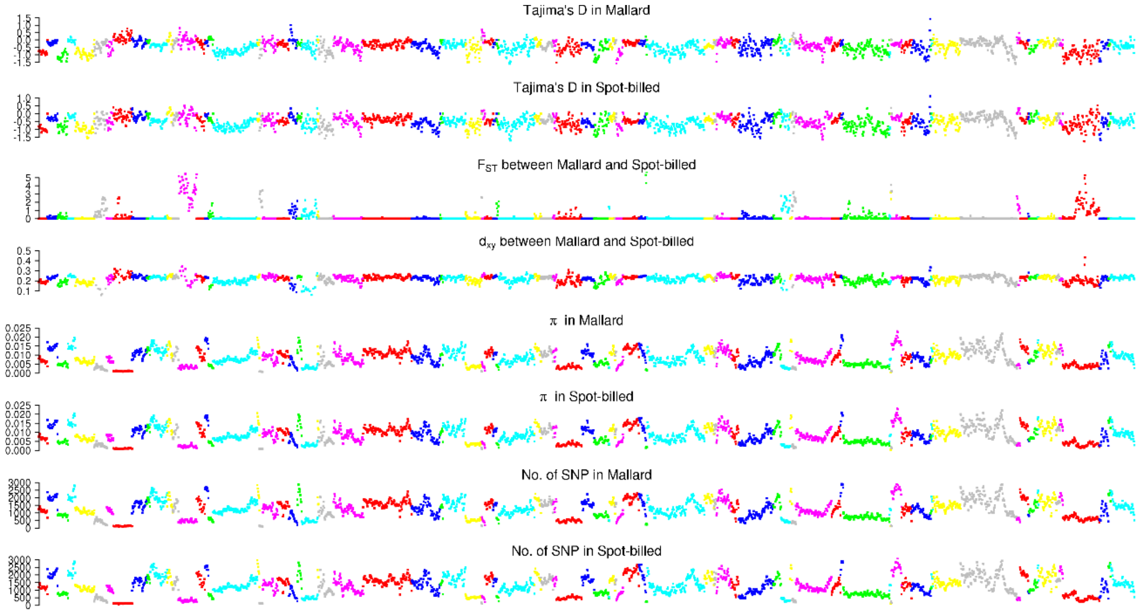

Scaffolds from KB743391.1 to KB743525.1



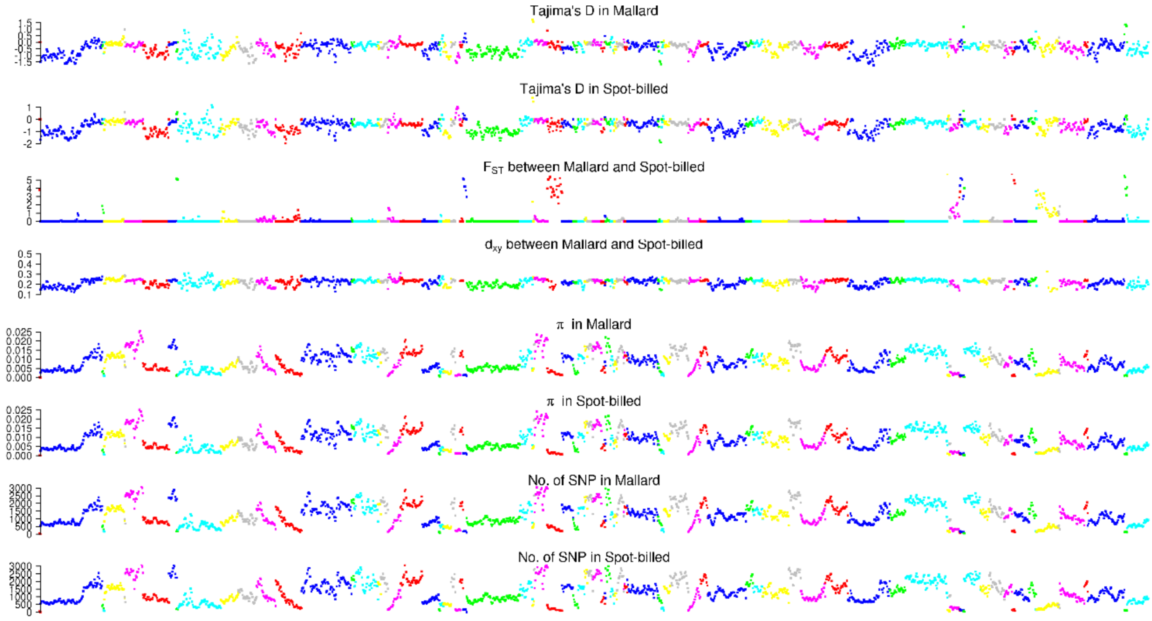

Scaffolds from KB743527.1 to KB743672.1



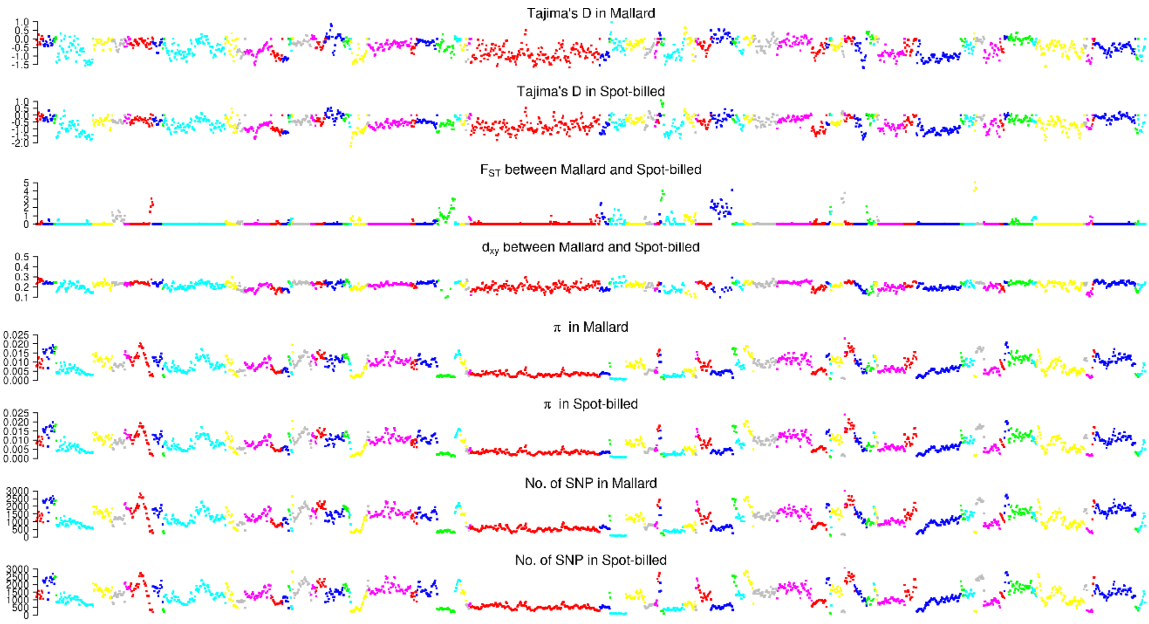

Scaffolds from KB743674.1 to KB743833.1



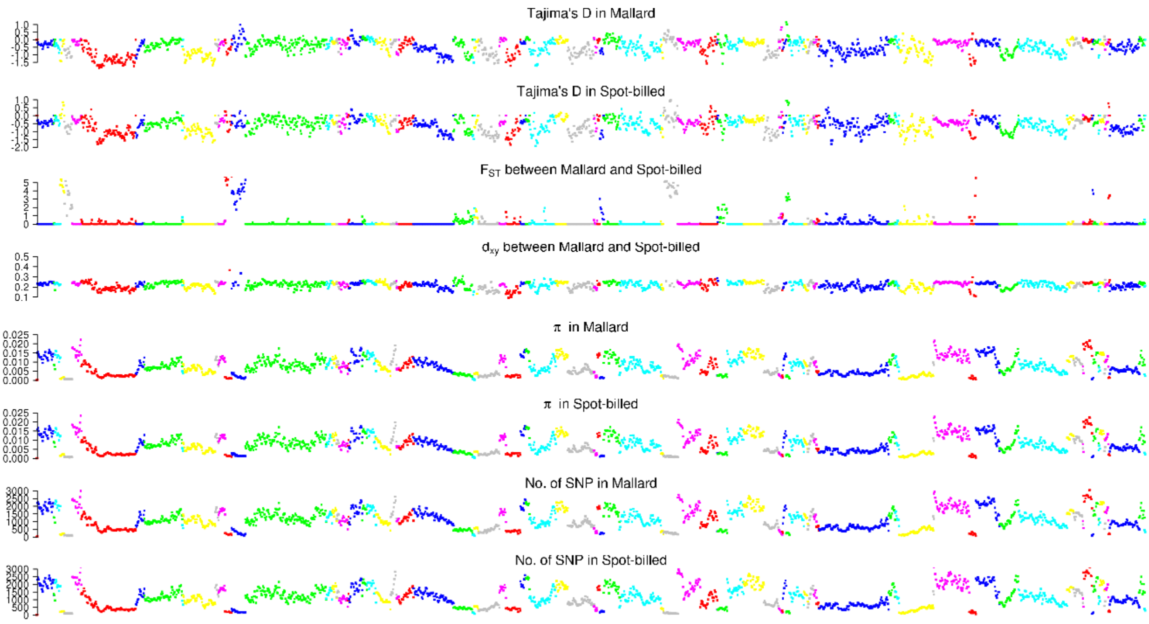

Scaffolds from KB743839.1 to KB744029.1



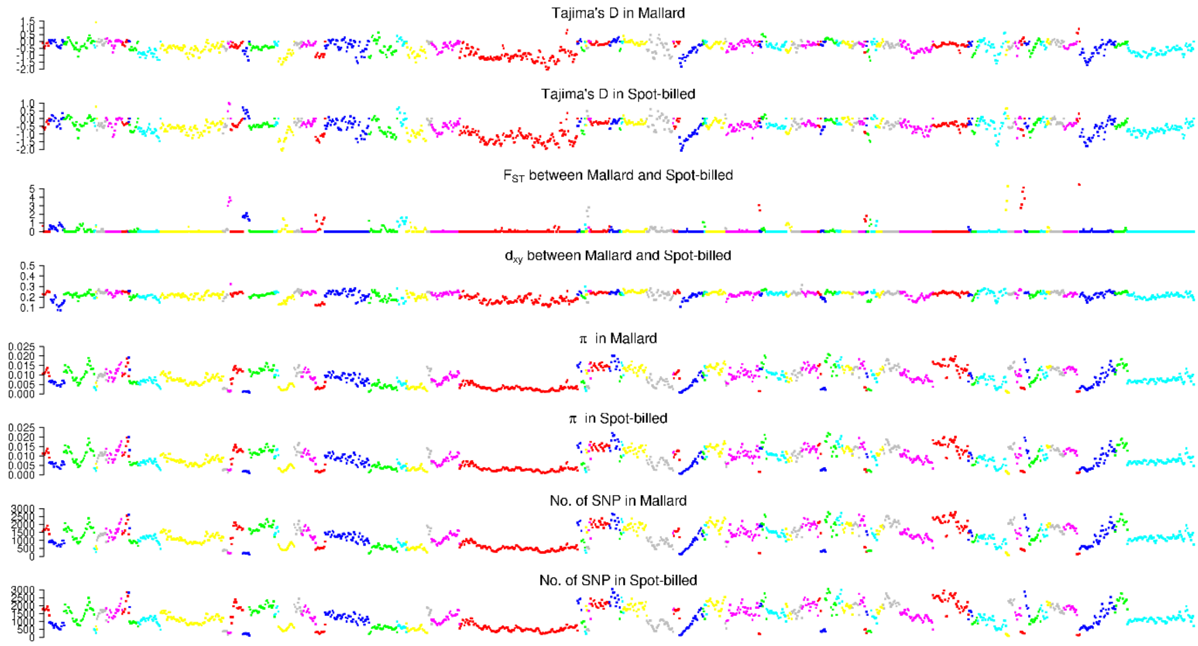

Scaffolds from KB744032.1 to KB744232.1



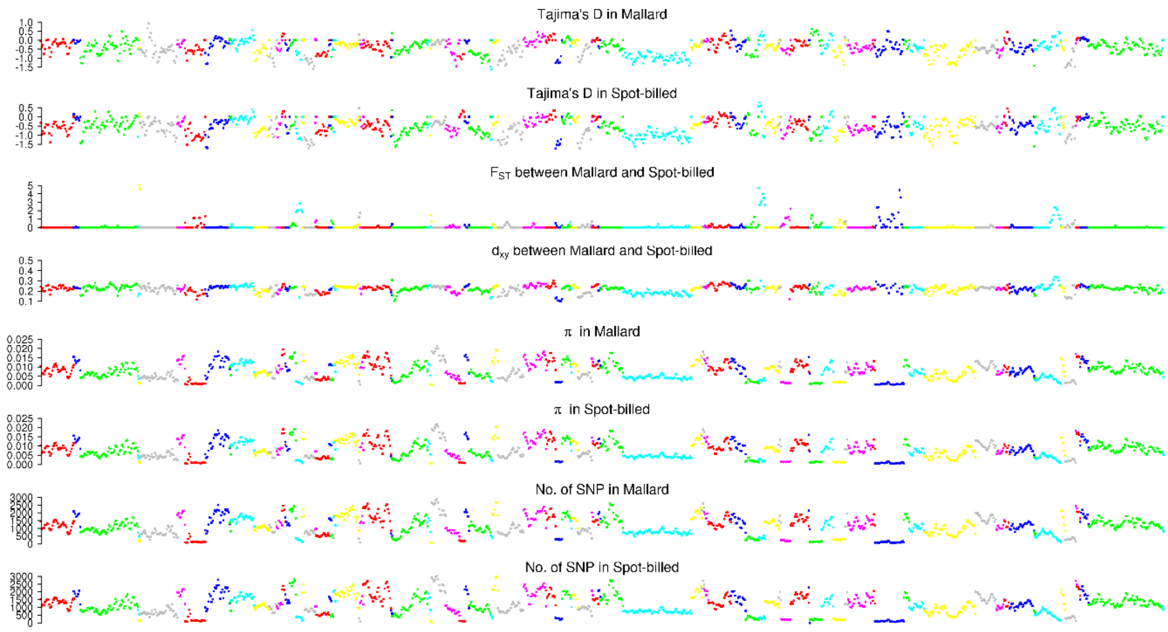

Scaffolds from KB744235.1 to KB744466.1



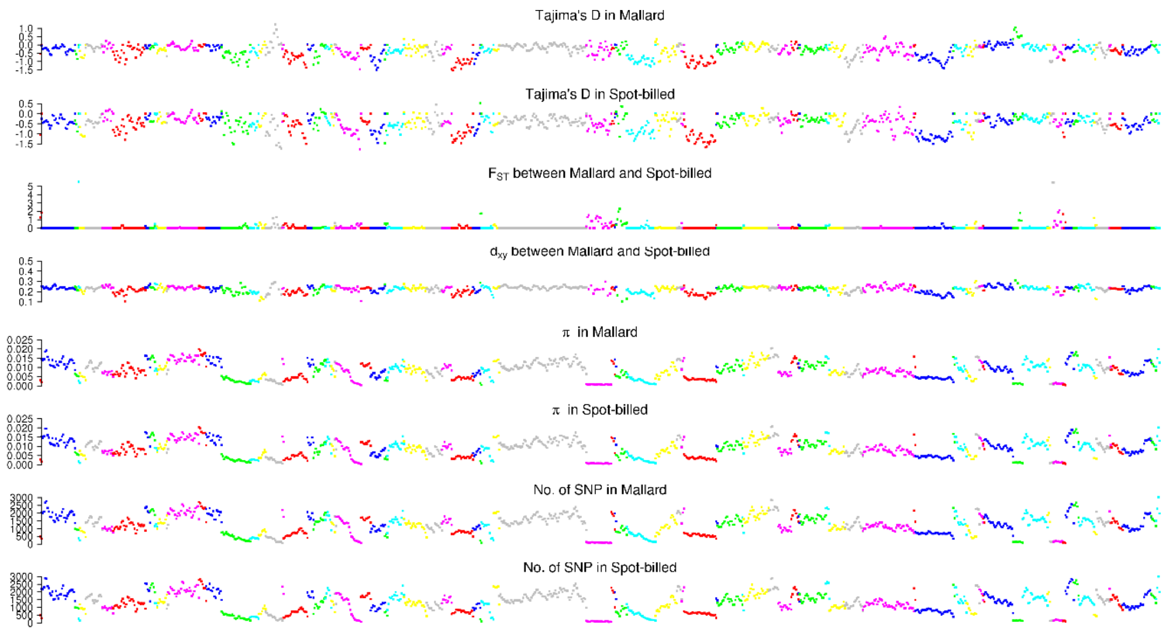

Scaffolds from KB744475.1 to KB744721.1




Scaffolds from KB744725.1 to KB745078.1




Scaffolds from KB745085.1 to KB745764.1




Scaffolds from KB745806.1 to KB747284.1

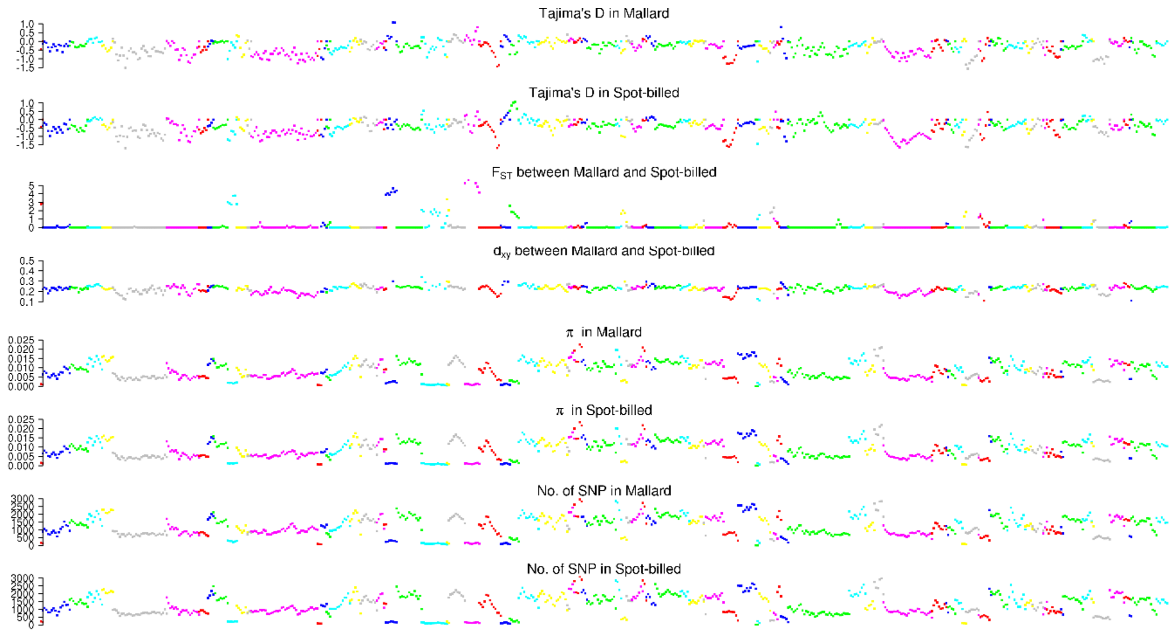



**Fig. 23. Distribution of Tajima'D, positive end of z$F_{ST}$, dxy, number of SNPs per window, nucleotide diversity (π) and zHp along scaffolds in ducks.**

These numbers were counted with 40 kb window by step size of 20 kb.

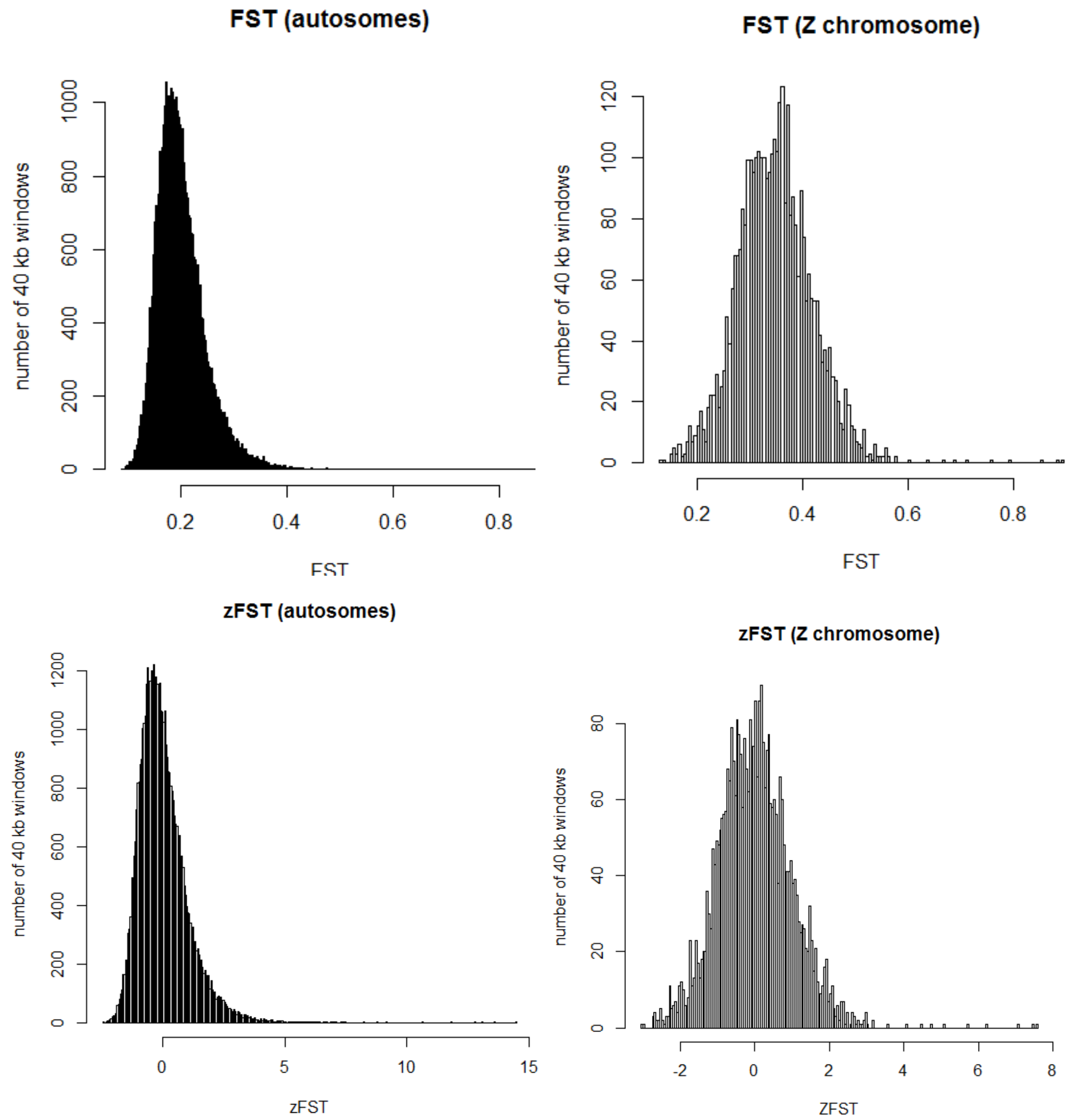



**Fig. S24. Distribution of $F_{ST}$/z$F_{ST}$ between the wild and the domestic duck lineage.**

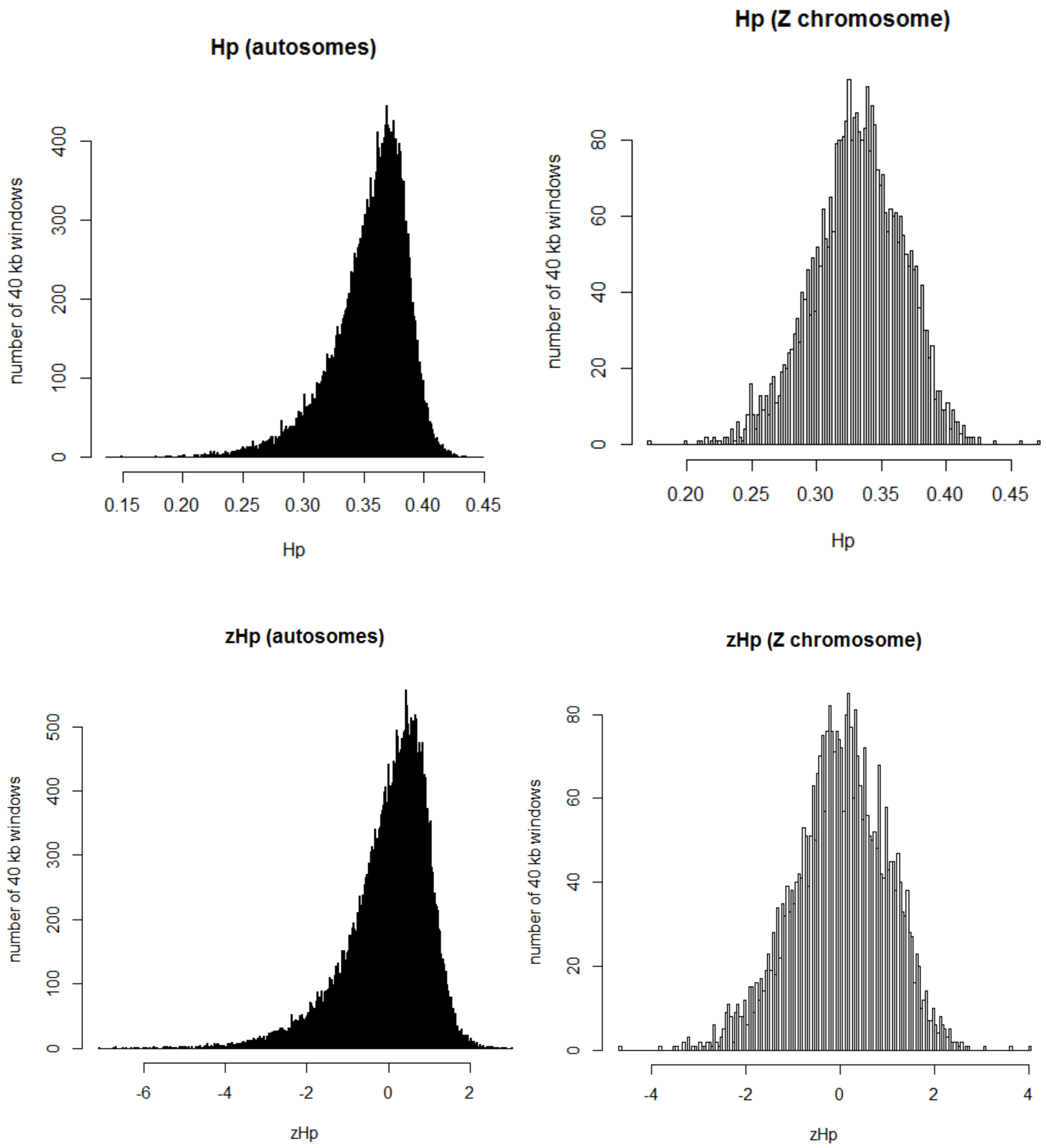



**Fig. S25. Distribution of Hp/zHp in the domestic duck lineage.**

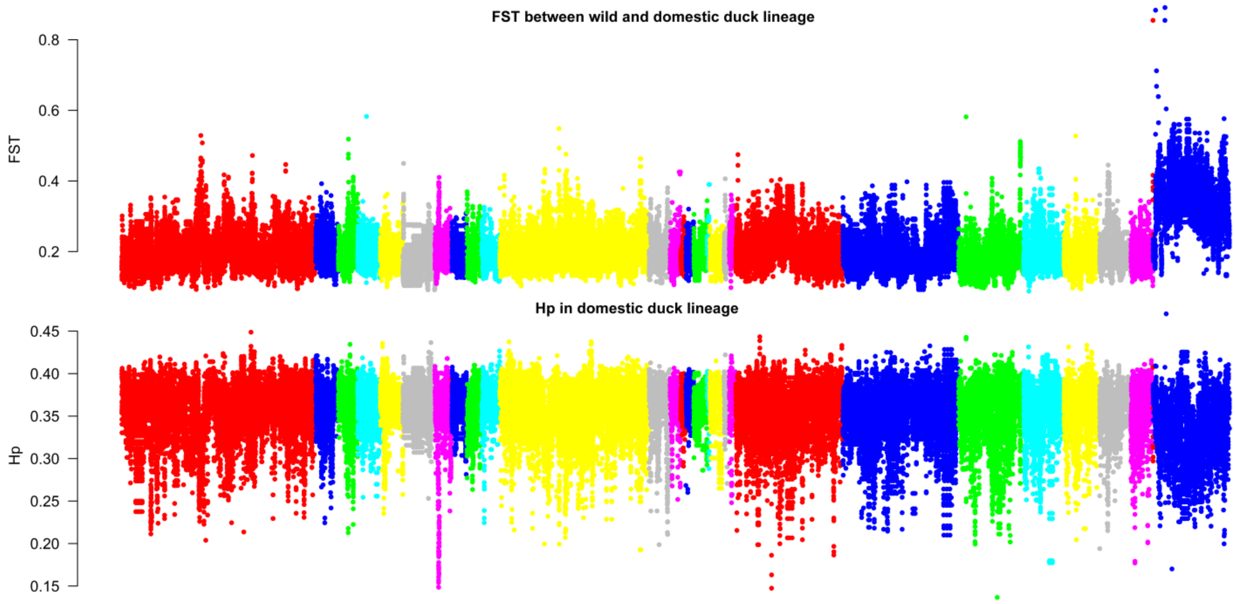



**Fig. S26. Distribution of $F_{ST}$ between the wild and the domestic duck lineage, and Hp in the domestic duck lineage.**

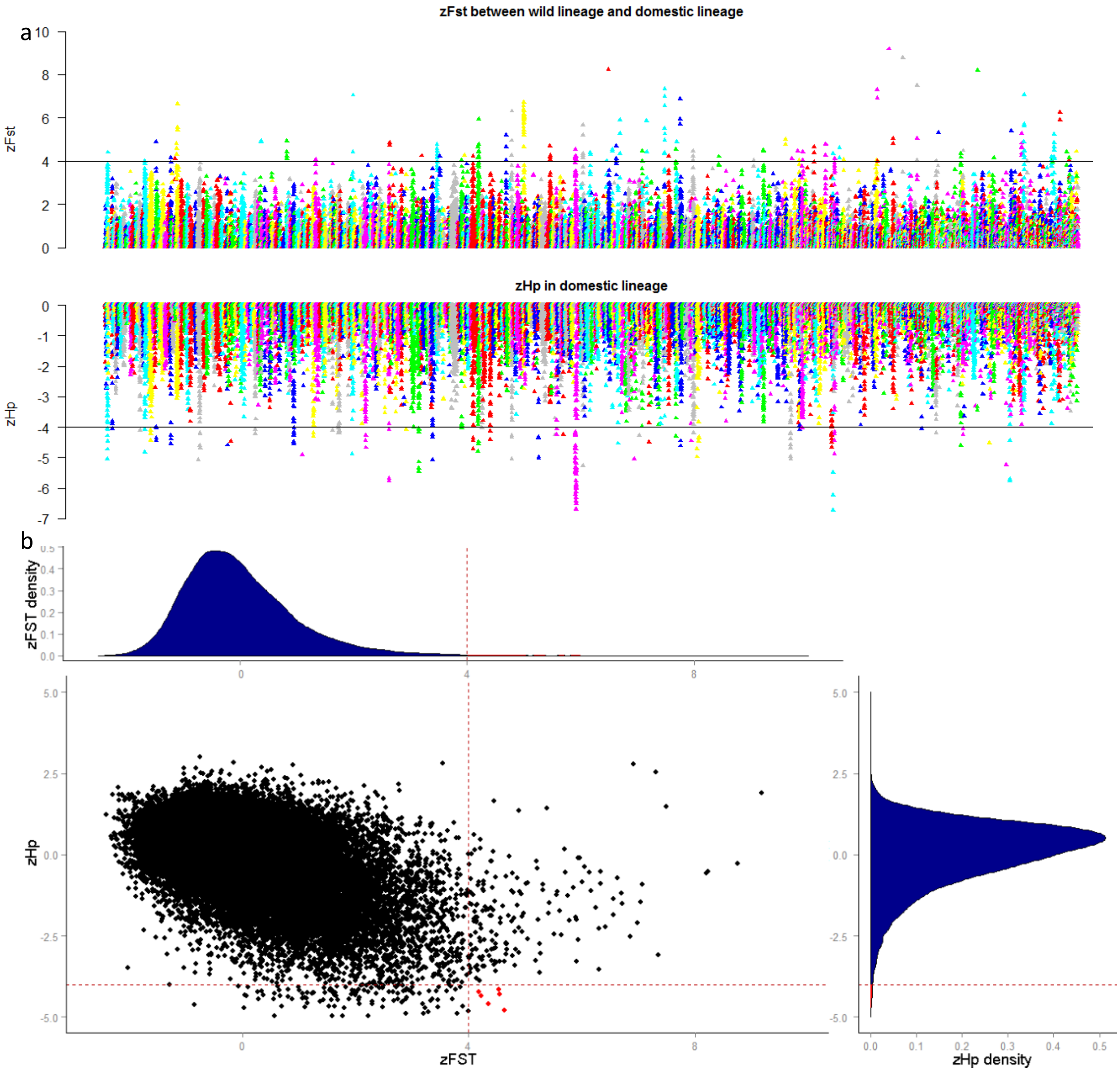



**Fig. S27. Distribution of z$F_{ST}$ and zHp in ducks.**

z$F_{ST}$ and zHp were counted with 40 kb window by a step size of 20 kb. (**a**) The positive end of the z$F_{ST}$ between wild lineage and domestic lineage and negative end of zHp in domestic lineage distribution plotted along scaffolds. (**b**) Distribution of z$F_{ST}$ and zHp. Red dash line indicate threshold and the windows identified as selection regions by both zHp and z$F_{ST}$ were show in red dots. (scaffolds homologous to chicken Z chromosome are not shown)

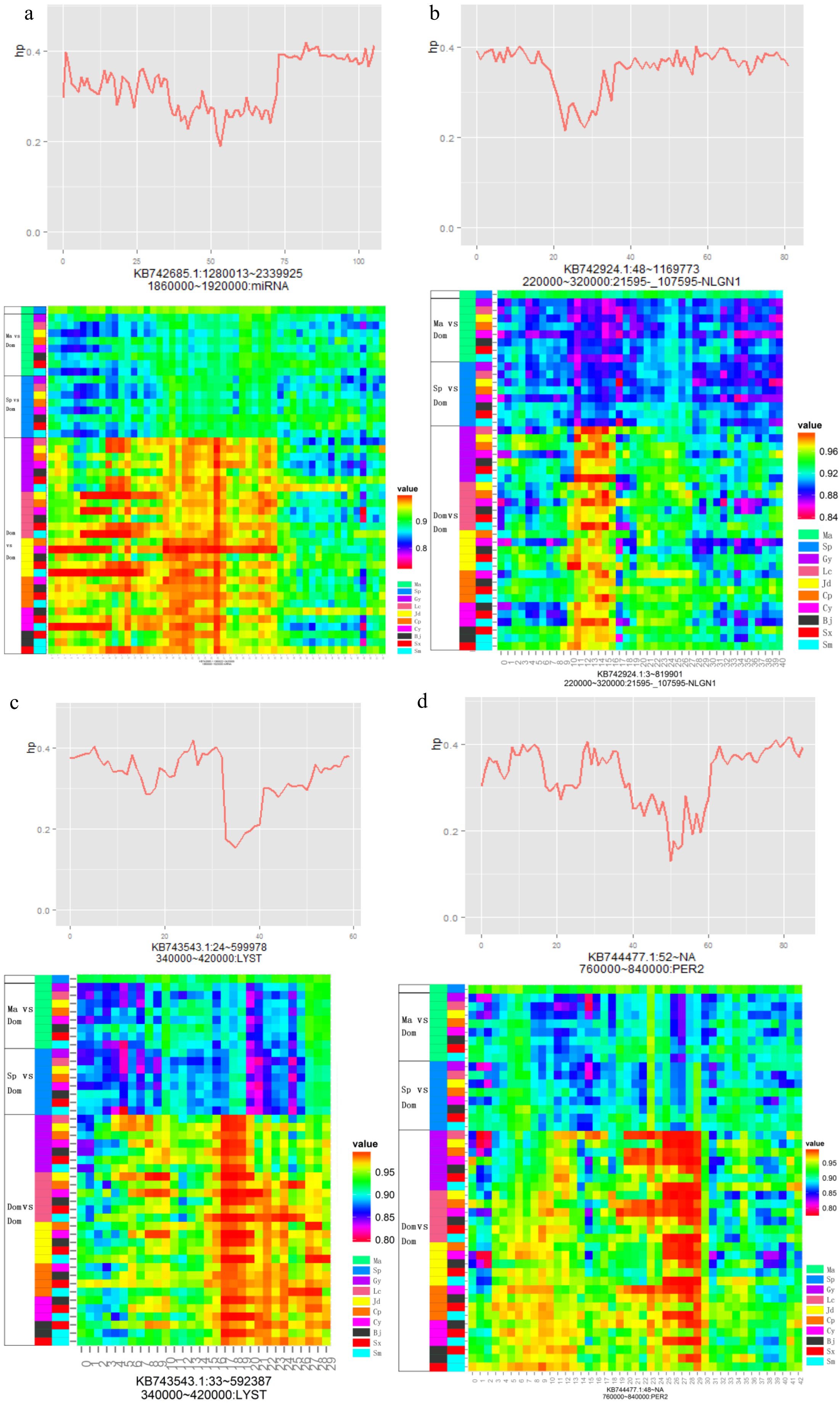

a
hp
KB742685.1:1280013~2339925
1860000~1920000:miRNA
b
hp
KB742924.1:48~1169773
220000~320000:21595-_107595-NLGN1
value
Ma vs Dom
Sp vs Dom
Dom vs Dom
Ma
Sp
Gy
Lc
Jd
Cp
Cy
Bj
Sx
Sm
KB742924.1:3~819901
220000~320000:21595-_107595-NLGN1
c
hp
KB743543.1:24~599978
340000~420000:LYST
KB743543.1:33~592387
340000~420000:LYST
d
hp
KB744477.1:52~NA
760000~840000:PER2
KB744477.1:48~NA
760000~840000:PER2

**Fig. S28. Details of four putative selective sweep regions in the domestic duck lineage.**

Each panel shows distribution of the hp value using non-overlap 10 kb window (upside) and scores between pairwise comparisons in 20 kb non-overlap windows (downside). (**a**) *miR-3598-5p* locus. (**b**) *NLGN1* locus. (**c**) *LYST* locus. (**d**) *PER2* locus.

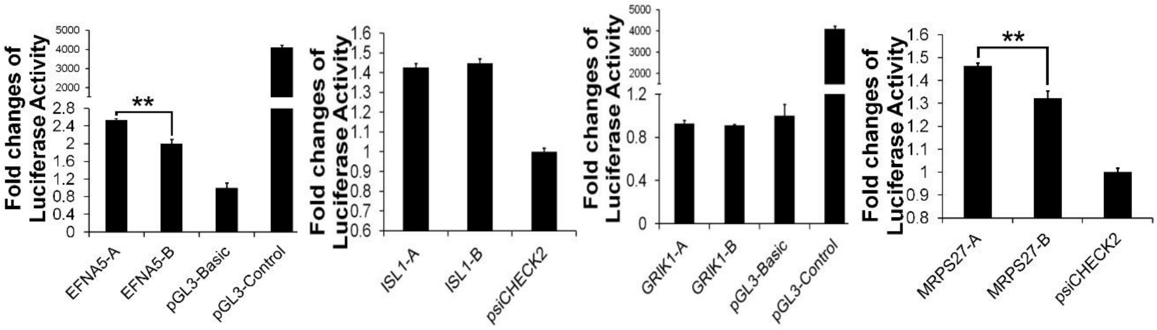
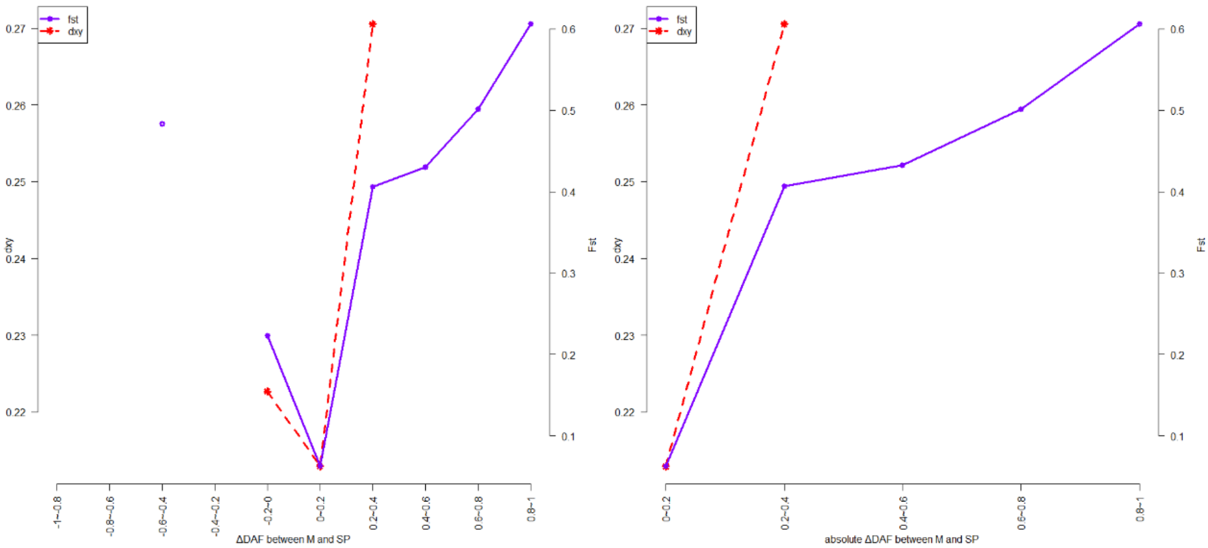



**Fig. S29. Effects of casual candidate mutation on luciferase reporter activity.**

Comparison in luciferase activity of alleles in wild (spot-billed and mallard) and domestic (Beijing) duck in DF1 (chicken embryonic fibroblast) cells. The *ISL1* was under positive selection in early stage of the wild lineage. The *GRIK1* and *LYST* were under positive selection in the domestic lineage. “-A”, “-B” represent wild mutation types and domestic mutation types respectively. DF1 cells were transfected with recombinant pGL3 luciferase reporter plasmids and luciferase activity was measured at 24 hours after transfection.

**a** **Mean $F_{ST}$, dxy of windows stratified by ΔDAF bin** **Mean $F_{ST}$, dxy of windows stratified by abs ΔDAF bin**




**b** **Mean $F_{ST}$ of windows stratified by ΔDAF bin** **Mean $F_{ST}$ of windows stratified by ΔDAF bin**

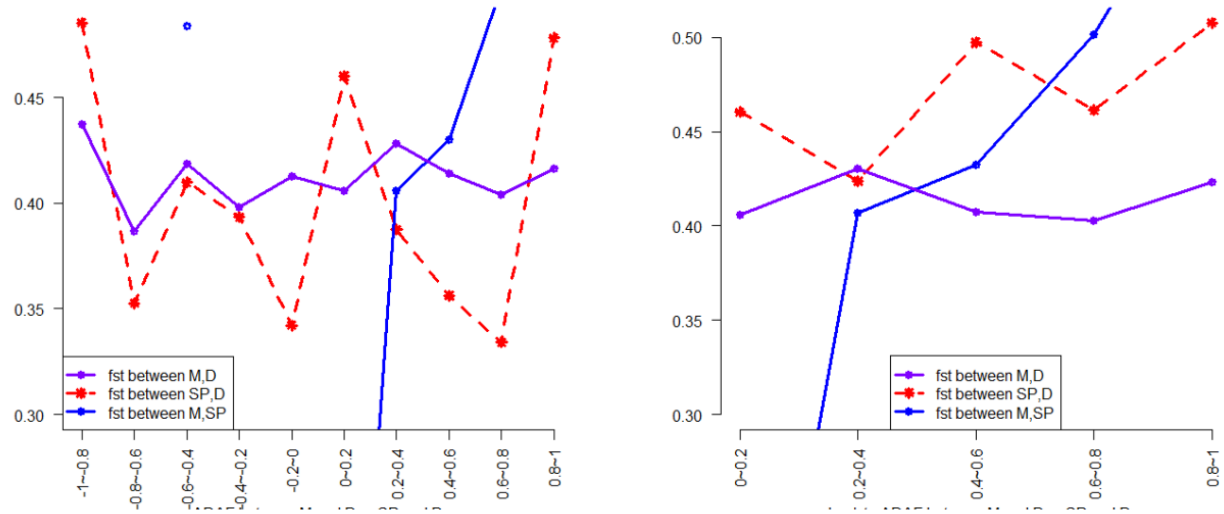
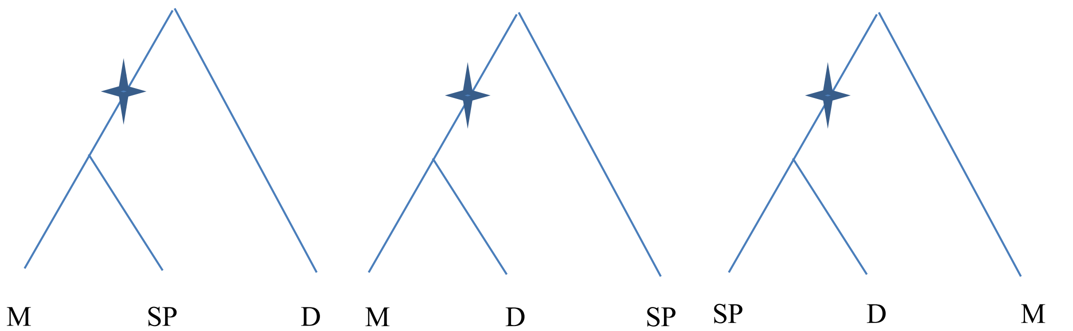



**Fig. S30. Observation on genomic divergence values stratified by ΔDAF provide evidence of introgression between domestic and wild ducks.**
**(a)** Mean $F_{ST}$ (purple lines), dxy (red dash line) of windows stratified by ΔDAF (absolute ΔDAF in right) bins between mallard and spot-billed duck populations. dxy is not available in higher ΔDAF bins due to fixed SNPs are exclude in dxy calculation. **(b)** Mean $F_{ST}$ between domestic and wild duck populations decreased in high ΔDAF bins may due to introgression. Blue lines (corresponding to purple line in (**a**)) indicate no evidence of introgression between mallard and spot-billed in compare with the measurement between wild (red dash line: spot-billed, purple line: mallard) and domestic ducks. "M", "SP" and "D" indicate mallard, spot-billed and domestic duck, respectively.

**a**




Number of sites supporting different tree:
BBAA:117315 (49%) BABA:70662 (29%) ABBA:52500 (22%)

**b**

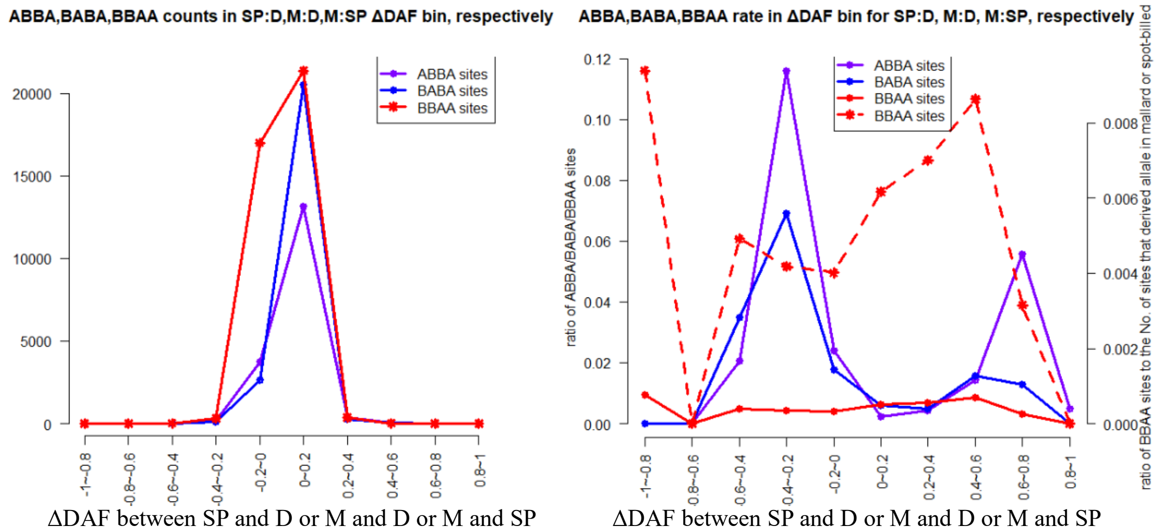



**Fig. S31. Absent evidence of introgression between mallard duck and spot-billed duck population based on ABBA-BABA analysis of mallard, spot-billed and domestic duck. (a)** Number of sites supporting different trees is indicated both as a percentage and as actual numbers. **(b)** Left panel, number of ABBA, BABA and BBAA sites were stratified by ΔDAF between spot-billed: domestic (purple line), mallard: domestic (blue line) and mallard: spot-billed (red line), respectively. Right panel, BBAA rate (red line) were measured as the BBAA counts divided by number of sites that derived allele present in either mallard or spot-billed population in each bins (ABBA (purple line), BABA (blue line) similarly). The red dash line is corresponding to right y-axis. “M”, “SP” and “D” indicate mallard, spot-billed and domestic duck, respectively.

## Supplementary Tables

**Table S1: Statistics of the reads in mallard and spot-billed genome assemblies**

| Paired-end libraries (bp) | Paired-end insert size | Number of libraries | Average read length (bp) | Total data (Gb) | Sequence coverage (X) | Physical coverage (X) |
|---|---|---|---|---|---|---|
| mallard | 250bp | 3 | 150_150 | 89.76 | 70.81 | 59.01 |
| | 500bp | 2 | 100_100 | 46.18 | 36.94 | 92.35 |
| | 800bp | 2 | 100_100 | 57.4 | 45.92 | 183.68 |
| | 2K | 3 | 49_49 | 27.37 | 21.9 | 446.94 |
| | 5K | 2 | 49_49 | 16.32 | 13.06 | 666.33 |
| | 10K | 3 | 49_49 | 11.67 | 9.34 | 953.06 |
| | 20K | 2 | 49_49 | 13.75 | 11 | 2,244.90 |
| Total | / | 17 | / | 262.45 | 208.97 | 4,646.27 |
| | | | | | | |
| spot-billed | 250bp | 3 | 150_150 | 95.64 | 76.51 | 63.76 |
| | 500bp | 2 | 100_100 | 63.39 | 50.71 | 126.78 |
| | 800bp | 2 | 100_100 | 45.31 | 36.25 | 145 |
| | 2K | 3 | 49_49 | 26.2 | 20.96 | 427.76 |
| | 5K | 2 | 49_49 | 17.11 | 13.69 | 698.47 |
| | 10K | 3 | 49_49 | 15.37 | 12.3 | 1,255.10 |
| | 20K | 2 | 49_49 | 12.17 | 9.74 | 1,987.76 |
| Total | / | 17 | / | 275.19 | 220.16 | 4,704.63 |

**Table S2: Statistics of the three *de novo* ducks assemblies**

| Category | Contig | | | Scaffold | | |
|---|---|---|---|---|---|---|
| | Beijing Duck | Mallard | Spot-billed | Beijing Duck | Mallard | Spot-billed |
| Total length(kb) | 1,069,961 | 1,176,969 | 1,187,333 | 1,105,049 | 1,270,808 | 1,317,385 |
| Total number | 227,597 | 172,189 | 194,934 | 78,487 | 61,591 | 67,685 |
| Longest size(bp) | 263,737 | 484,574 | 673,122 | 5,998,093 | 17,658,973 | 25,320,044 |
| N50 size(bp) | 26,138 | 38,271 | 33,061 | 1,233,631 | 2,489,142 | 2,054,876 |
| N50 number | 11,193 | 7,876 | 9,194 | 268 | 125 | 154 |
| N90 size(bp) | 3,085 | 5,381 | 4,386 | 195,458 | 145,390 | 102,224 |

| | | | | | | |
|---|---|---|---|---|---|---|
| N90 number | 53,850 | 38,114 | 45,269 | 1,097 | 870 | 1,035 |

**Table S3: Comparison in length and coverage of the mallard and the spot-billed genome using seven Beijing duck BAC sequences**

| BACs name | Length of BACs (bp) | mallard coverage | spot-billed coverage |
|---|---|---|---|
| ndae_TLR1 | 67,954 | 85.91% | 70.03% |
| ndbaa_TLR15 | 79,704 | 98.50% | 98.25% |
| ndbab_TLR5 | 72,128 | 98.41% | 98.32% |
| ndbac_1_TLR3 | 76,489 | 97.53% | 96.49% |
| ndbad_2_TLR7 | 97,799 | 95.64% | 87.51% |
| ndbae_TLR2 | 75,080 | 98.82% | 93.59% |
| ndbag_TLR4 | 77,042 | 97.79% | 97.98% |

**Table S4: Evaluation of the mallard genome coverage using duck EST sequences**

| Dataset | Number | Total length (bp) | Covered by assembly (%) | with >90% sequence in one scaffold | | with >50% sequence in one scaffold | |
|---|---|---|---|---|---|---|---|
| | | | | Number | Percent (%) | Number | Percent (%) |
| >0bp | 319,996 | 98,257,350 | 99.02 | 294,845 | 92.14 | 315,689 | 98.65 |
| >200bp | 144,337 | 73,939,509 | 99.53 | 135,799 | 94.08 | 143,128 | 99.16 |
| >500bp | 43,896 | 43,155,260 | 99.84 | 41,595 | 94.76 | 43,687 | 99.52 |
| >1000bp | 13,652 | 22,939,026 | 99.99 | 12,831 | 93.99 | 13,613 | 99.71 |

**Table S5: Evaluation of the spot-billed genome coverage using duck EST sequences**

| Dataset | Number | Total length (bp) | Covered by assembly (%) | with >90% sequence in one scaffold | | with >50% sequence in one scaffold | |
|---|---|---|---|---|---|---|---|
| | | | | Number | Percent (%) | Number | Percent (%) |
| >0bp | 319,996 | 98,257,350 | 99.01 | 294,324 | 91.98 | 315,566 | 98.62 |
| >200bp | 144,337 | 73,939,509 | 99.53 | 135,565 | 93.92 | 143,053 | 99.11 |
| >500bp | 43,896 | 43,155,260 | 99.85 | 41,531 | 94.61 | 43,670 | 99.49 |
| >1000bp | 13,652 | 22,939,026 | 99.99 | 12,802 | 93.77 | 13,606 | 99.66 |

**Table S6: Type and proportion of transposable elements(TEs) in the Mallard, Spot-billed, Beijing duck and chicken genome**

| TE type | mallard length (bp) | % genome | Spot-billed length (bp) | % genome | Beijing duck length (bp) | % genome | Chicken length (bp) | % genome |
|---|---|---|---|---|---|---|---|---|
| DNA | 9,479,699 | 0.75 | 9,438,606 | 0.71 | 5,590,903 | 0.51 | 13,872,990 | 1.33 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| LINE | 98,060,422 | 7.71 | 101,177,399 | 7.68 | 63,146,176 | 5.71 | 116,778,031 | 11.15 |
| SINE | 1,389,649 | 0.1 | 1,274,941 | 0.09 | 892,886 | 0.08 | 573,610 | 0.05 |
| LTR | 52,195,974 | 4.1 | 48,381,075 | 3.67 | 21,255,312 | 1.92 | 47,374,647 | 4.53 |
| Other | 21,136 | 0 | 12,303 | 0 | 3,201 | 0 | 3,338 | 0 |
| Unknown | 1,100,812 | 0.08 | 3,402,377 | 0.25 | 4,600,535 | 0.41 | 3,351,756 | 0.32 |
| Total | 136,134,153 | 10.71 | 139,051,537 | 10.55 | 80,740,993 | 7.31 | 149,208,388 | 14.25 |

**Table S7: Gene annotation of the mallard and spot-billed reference gene sets using five protein databases**

| Database | Mallard | | Spot-billed | |
|---|---|---|---|---|
| | Number | Percent (%) | Number | Percent (%) |
| Total | 21,056 | 100 | 21,123 | 100 |
| InterPro | 15,335 | 72.83 | 15,308 | 72.47 |
| GO | 12,817 | 60.87 | 12,815 | 60.67 |
| KEGG | 15,878 | 75.41 | 15,803 | 74.81 |
| Swissprot | 17,551 | 83.35 | 17,531 | 82.99 |
| TrEMBL | 18,371 | 87.25 | 18,327 | 86.76 |
| All annotated | 18,467 | 87.70 | 18,426 | 87.23 |
| Unannotated | 2,589 | 12.29 | 2,697 | 12.77 |

**Table S8: Gene family of two mammals and seven birds clustering by TreeFam**

| Species | Total genes | Unclustered genes | Families | Unique families | Ave. genes per family |
|---|---|---|---|---|---|
| mallard | 21,056 | 2,478 | 14,541 | 30 | 1.28 |
| Spot-billed | 21,123 | 2,604 | 14,530 | 42 | 1.27 |
| Beijing duck | 20,629 | 5,381 | 13,498 | 18 | 1.13 |
| *G. gallus* | 15,508 | 493 | 13,351 | 6 | 1.12 |
| *S. camelus* | 16,178 | 326 | 11,579 | 105 | 1.37 |
| *M. gallopavo* | 14,123 | 455 | 12,414 | 7 | 1.10 |
| *F. albico* | 15,303 | 716 | 13,284 | 33 | 1.10 |
| *H. sapiens* | 20,238 | 1,825 | 15,036 | 117 | 1.22 |
| *M. musculus* | 22,627 | 1,549 | 15,407 | 139 | 1.37 |

**Table S9: Over represented GO terms (p-value <0.001) of wild duck expanded gene families**

See file “Table_S9.xlsx”

**Table S10: Summary for whole-genome resequencing**

| Group | Species/breeds | Sample number | Sequence depth | Total data (Gb) | Read length | Sequence strategy | SNP number |
|---|---|---|---|---|---|---|---|
| Wild duck | Mallard | 14 | 14.1× | 218.2 | 125 | Individual | 32,982,331 |
| | Spot-billed | 13 | 14.5× | 209.5 | 125 | Individual | 33,581,513 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | Wigeon | 3 | 20× | 75.6 | 125 | Individual | 21,393,824 |
| | Gadwall | 1 | 20× | 17.0 | 125 | Individual | 10,756,767 |
| | Pintail | 1 | 20× | 25.7 | 125 | Individual | 8,551,170 |
| | Falcata teal | 1 | 20× | 22.3 | 125 | Individual | 7,658,519 |
| | Common teal | 1 | 20× | 22.1 | 125 | Individual | 9,146,292 |
| | Baikal teal | 1 | 20× | 23.2 | 125 | Individual | 8,500,280 |
| Indigenous Chinese duck breeds | Beijing | 27 | 13.5× | 400.7 | 125 | Individual | 14,748,500 |
| | Shaoxing | 27 | 2~3.5× | 95.1 | 100 | Individual | 15,414,890 |
| | Gaoyou | 10 | 4.7× | 56.9 | 100 | Pooling | 9,190,980 |
| | Shanma | 10 | 4.0× | 50.4 | 100 | Pooling | 10,185,166 |
| | Lianchengbai | 10 | 4.7× | 57.6 | 100 | Pooling | 8,978,662 |
| | Jinding | 10 | 4.4× | 55.5 | 100 | Pooling | 8,617,344 |
| Commercial breeds | Cherry valley | 10 | 7.5× | 94.5 | 100 | Pooling | 9,615,871 |
| | Campbell | 10 | 4.5× | 57.3 | 100 | Pooling | 9,821,965 |
| Outgroup | Muscovy | 10 | 5.24× | 57.9 | 100 | Pooling | -- |
| | Taihu goose | 10 | 3.5× | 39.7 | 100 | Pooling | -- |

**Table S11: Tests for population mixture of mallard, spot-billed, Beijing and Shaoxing ducks**

| Source 1 | Source 2 | Target | f3 | Standard error | Z | SNPs |
|---|---|---|---|---|---|---|
| Mallard | Spot-billed | Beijing | 0.007886 | 0.000748 | 10.546 | 9,997 |
| Mallard | Spot-billed | Shaoxing | 0.008662 | 0.000814 | 10.643 | 9,943 |
| Mallard | Spot-billed | Mix Beijing, Shaoxing | 0.007279 | 0.000305 | 23.834 | 2,930,914 |
| Mallard_hz | Spot-billed_hz | Beijing | 0.008656 | 0.001076 | 8.045 | 9,975 |
| Mallard_hz | Spot-billed_hz | Shaoxing | 0.010200 | 0.000683 | 14.930 | 9,907 |
| Mallard_fh | Spot-billed_fh | Beijing | 0.007320 | 0.000603 | 12.147 | 9,986 |
| Mallard_fh | Spot-billed_fh | Shaoxing | 0.007380 | 0.000988 | 7.470 | 9,921 |

Considering the extensive substructure exist in wild ducks which reflect those individuals may trace their recent ancestry to different populations even species, we assume different clusters combinations to source 1 and source 2, and combine different domestic breeds to one target population, to minimize the population effect. All the Z were significantly greater than zero.

**Table S12: Divergence without migration inferred parameters.**

| combinations | Parameter[a] | Conventional bootstrap 95% confidence interval | Conventional bootstrap 95% confidence interval (convert to real) | Maximum likelihood |
|---|---|---|---|---|
| Domestic, mallard duck | s (Nmallard) | 0.85658-0.874169 | 1567694-1637111 | 1595689 |
| | 1-s(Ndomestic) | 0.125831-0.143419 | 233297-265236 | - |
| | Ts | 0.0272-0.031008 | 100818-114733 | 102427 |
| Domestic, spot-billed duck | s(Nspot-bill) | 0.8776821-0.893130 | 1660302-1731619 | 1659603 |
| | 1-s(Ndomestic) | 0.10687-0.122315 | 205404-233549 | - |

| | | | | |
|---|---|---|---|---|
| | Ts | 0.02321-0.02653 | 89220-101346 | 91700 |
| Spot-billed, mallard duck | s(Nmallard) | 0.4352336-0.472079 | 949555-1031148 | 969384 |
| | 1-s(Nspot~) | 0.52792-0.564766 | 1146439- 1239046 | - |
| | Ts | 0.015755-0.0174258 | 68583-76294 | 69432 |

[a] see model code above.

**Table S13: Summary of selective sweep regions in the wild duck lineage**

These selective sweep regions occurred during their population in conjunction with or shortly after, their population divergence from domestic lineage (S<-4 for autosomes, S<-1.5 for Z-chromosomes). See file "Table_S13.xlsx"

**Table S14: Enriched gene ontology of positively selected genes in the wild duck lineage.**

See file "Table_S14.xlsx"

**Table S15: Enriched gene ontology in biological process terms of positively selected genes in wild duck lineage.**

See file "Table_S15.xlsx"

**Table S16: Mean values of population genomic parameters for Z-chromosome and autosomes in mallard and spot-billed**

| Index | | Autosomes | | Z-chromosome | |
|---|---|---|---|---|---|
| | | mean | SD | Mean | SD |
| Pi | Mallard | 0.007520 | 0.004199 | 0.003397 | 0.002495 |
| | Spot-billed | 0.007607 | 0.004321 | 0.003020 | 0.002625 |
| SNPs | Mallard | 1,123 | 572 | 494 | 348 |
| | Spot-billed | 1,150 | 600 | 440 | 376 |
| Tajima's D | Mallard | -0.525593 | 0.435471 | -0.257150 | 0.445641 |
| | Spot-billed | -0.625759 | 0.420070 | -0.348016 | 0.492465 |
| $F_{ST}$ | | 0.049303 | 0.034864 | 0.259623 | 0.150430 |

**Table S17: Mean values of genomic parameters inside/outside high $F_{ST}$ (between mallard and spot-billed ducks) regions in autosomes**

| Species | | pi (π) | | No. of SNPs | | Tajima's D | |
|---|---|---|---|---|---|---|---|
| | | mean | sd | mean | sd | mean | sd |
| Mallard | Inside | 0.002421 | 0.00217898 | 360 | 305 | -0.61100 | 0.579419 |
| | Outside | 0.007595 | 0.00418171 | 1,134 | 569 | -0.52064 | 0.433172 |
| Spot-billed | Inside | 0.002331 | 0.00196792 | 334 | 280 | -0.47211 | 0.656325 |
| | Outside | 0.007683 | 0.00430549 | 1,161 | 597 | -0.6228 | 0.415501 |

**Table S18: Summary of domestic candidate regions (zHp<-4 for autosomes, zHp<-2 for Z-chromosomes)**

See file "Table_S18.xlsx"

**Table S19: Summary of high $F_{ST}$ regions between the wild and domestic duck lineage.**

See file "Table_S19.xlsx"

**Table S20: Enriched gene ontology of positively selected genes in/close to domestic candidate regions.**

See file "Table_S20.xlsx"

**Table S21: Enriched gene ontology in biological process terms of positively selected genes in/close to domestic candidate regions.**

See file "Table_S21.xlsx"

**Table S22: Information of variant sites being performed luciferase assay to evaluate the activity.**

| Gene symbol | Variant position | Catalog of variant position | ΔAF | AF in wild | AF in domestic |
|---|---|---|---|---|---|
| *EFNA5* | KB744645.1: 114292,114293,114301 | 10 kb* | 0.949,0.916,0.740 | 0,0.916,0 | 0.949,0,0.741 |
| *EFNA5* | KB744645.1:58363 | 1$^{th}$ intron | 0.809 | 0.835 | 0.026 |
| *ISL1* | KB743873.1:25803 (indel) | 3' UTR | 0.875 | 0.875 | 0 |
| *MRPS27* | KB742554.1: 2151394,2151434,2151445 | 3' UTR | 0.960,0.419,0.417 | 0,0.419,0.419 | 0.960,0.419,0.419 |
| *GRIK1* | KB742951.1:543269 | 2$^{th}$ intron | 0.730 | 0.151 | 0.881 |
| *LYST* | KB743543.1:403502 | 5$^{th}$ intron | 0.611 | 0.685 | 0.074 |

Note: 10 kb* means 10 kb downstream of the gene.

**Table S23. Averaged all ΔDAF (absolute ΔDAF) of windows inside/outside high differentiation regions.**

| Mean value across SNPs | high differentiation regions | Genomic background |
|---|---|---|
| absolute ΔDAF | 0.0385672 | 0.03672404 |
| ΔDAF | 0.0370516 | 0.03497989 |

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